

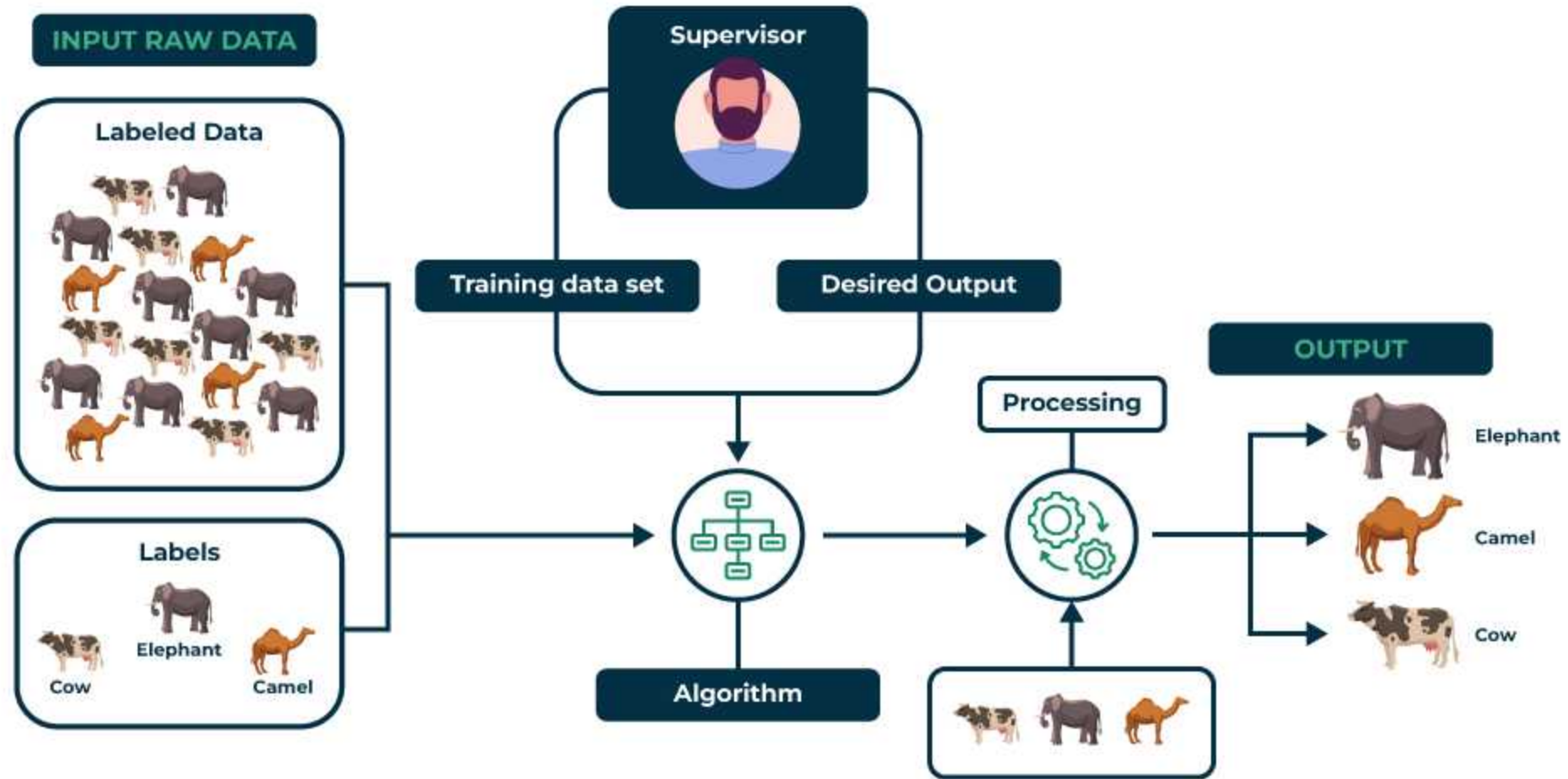
Machine Learning



- ◇ Chapter VII:
- ◇ Supervised Learning: Part II

◇ Prof. Dr. Aderval S. Luna

Supervised Learning



Supervised Learning

```
graph TD; A[Supervised Learning] --> B[Regression]; A --> C[Classification];
```

The diagram illustrates the structure of Supervised Learning. At the top, a blue rectangular box contains the text 'Supervised Learning'. Two arrows originate from the bottom of this box, pointing downwards to two separate ovals. The oval on the left is orange and contains the text 'Regression'. The oval on the right is light blue and contains the text 'Classification'.

Regression

Classification

Supervised Learning: Regression Tasks

- ◆ Regression in machine learning consists of mathematical methods that allow data scientists to predict a continuous outcome (y) based on the value of one or more predictor variables (x).
- ◆ <https://www.google.com/search?q=regression+in+machine+learning>

Supervised Learning Algorithms: Regression Tasks

- ◆ Ordinary Least Squares, OLS
- ◆ Weighted Least Squares, WLS
- ◆ Locally Weighted Linear Regression
- ◆ Multiple Linear Regression, MLR
- ◆ Principal Component Regression, PCR
- ◆ Partial Least Squares Regression, PLS
- ◆ Regularized Regression
 - Lasso
 - Ridge
 - Elastic Net
- ◆ Multivariate Adaptive Regression by Splines, MARS
- ◆ Model Trees, MT

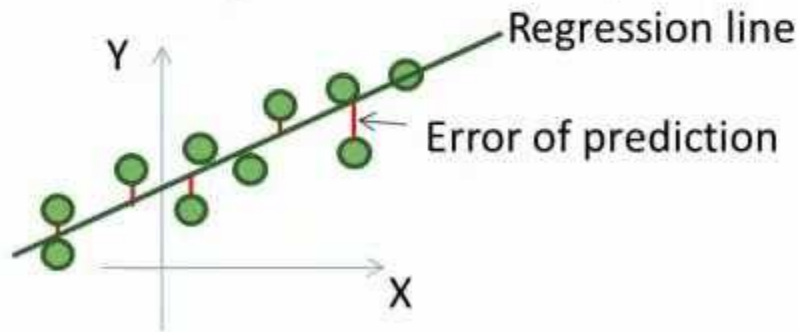
What is Ordinary Least Squares regression?

- X_i are the k explanatory variables and Y is a dependent variable.
- The model is linear - for each sample n , the value y_n is:

$$y_n = \sum_{i=0}^k \beta_i x_{ni} + \varepsilon_n$$

- The coefficients β are found by minimizing the error of prediction:

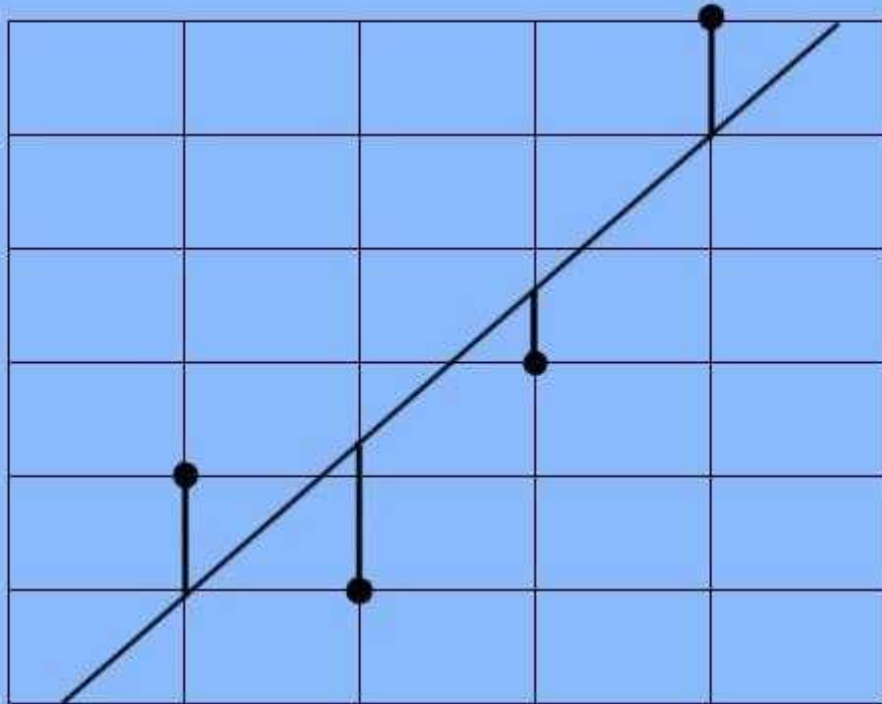
Simple linear
regression



Least Squares Method

['lēst 'skwers 'me-thəd]

A form of mathematical regression analysis used to determine the line of best fit for a set of data, providing a visual demonstration of the relationship between the data points.



Ordinary Least Squares, OLS

The diagram illustrates the Ordinary Least Squares (OLS) regression equation, $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$, with labels and arrows identifying each component:

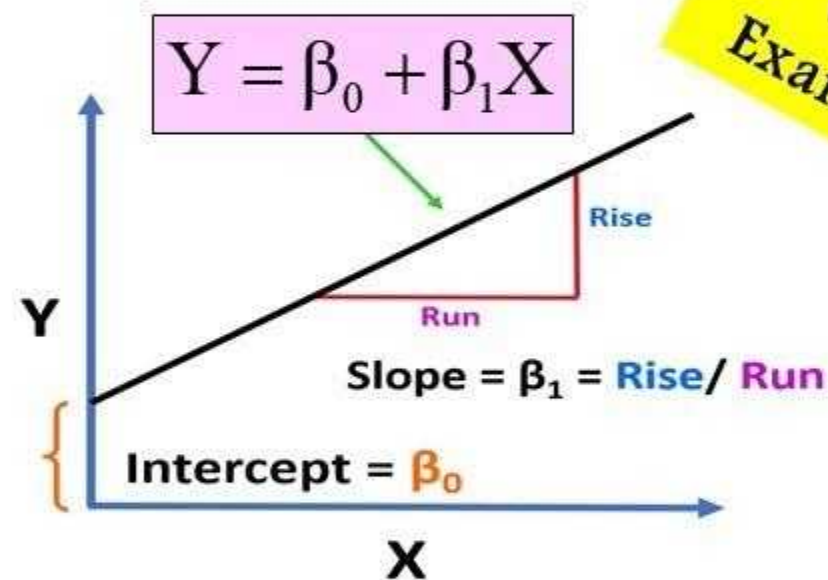
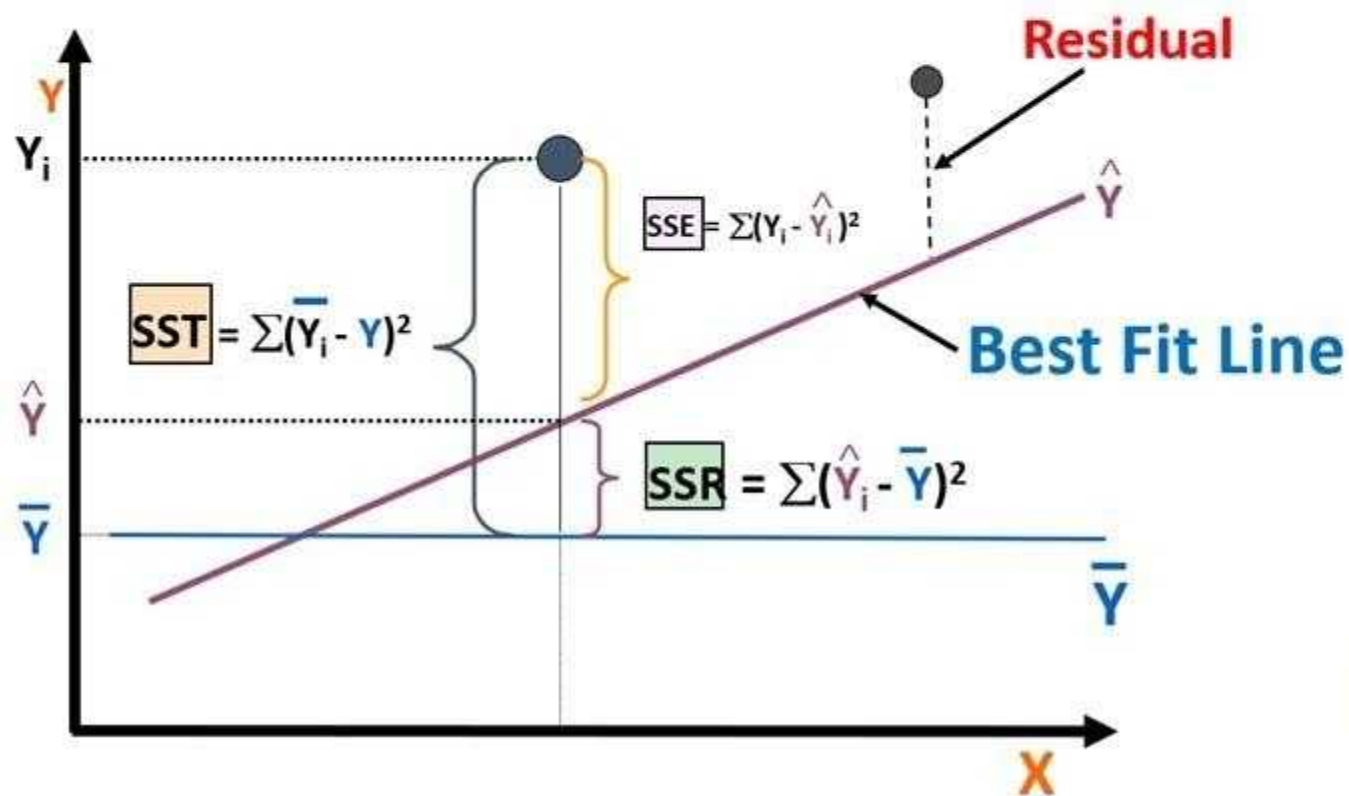
- Dependent Variable**: Points to Y_i .
- Population Y intercept**: Points to β_0 .
- Population Slope Coefficient**: Points to β_1 .
- Independent Variable**: Points to X_i .
- Random Error term**: Points to ϵ_i .

Below the equation, two blue curly braces group the terms into components:

- Linear component**: A brace under $\beta_0 + \beta_1 X_i$.
- Random Error component**: A brace under ϵ_i .

Linear Regression

Least Square Method



Explained
with
Examples

$$SST = SSR + SSE$$

$$SST = \sum (Y_i - \bar{Y})^2 = SSR = \sum (\hat{Y}_i - \bar{Y})^2 + SSE = \sum (Y_i - \hat{Y}_i)^2$$

Ordinary Least Squares, OLS

$$RSS(\beta_0, \beta_1) = \sum_{i=1}^n [Y_i - (\beta_0 + \beta_1 X_i)]^2 = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2.$$

Linear Least Squares

- **Linear Least Squares:**

$$0 = \frac{\partial \phi}{\partial a} = \sum_{k=0}^n 2x_k (ax_k + b - y_k)$$
$$0 = \frac{\partial \phi}{\partial b} = \sum_{k=0}^n 2 (ax_k + b - y_k)$$

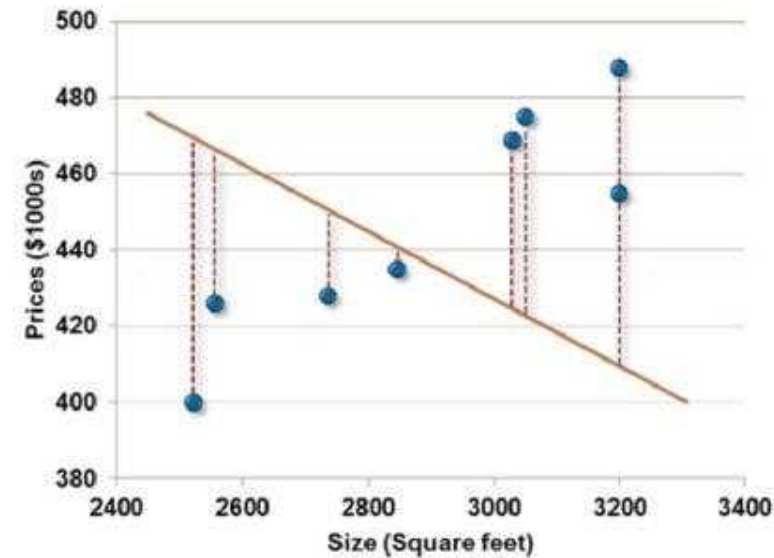
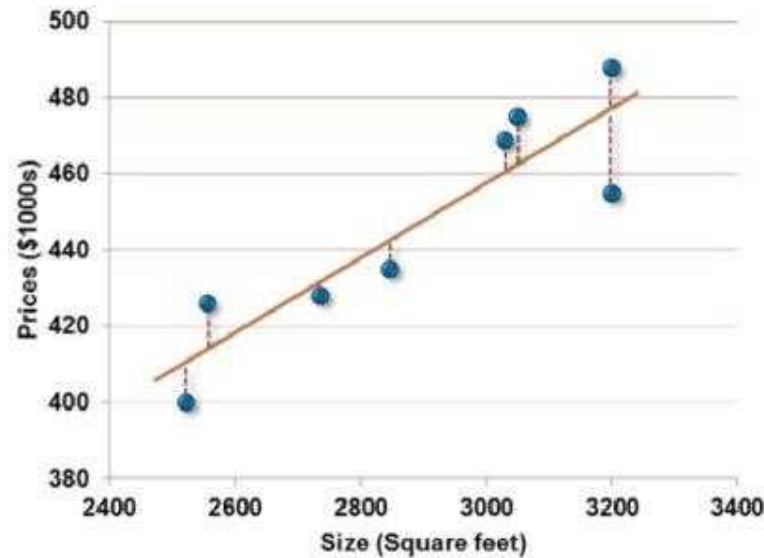
- These equations can be written as:

$$\begin{aligned}\sum x_k^2 a + \sum x_k b &= \sum x_k y_k \\ \sum x_k a + (n+1)b &= \sum y_k\end{aligned}$$

- These last two equations are called the **Normal Equations** for the best line fit to the data.

Least-Squares Regression Line

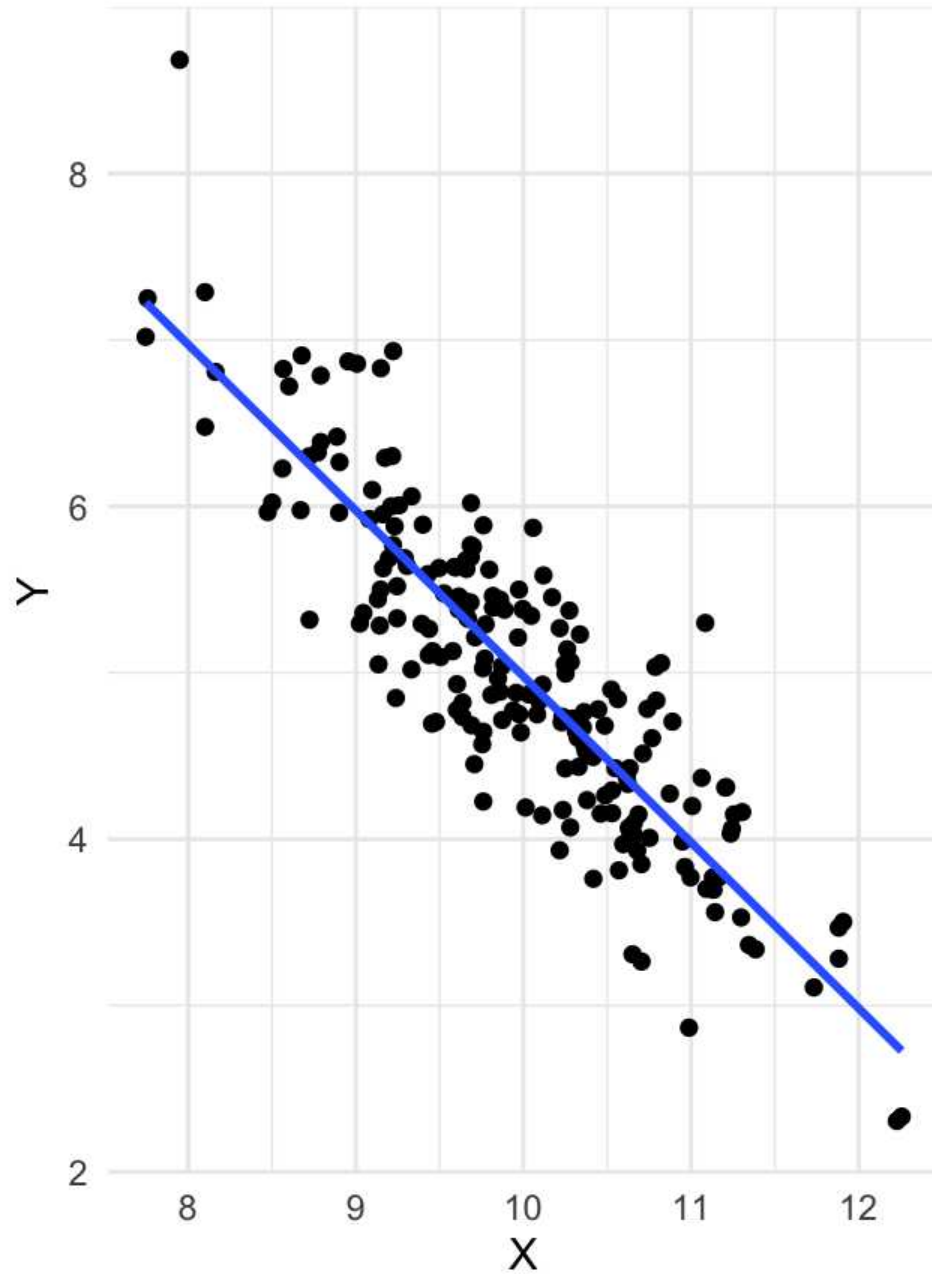
The figures present scatterplots of the previous data, each with a different line superimposed. It is clear that the line in the figure on the left fits better than the line in the figure on the right. The reason is that the vertical distances are, on the whole, smaller. The line that fits best is the line for which the sum of squared vertical distances is as small as possible. This line is called the **least-squares regression line**.



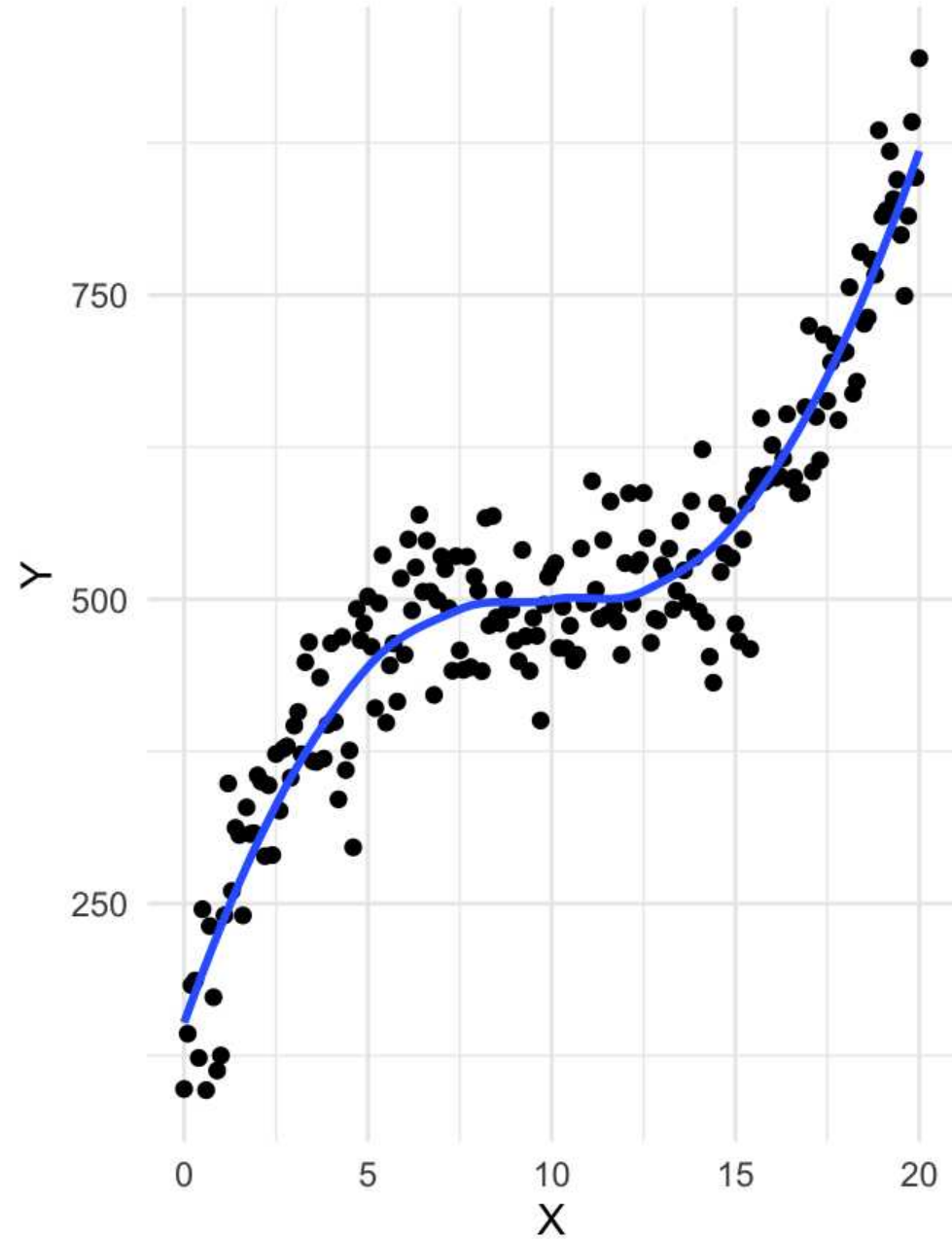
Limitations of the Least Squares Method

- ◊ Sensitivity to Outliers
- ◊ Overfitting
- ◊ Non-linearity
- ◊ Inefficient in the Presence of Heteroskedasticity
- ◊ Violation of Assumptions

Linearity is respected



Linearity is **not** respected

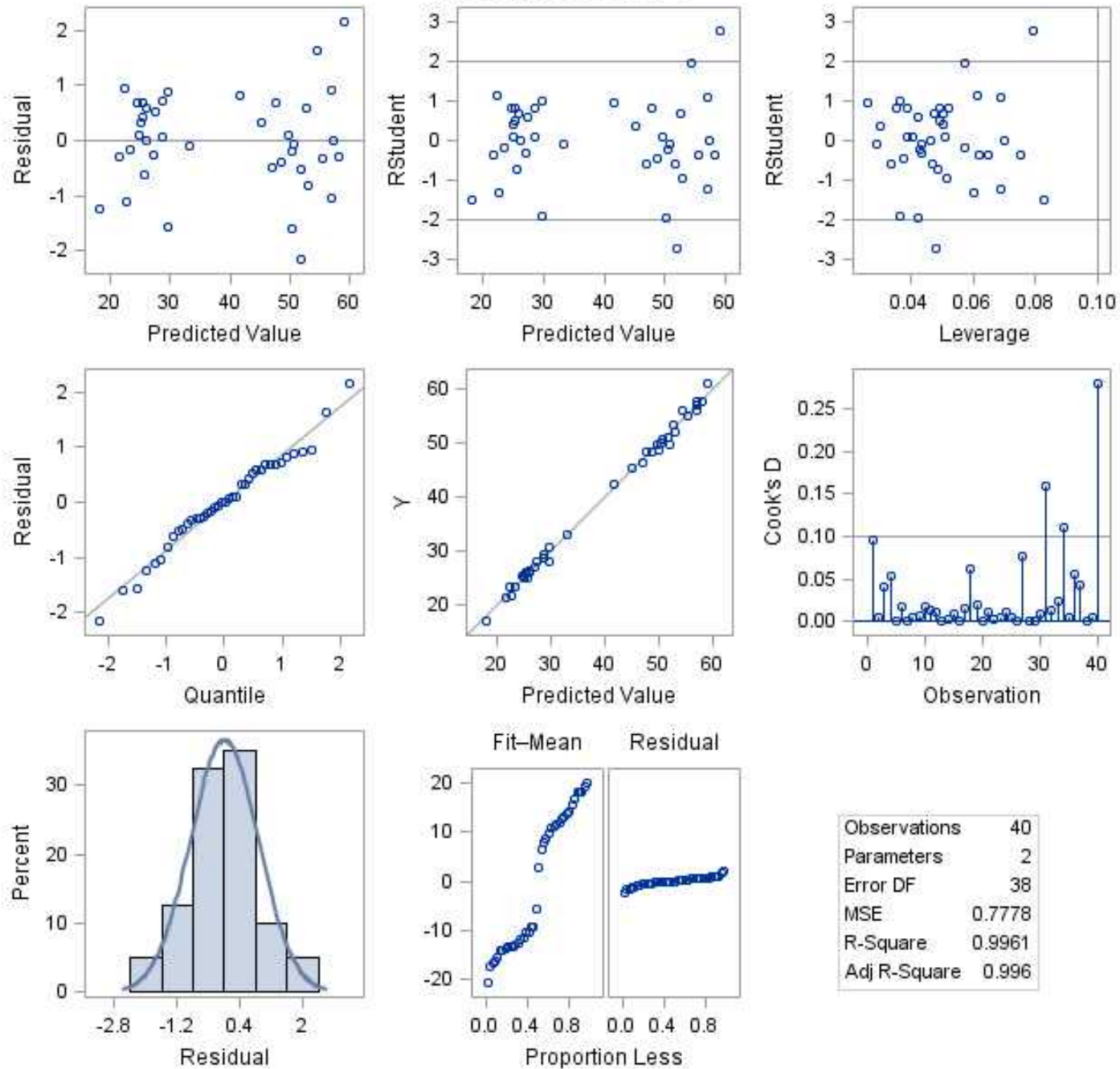


VIOLATION OF THE ASSUMPTIONS OF THE OLS

- ◆ The error term has an expected value of zero, $E(\varepsilon_i) = 0$
- ◆ The variance of the error term is constant (homoskedasticity)
- ◆ The covariance between the residuals is zero, $Cov(\varepsilon_i, \varepsilon_j) = 0$
- ◆ The covariance between the residual and the explanatory variables should be zero, i. e., $Cov(\varepsilon_i, X_j) = 0$
- ◆ The correlation among explanatory variables is zero, $Corr(X_i, X_j)$.

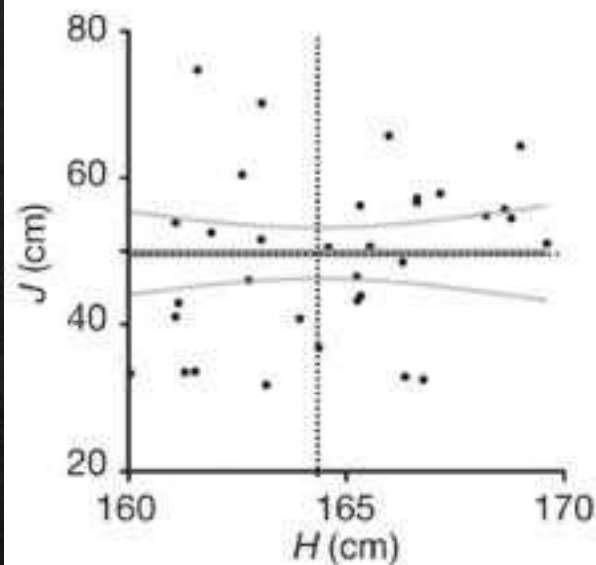
Violation	Consequence	Detection method
Non-normality of errors	F - and t -tests unreliable	-Normal Q-Q plots -Formal tests of normality
Non-linearity	$s_{\theta i}$ unreliable	-Partial residual plots -Scatter plots of Y on X_i
Heteroscedasticity	$s_{\theta i}$ unreliable	-Residual plots -White's test
Autocorrelation	$s_{\theta i}$ unreliable	-Residual plots -Durbin-Watson test
Multicollinearity	$s_{\theta i}$ unreliable t -test non-significant	-Variance inflation factor (VIF) -Tolerance
Measurement errors	Errors in X_i bias $\bar{\theta}_i$ towards 0 Errors in Y inflate $s_{\theta i}$	-Small t -values; $P > 0.05$ -Inflated $s_{\theta i}$ values
Outliers	$s_{\theta i}$ unreliable	-Studentized residuals -Scatter plots of Y on X_i

Fit Diagnostics for Y



a Uncorrelated predictors

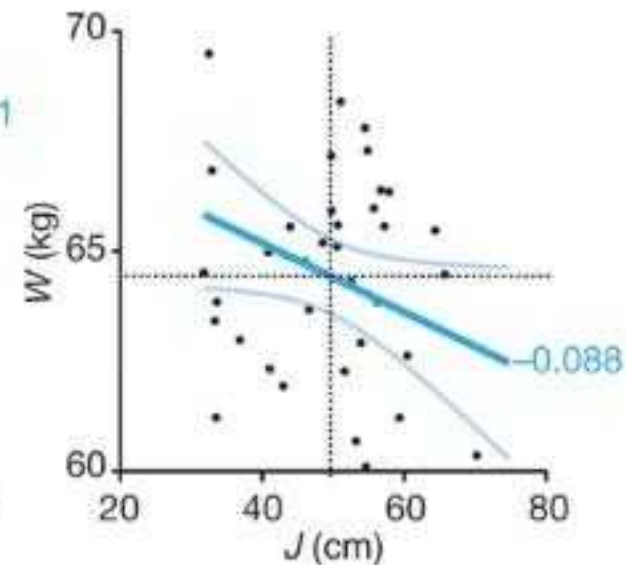
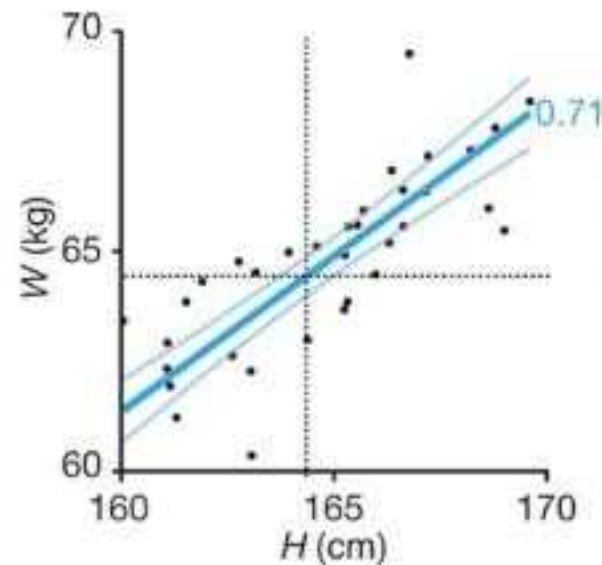
$$r(H,J) = 0$$

**b**

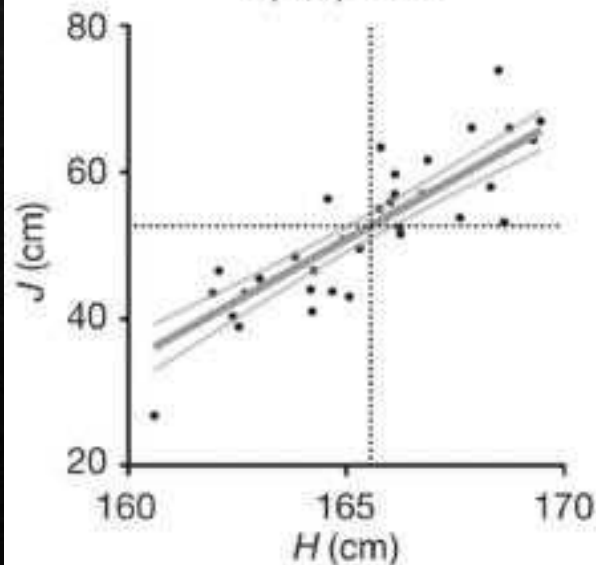
Regression for uncorrelated predictors

Weight on height

Weight on jump height

**c** Correlated predictors

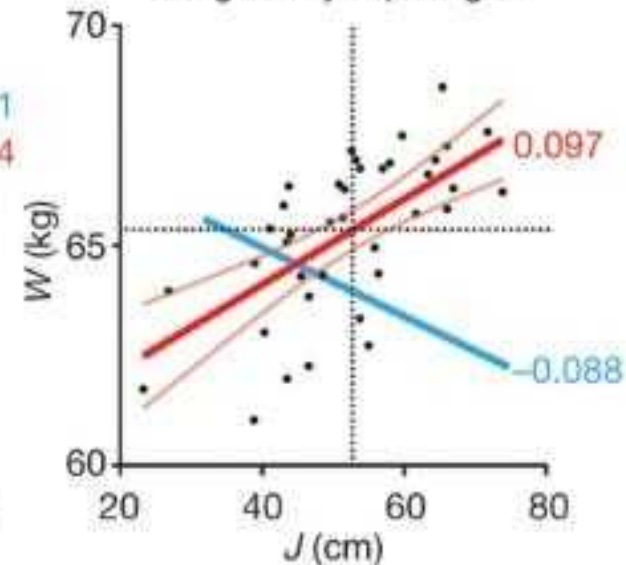
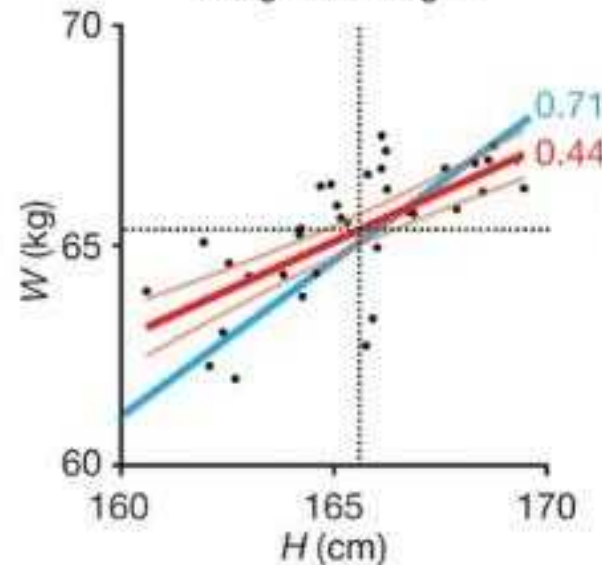
$$r(H,J) = 0.9$$

**d**

Regression for correlated predictors

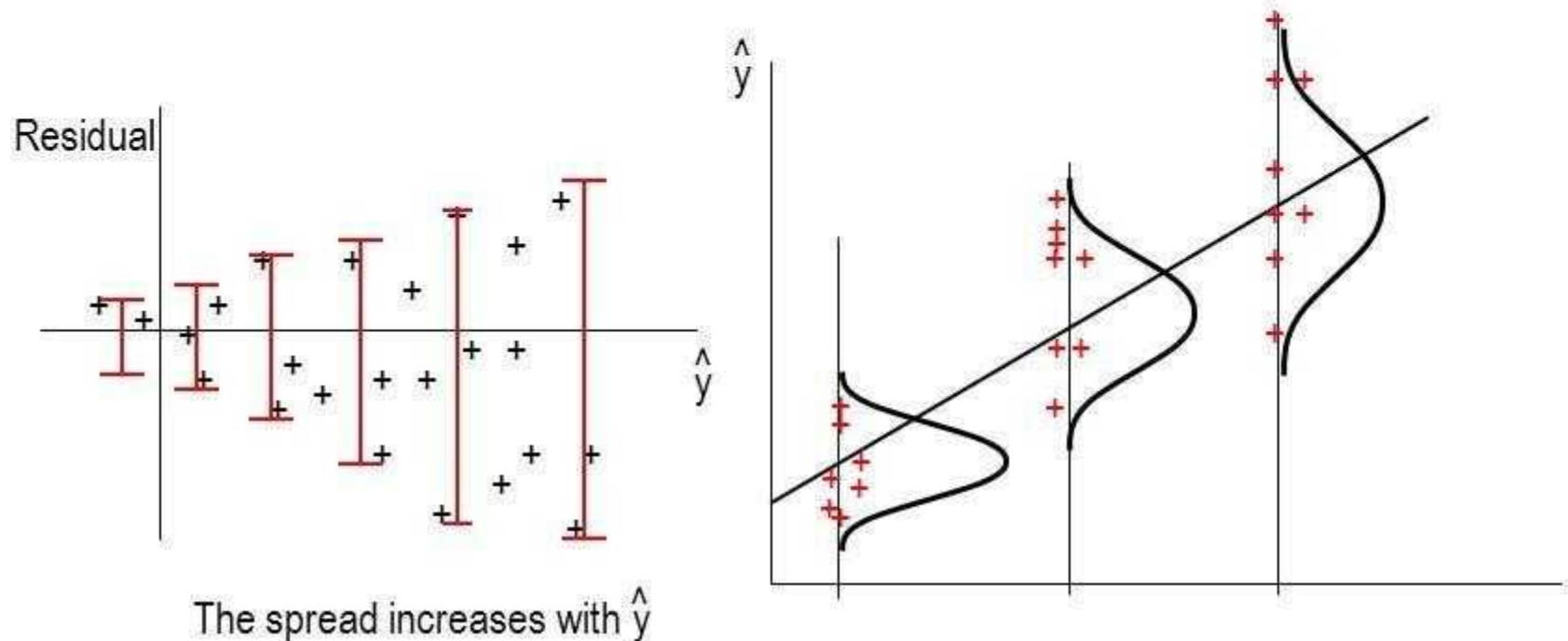
Weight on height

Weight on jump height



Heteroscedasticity

- When the requirement of a constant variance is violated we have a condition of heteroscedasticity.
- Diagnose heteroscedasticity by plotting the residual against the predicted y .



Weighted Least Squares, WLS

- ♦ WLS, also known as weighted linear regression, is a generalization of ordinary least squares and linear regression in which knowledge of the unequal variance observations (*heteroscedasticity*) is incorporated into the regression.
- ♦ https://en.wikipedia.org/wiki/Weighted_least_squares

Consider the weighted simple linear regression with responses $y_1, \dots, y_n$, covariate values $x_1, \dots, x_n$ and known weights $w_1, \dots, w_n$.

(a) Define the weighted means of x and y to be

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i} \quad \bar{y}_w = \frac{\sum w_i y_i}{\sum w_i}.$$

Show that $\sum_{i=1}^n w_i (x_i - \bar{x}_w) = 0$, and hence that $\sum_{i=1}^n w_i x_i (x_i - \bar{x}_w) = \sum_{i=1}^n w_i (x_i - \bar{x}_w)^2$ and that

$$\sum_{i=1}^n w_i x_i (y_i - \bar{y}_w) = \sum_{i=1}^n w_i (x_i - \bar{x}_w) (y_i - \bar{y}_w)$$

(b) Derive the weighted least squares estimates in the weighted simple linear regression setting:

$$\hat{\beta}_0 = \bar{y}_w - \hat{\beta}_1 \bar{x}_w$$

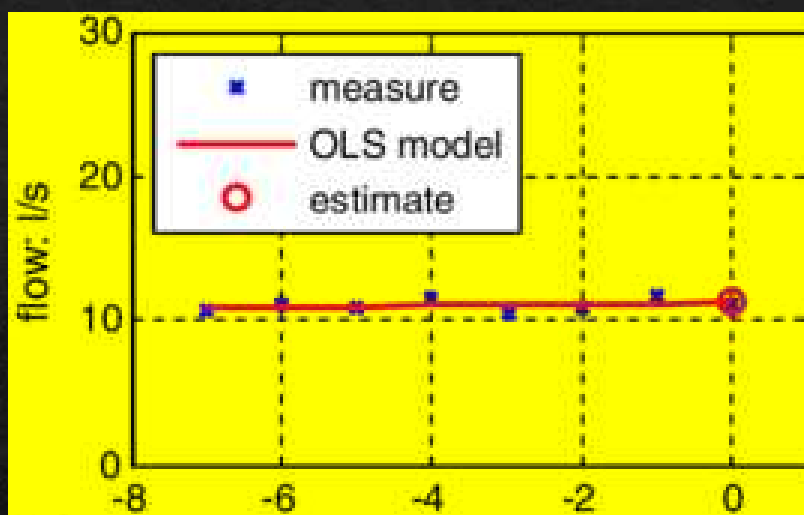
$$\hat{\beta}_1 = \frac{\sum w_i (x_i - \bar{x}_w) (y_i - \bar{y}_w)}{\sum w_i (x_i - \bar{x}_w)^2}$$

(c) Show that these estimators can be written as

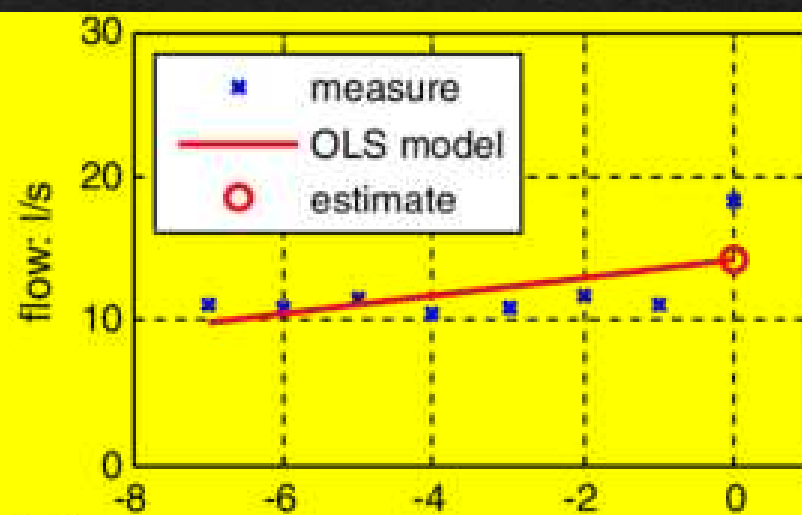
$$\hat{\beta}_1 = \sum_{i=1}^n \frac{w_i (x_i - \bar{x}_w)}{\sum w_j (x_j - \bar{x}_w)^2} Y_i$$

$$\hat{\beta}_0 = \sum_{i=1}^n \left(\frac{w_i}{\sum w_j} - \frac{w_i \bar{x}_w (x_i - \bar{x}_w)}{\sum w_j (x_j - \bar{x}_w)^2} \right) Y_i$$

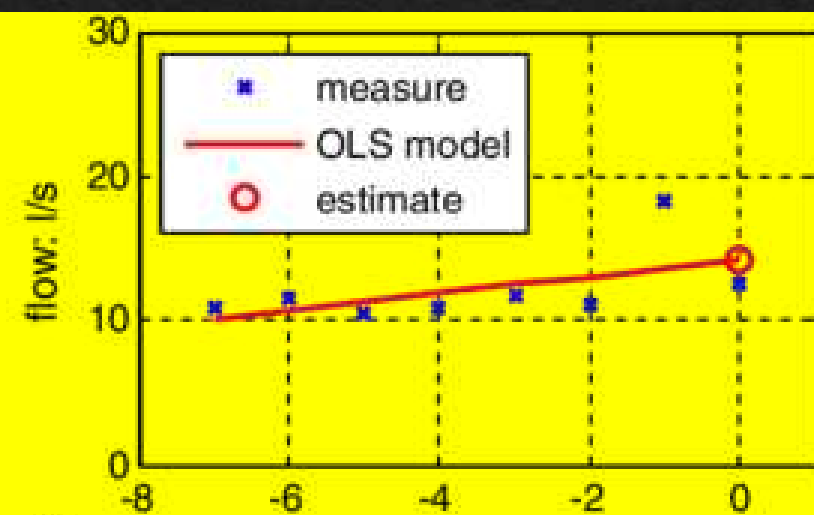
and hence that the estimators are unbiased.



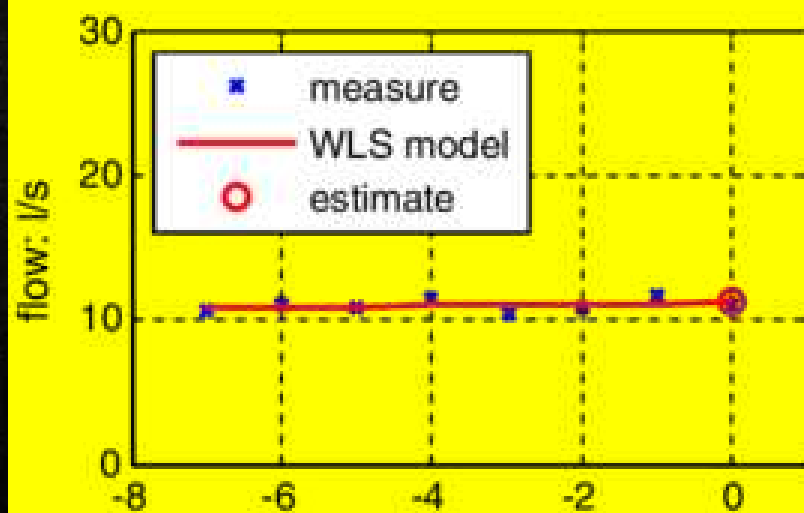
(a) Weeks from-Jul/31/2008 11am



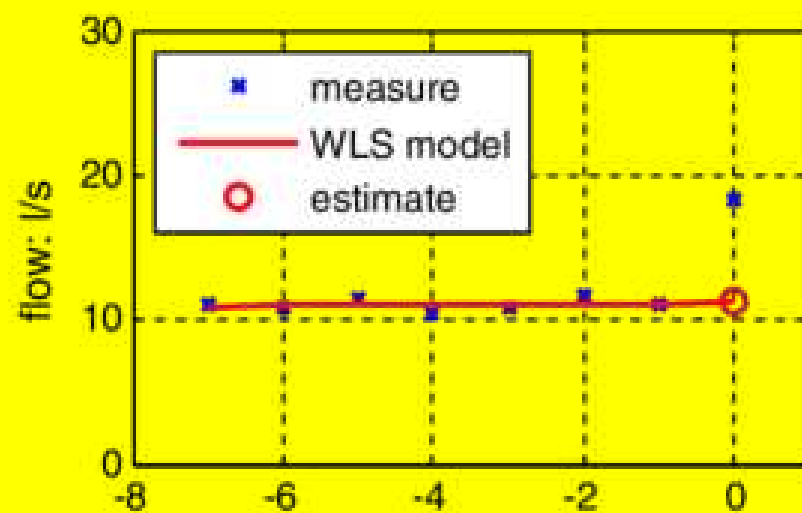
(b) Weeks from-Aug/07/2008 11am



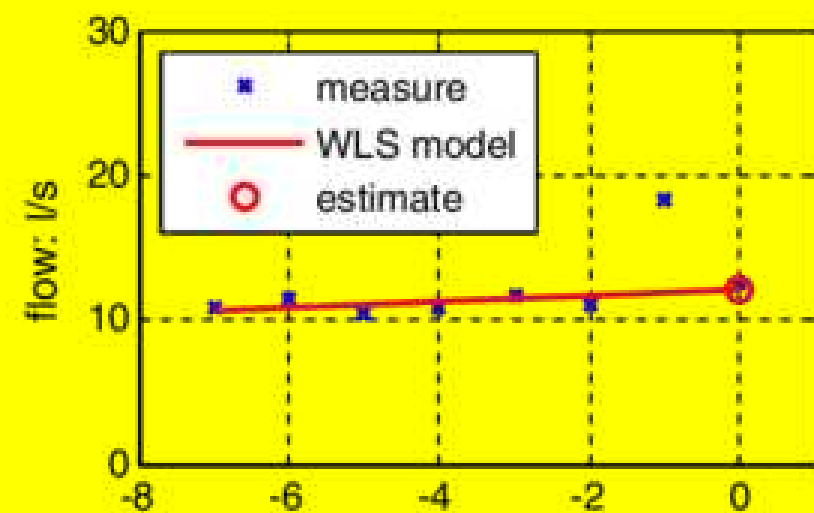
(c) Weeks from-Aug/14/2008 11am



(d) Weeks from-Jul/31/2008 11am



(e) Weeks from-Aug/07/2008 11am



(f) Weeks from-Aug/14/2008 11am

Application of Weighted Least Squares Regression in Forecasting

Sulaimon Mutiu O.

Department of Statistics & Mathematics Moshood Abiola Polytechnic, Abeokuta, Ogun State, Nigeria

Abstract: This work models the loss of properties from fire outbreak in Ogun State using Simple Weighted Least Square Regression. The study covers (secondary) data on fire outbreak and monetary value of properties loss across the twenty (20) Local Government Areas of Ogun state for the year 2010. Data collected were analyzed electronically using SPSS 21.0. Results from the analysis reveal that there is a very strong positive relationship between the number of fire outbreak and the loss of properties; this relationship is significant. Fire outbreak exerts significant influence on loss of properties and it accounts for approximately 91.2% of the loss of properties in the state.

Keywords: Fire Outbreak, Heteroscedasticity, Properties.

I. INTRODUCTION

The method of least squares is a standard approach in regression analysis to the approximate solution of overdetermined systems, i.e., sets of equations in which there are more equations than unknowns. "Least squares" means that the overall solution minimizes the sum of the squares of the errors made in the results of every single equation.

The most important application is in data fitting. The best fit in the least-squares sense minimizes the sum of squared residuals, a residual being the difference between an observed value and the fitted value provided by a model. When the problem has substantial uncertainties in the independent variable (the x variable), then simple regression and least squares methods have problems; in such cases, the methodology required for fitting errors-in-variables models may be considered instead of that for least squares.

Least squares problems fall into two categories: linear or ordinary least squares and non-linear least squares, depending on whether or not the residuals are linear in all unknowns. The linear least-squares problem occurs in statistical regression analysis; it has a closed-form solution. The non-linear problem is usually solved by iterative refinement; at each iteration the system is approximated by a linear one, and thus the core calculation is similar in both cases.

One of the common assumptions underlying most process modelling methods, including linear and nonlinear least squares regression, is that each data point provides equally precise information about the deterministic part of the total process variation. In other words, it is assumed that the standard deviation of the error term is constant over all values of the predictor or explanatory variables. This assumption, however, clearly does not hold, even approximately, in every modelling application.

The usual linear regression model, $y = a + bx$ assumes that all the random error components are identically and independently distributed with constant variance. When this assumption is violated, then ordinary least squares estimator of regression coefficient loses its property of minimum variance in the class of linear and unbiased estimators. The violation of such assumption can arise in anyone of the following situations:

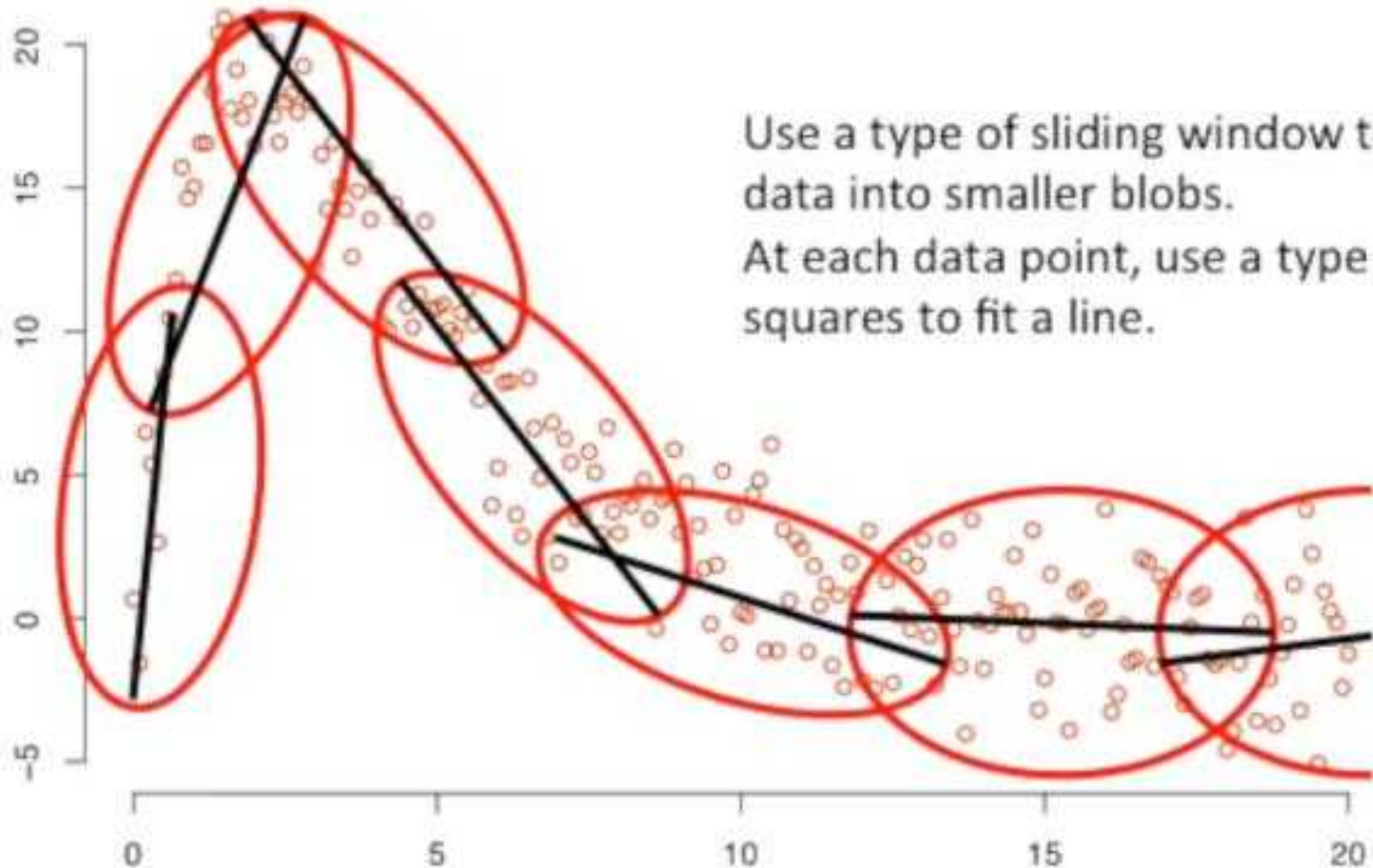
1. The variance of random error components is not constant.

Locally Weighted Regression

Local means using nearby points, model is only trained on data near the query point

Weighted means we value points based on how far they are from the query point

Regression means approximating a function
(i. e. linear regression approximates $y = mx + b$)



Use a type of sliding window to divide the data into smaller blobs.
At each data point, use a type of least squares to fit a line.

Locally Weighted Regression

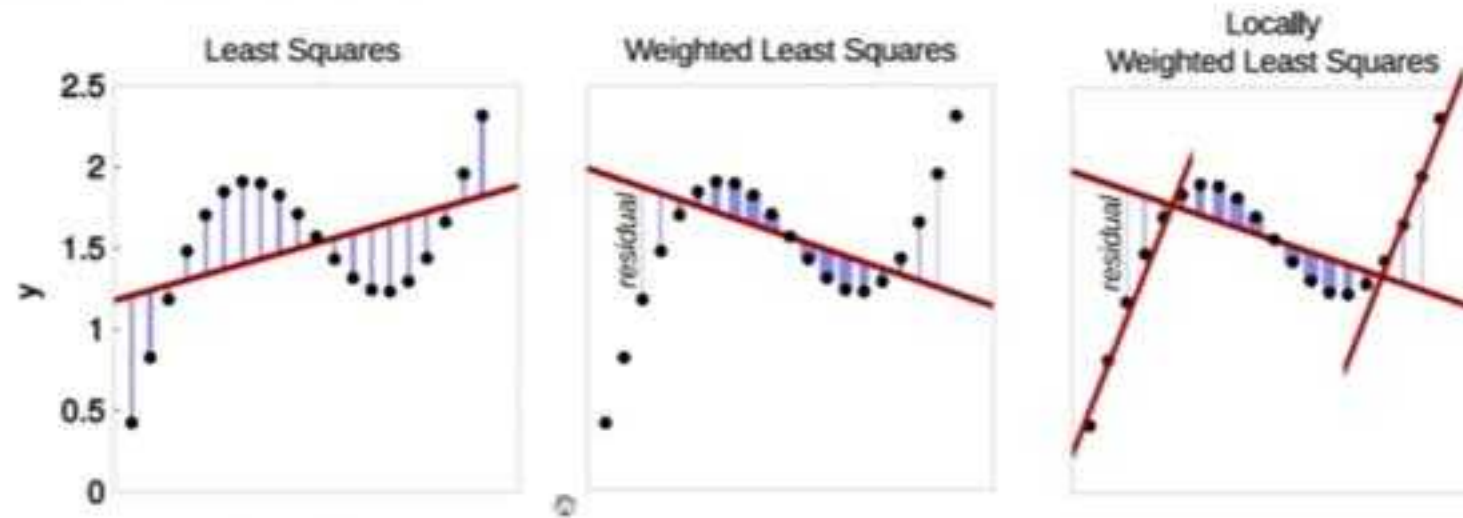


Figure: The thickness of the lines indicates the weights.

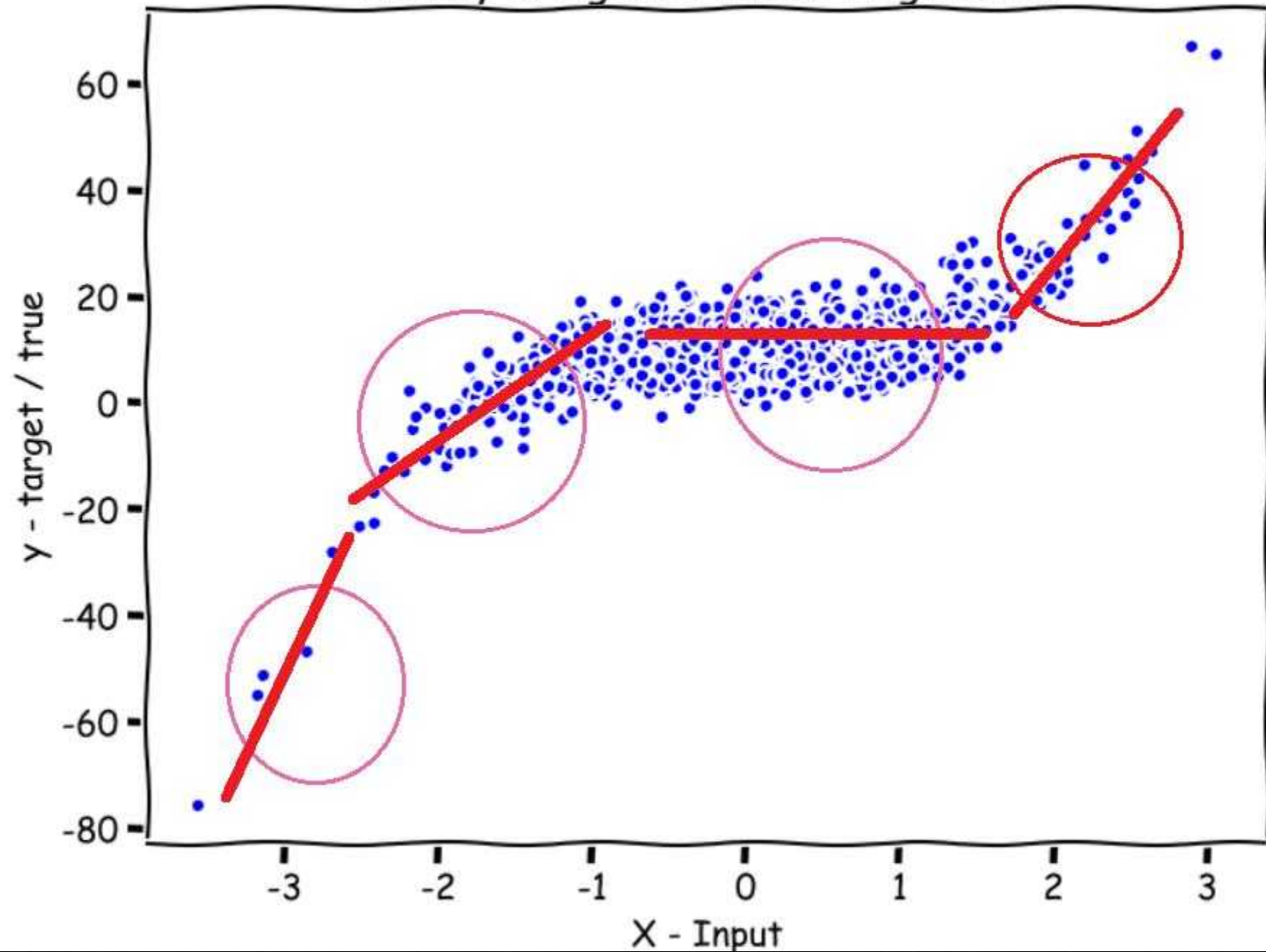
- ▶ General idea:
 - ▶ Split the function domain into linear parts
 - ▶ Give more weight to the centers
- ▶ Local linear models are tuned with Least Squares
- ▶ The importance of datapoints is represented by a Gaussian function

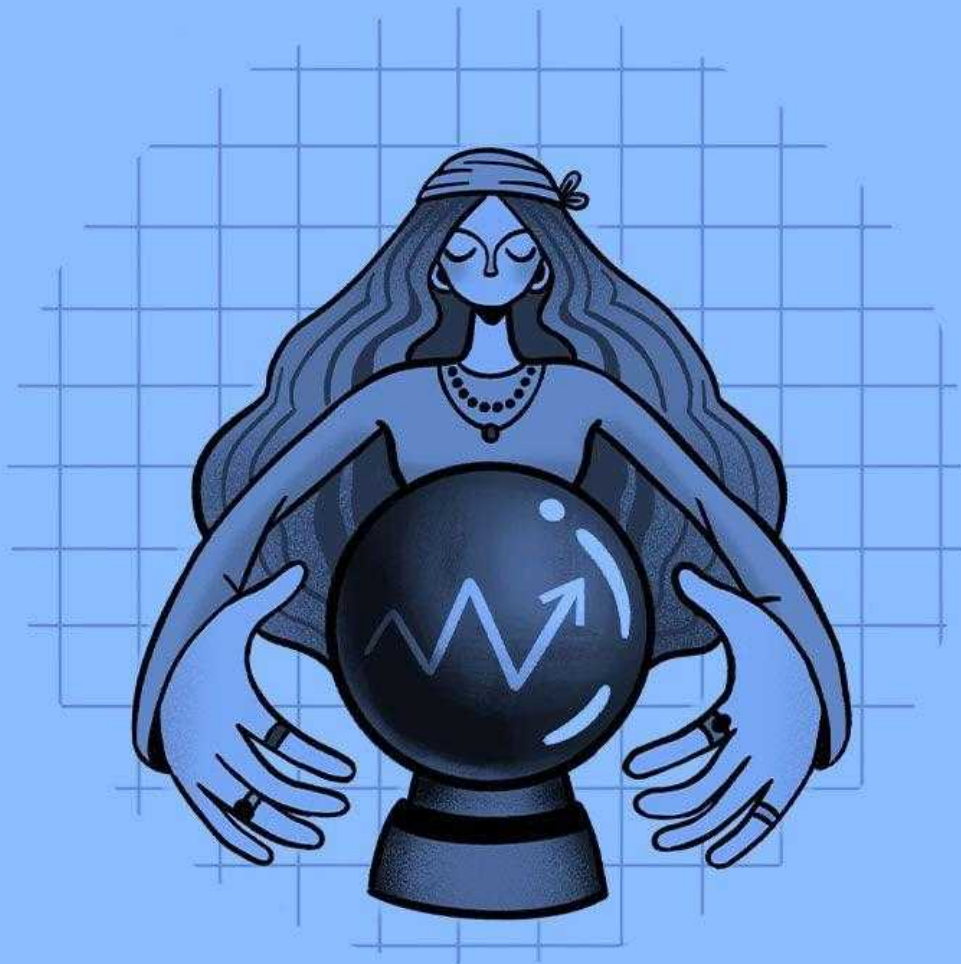


William S Cleveland and Susan J Devlin (1988) Locally weighted regression: an approach to regression analysis by local fitting.
Journal of the American statistical association, 83(403):596-610.



Locally Weighted Linear Regression





Multiple Linear Regression (MLR)

['mæl-tə-pəl 'li-nē-ər ri-'gre-shən]

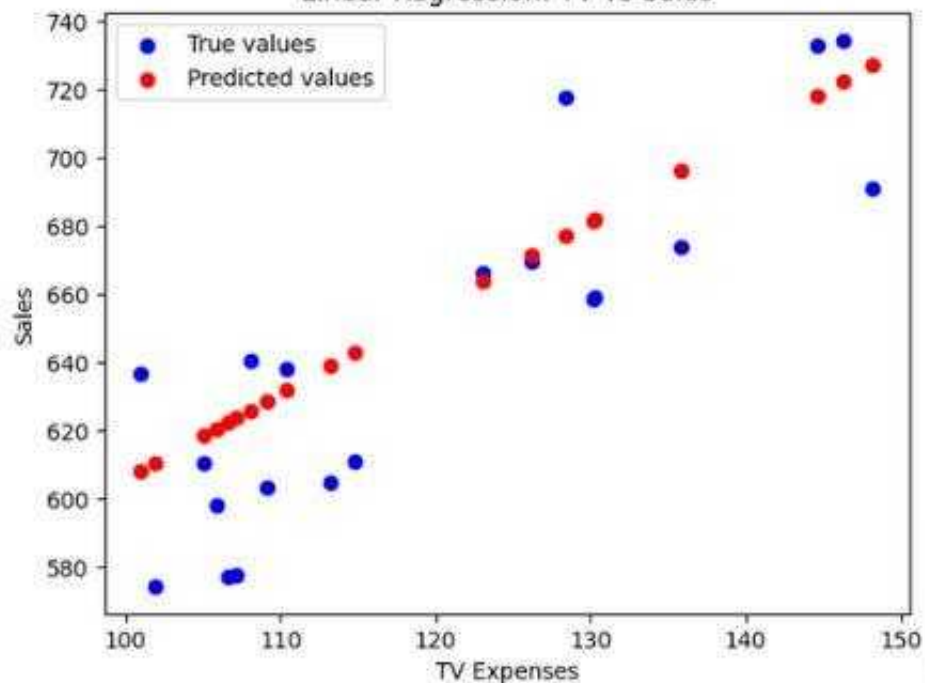
A statistical technique that uses several explanatory variables to predict the outcome of a response variable.

LINEAR REGRESSION

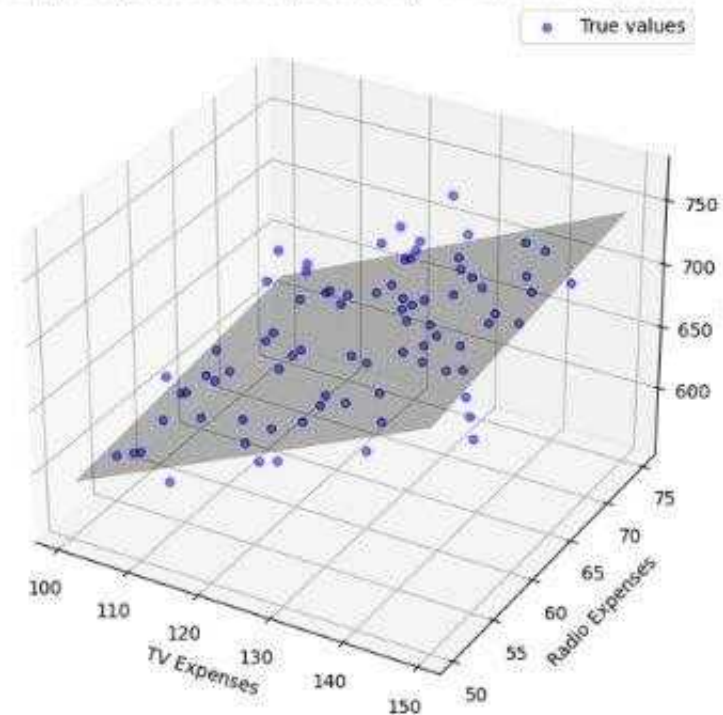


MULTIPLE REGRESSION

Linear Regression: TV vs Sales



Multiple Regression: Sales predicted by TV and Radio Expenses



What is Multiple Linear Regressions?

Multiple Linear Regressions (MLR)

A statistical means used extensively to model relationship between variables

It determines the relationship between one dependent and several independent variables

Is an extension of simple linear regression

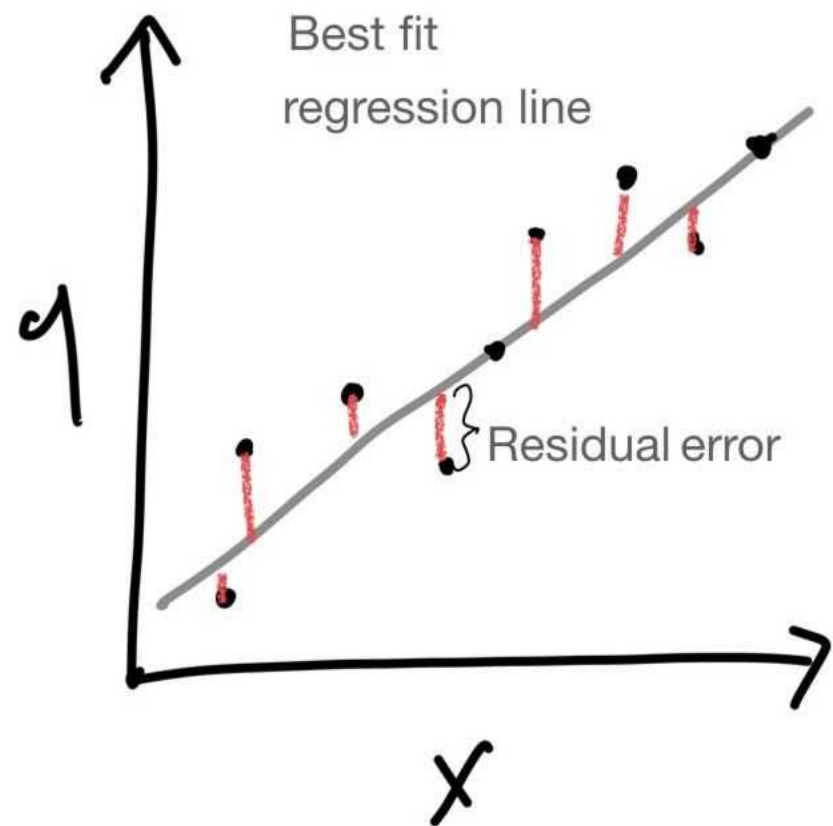
It aids in making prediction and conclusion regarding the involved components

The MLR model is based on ordinary least squares

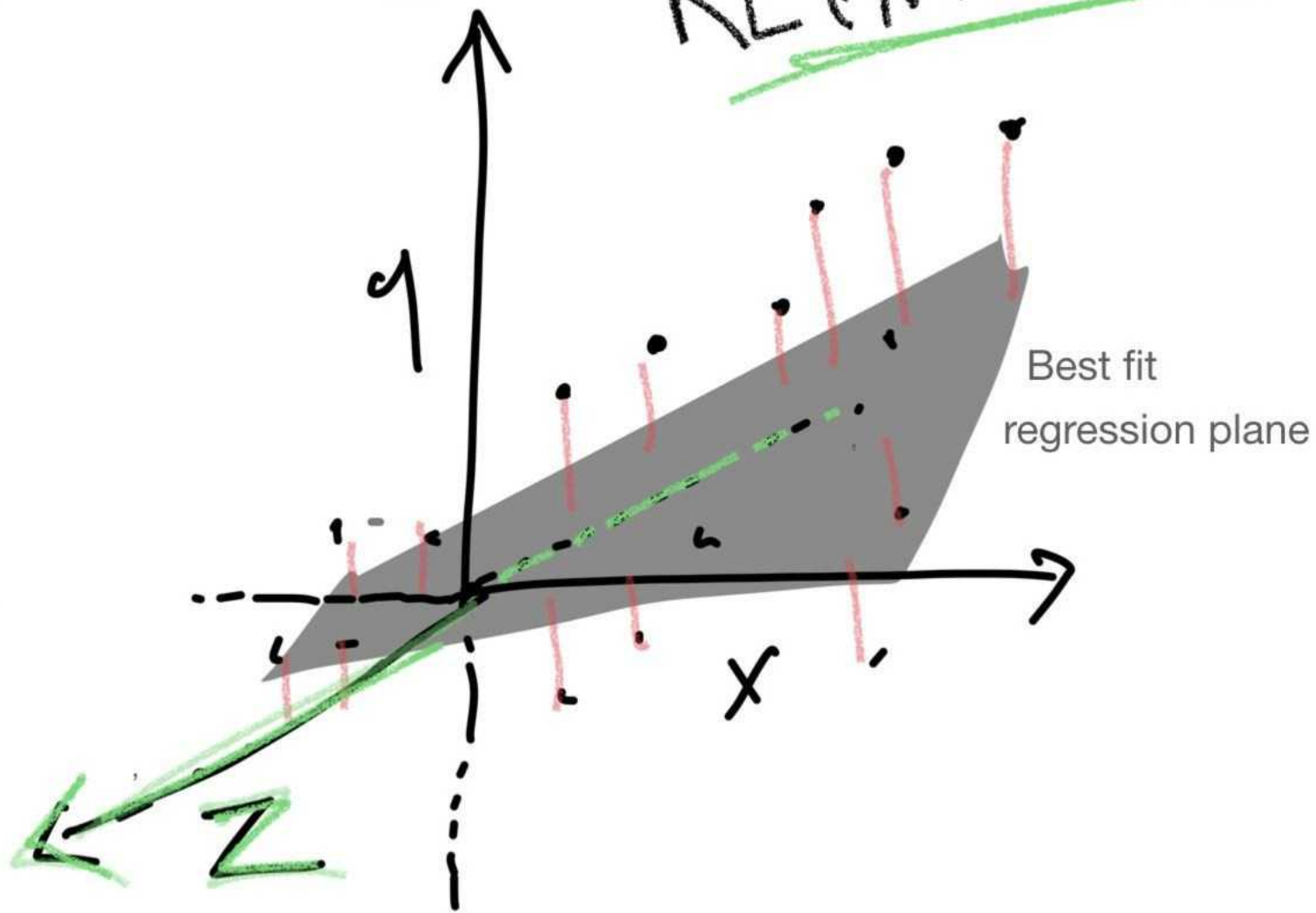
Helps making statistical prediction of components

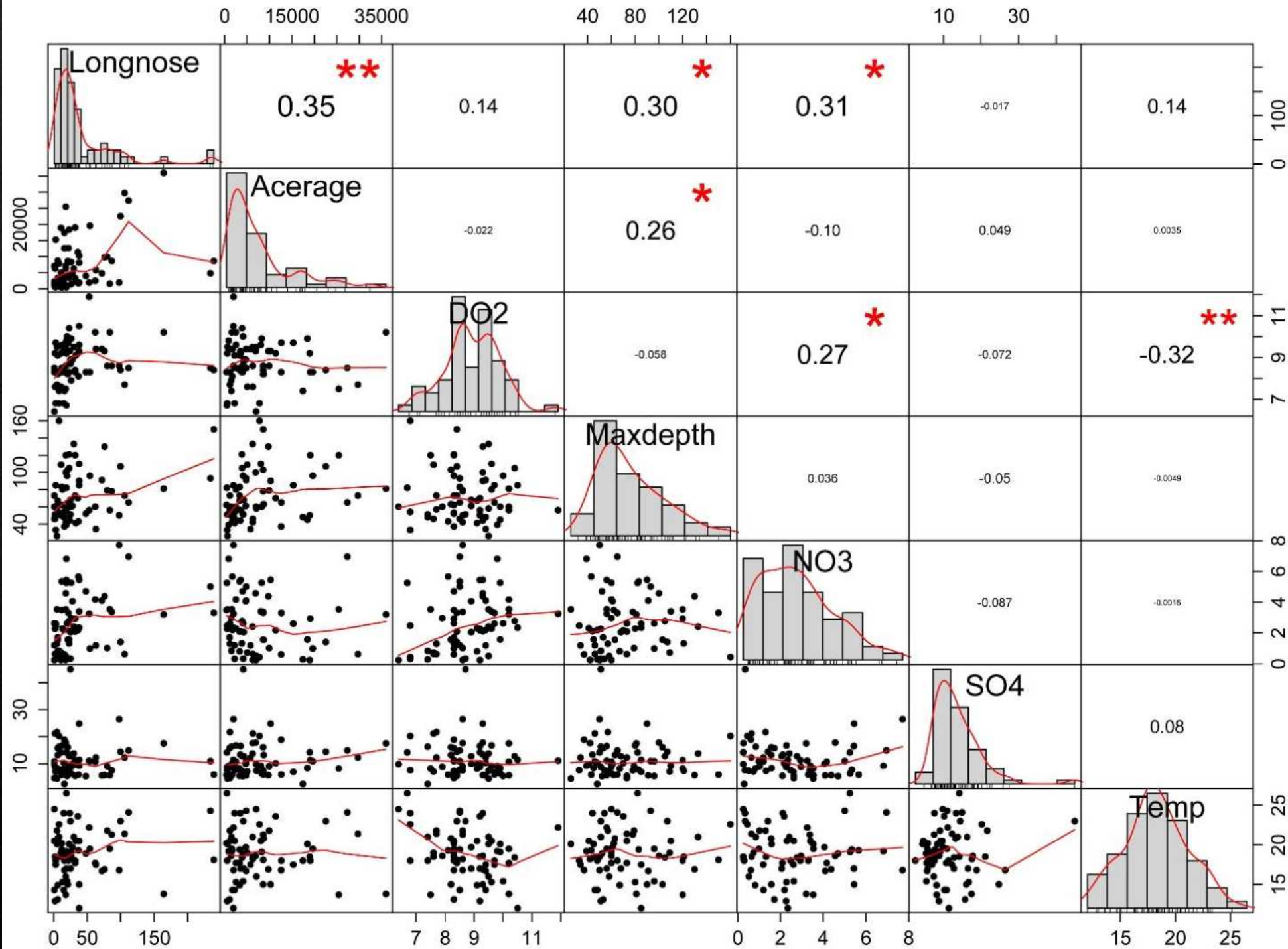


SIMPLE LINEAR REGRESSION

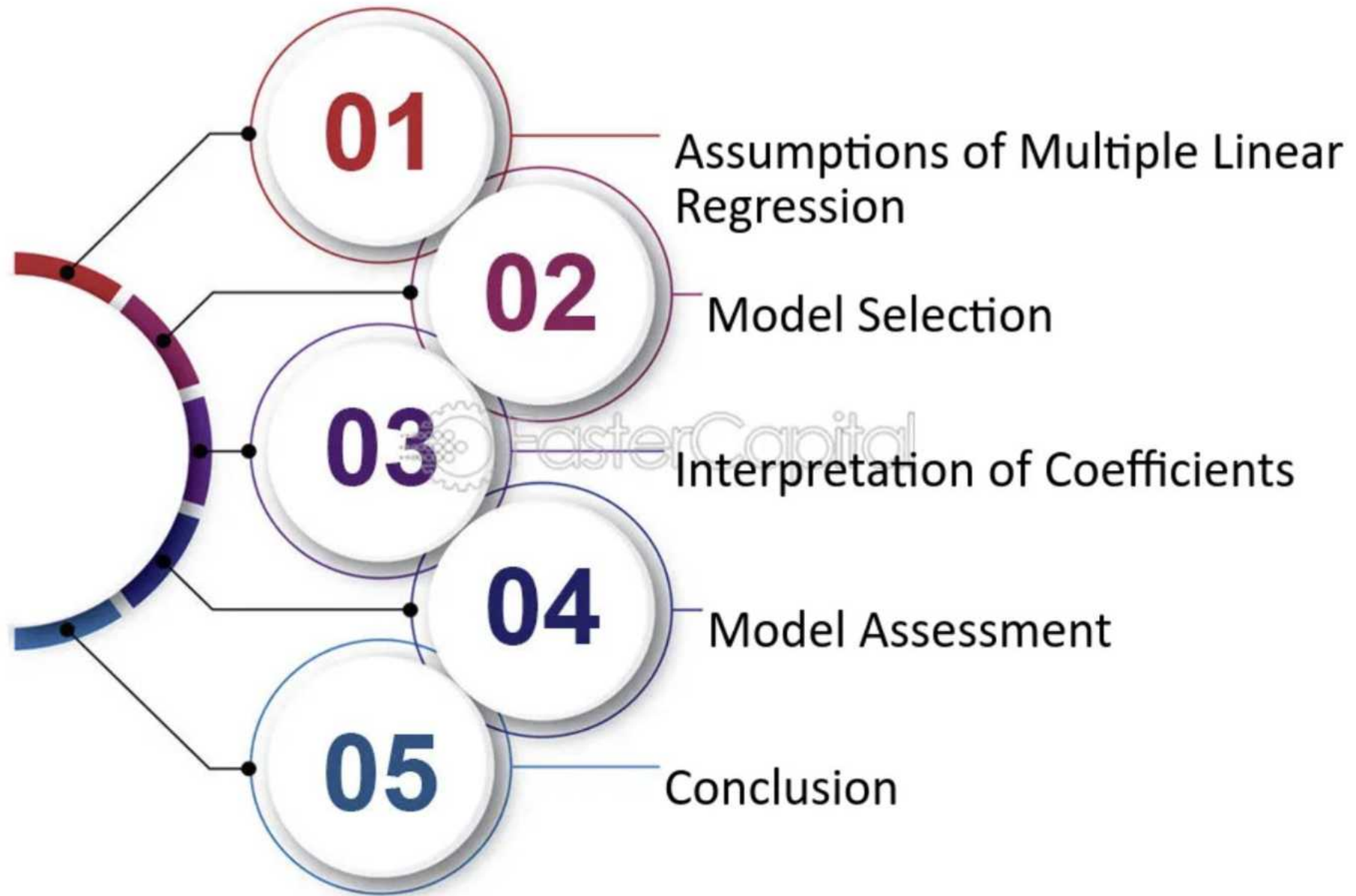


MULTIPLE REGRESSION



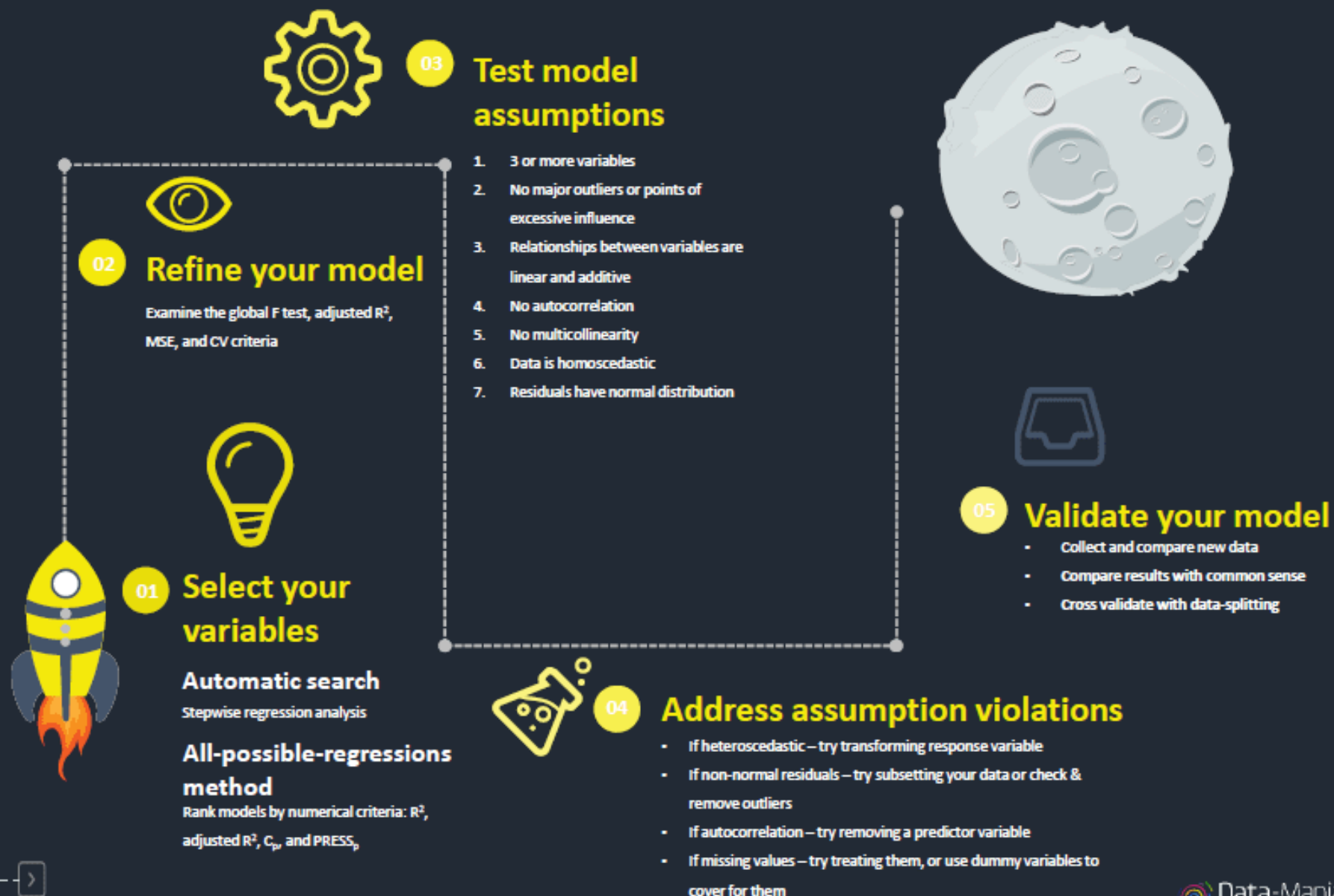


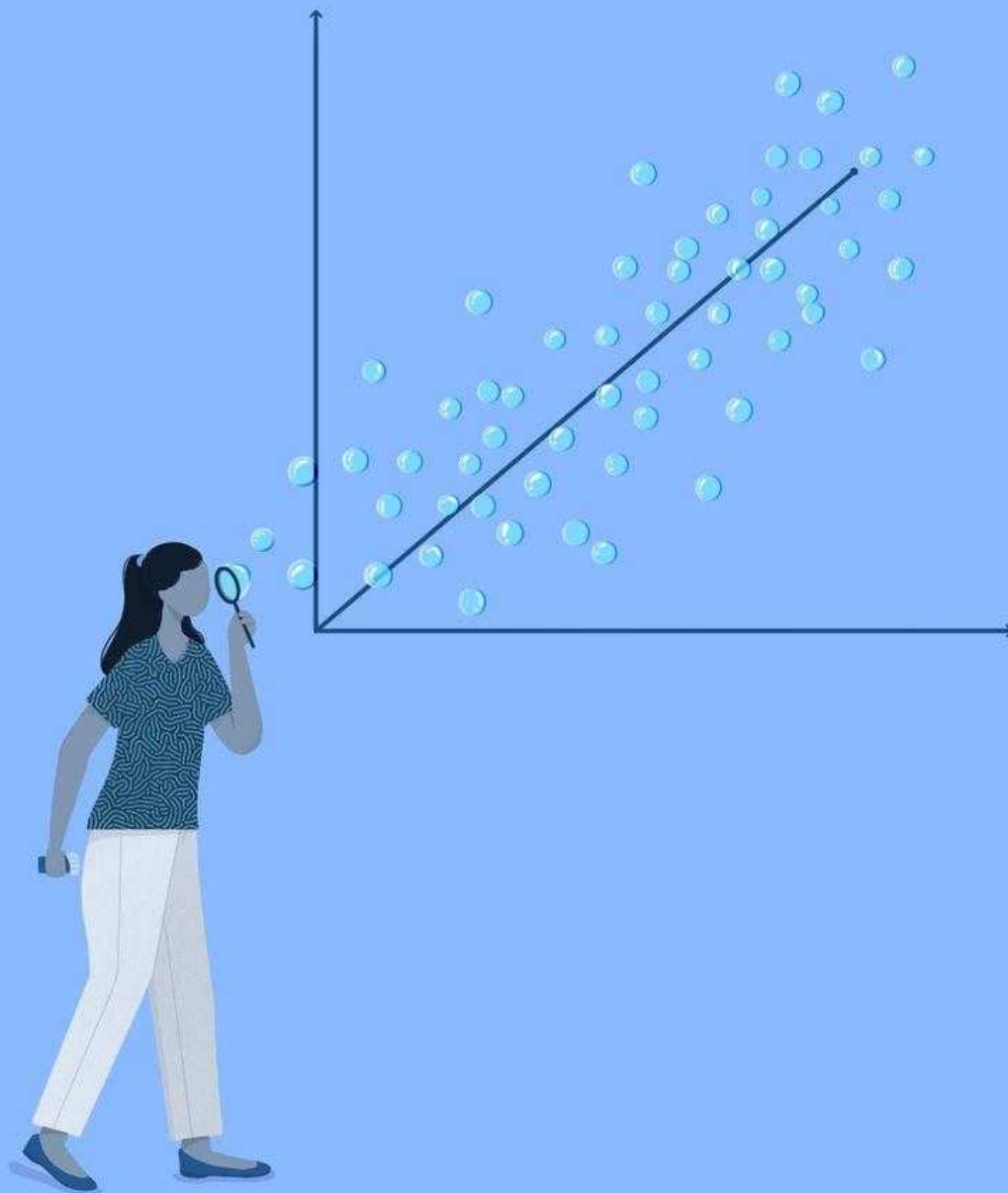
Multiple Linear Regression



5 STEP WORKFLOW FOR MULTIPLE LINEAR REGRESSION

1





Multicollinearity

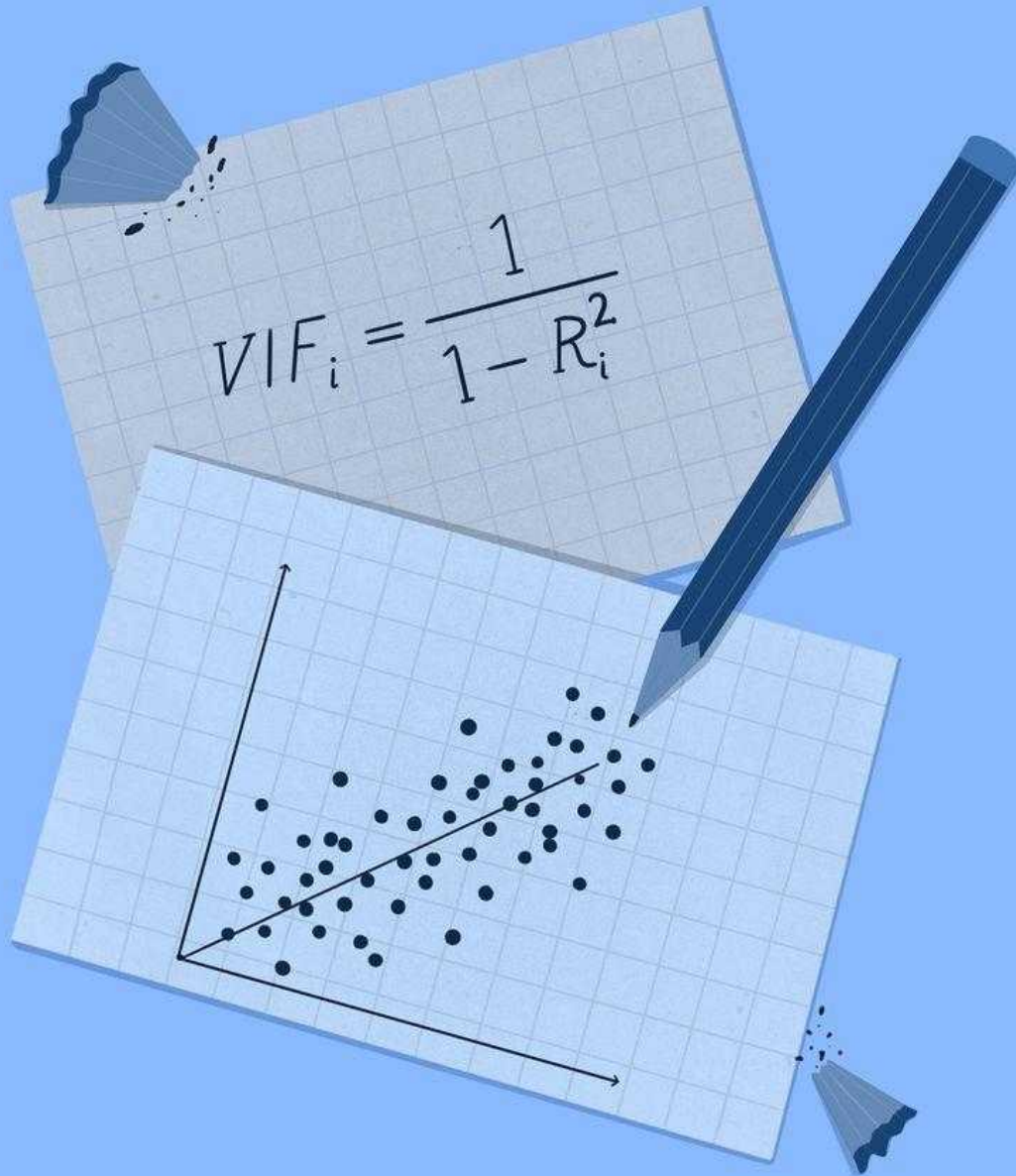
[,məl-tē-kə-,li-nē-'er-ə-tē]

When there is strong correspondence among two or more independent variables in a multiple regression model.

Variance Inflation Factor (VIF)

['ver-ē-ən(t)s in-'flā-shən 'fak-tər]

A measure of the amount of multicollinearity in a set of multiple regression variables.



Techniques for Selecting the "Best"

Regression Equation

- The best Regression equation is not necessarily the equation that explains most of the variance in Y (the highest R^2).
- This equation will be the one with all the variables included.
- The best equation should also be simple and interpretable. (i.e. contain a small no. of variables).
- Simple (interpretable) & Reliable - opposing criteria.
- The best equation is a compromise between these two.

Methods of Regression

```
graph TD; A[Methods of Regression] --> B[Enter (Forced entry)]; A --> C[Stepwise]; A --> D[Remove (Forced Remove)]; C --> E[Forward Selection]; C --> F[Backward Elimination];
```

Enter
(Forced entry)

Stepwise

Remove
(Forced Remove)

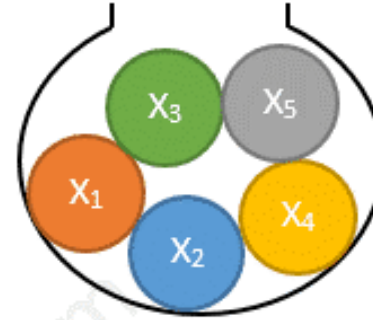
Forward
Selection

Backward
Elimination

Forward stepwise selection example with 5 variables:

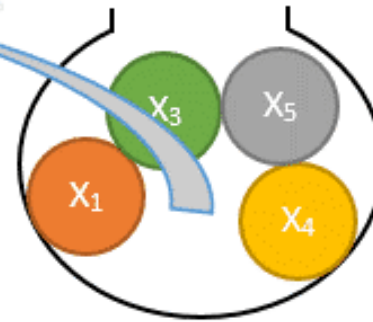
Start with a model with no variables

Null Model



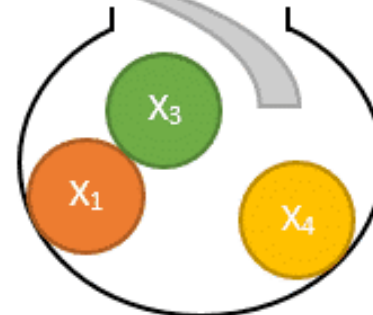
Add the most significant variable

Model with 1 variable

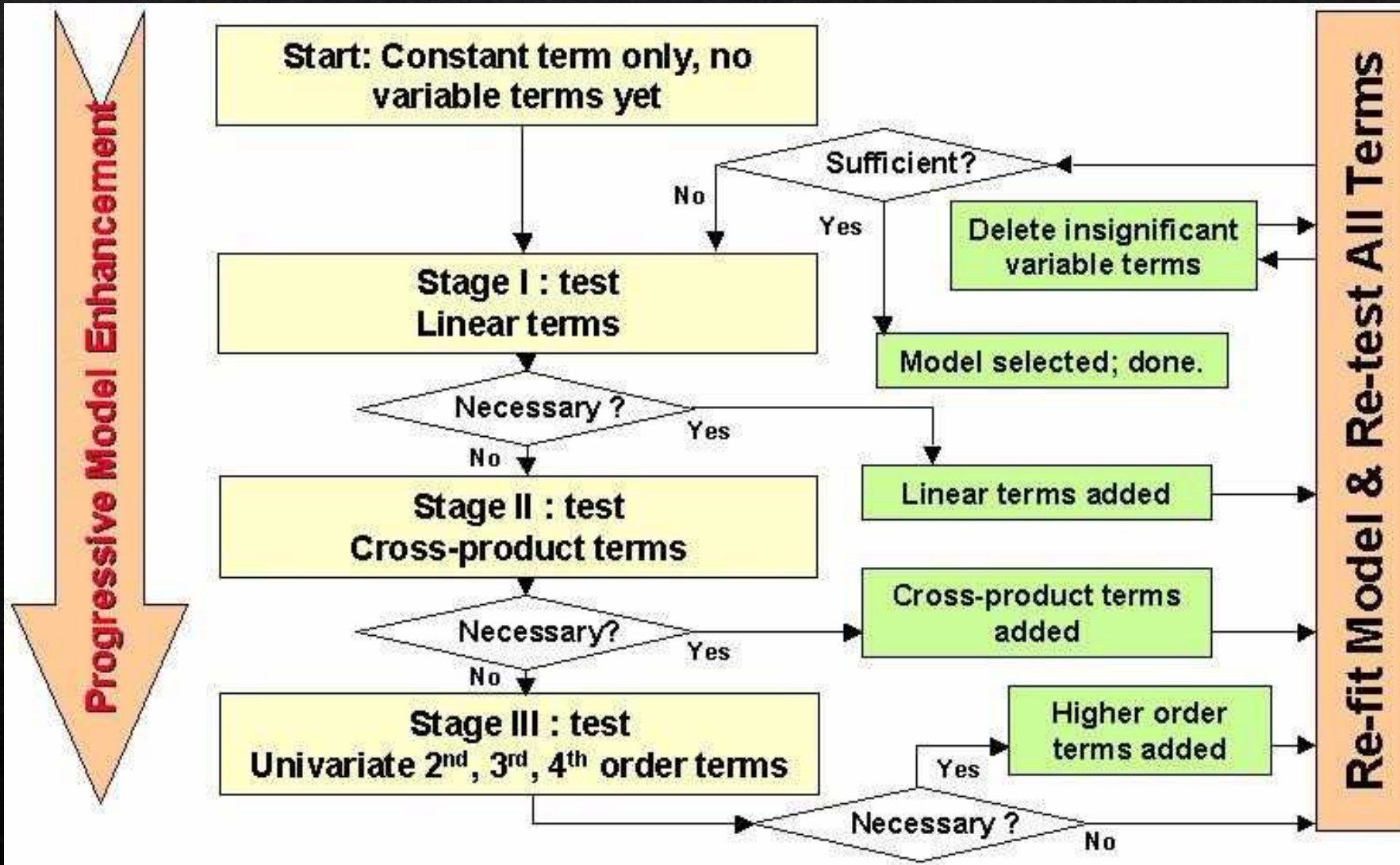


Keep adding the most significant variable until reaching the stopping rule or running out of variables

Model with 2 variables



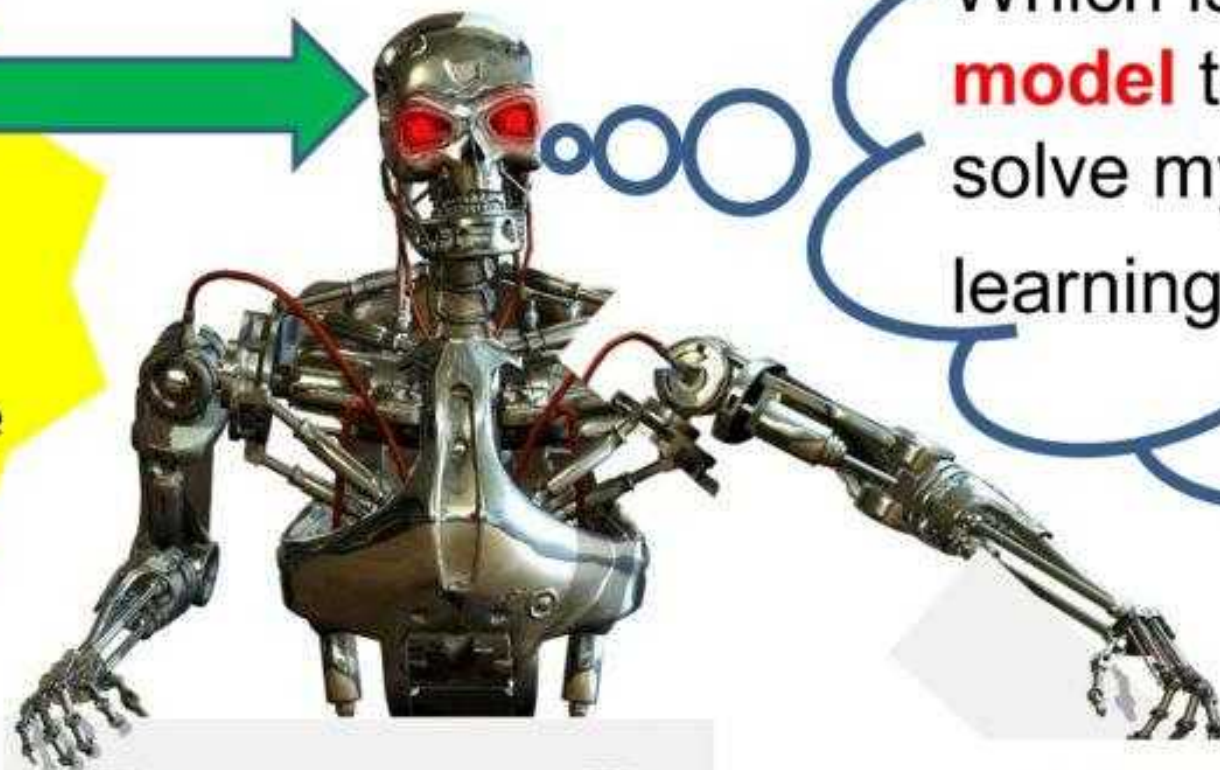
Stepwise Regression





Dr. Hirotugu Akaike

Hirotugu Akaike image by The Institute of Statistical Mathematics
is licensed under a Creative Commons Attribution-ShareAlike 4.0
International License. commons.wikimedia.org/wiki/File:Akaike-hirotugu.jpg



Which is the **best model** to help me solve my machine learning problem?



Model 1



Model 2



Model 3

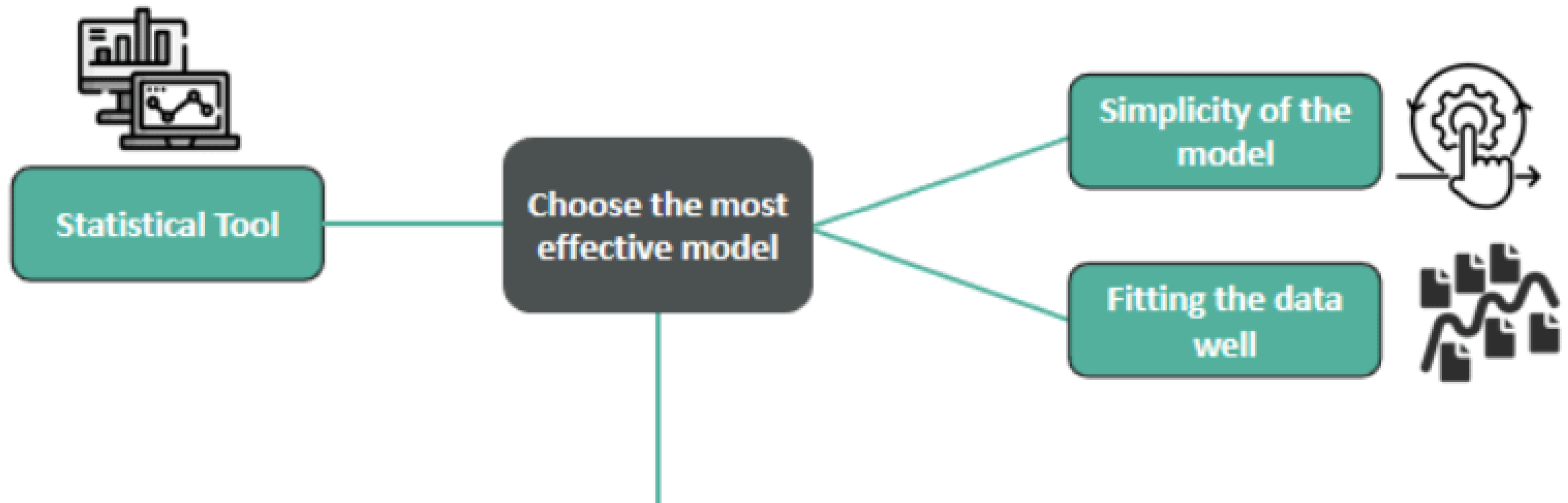


Model 4



Model 5

Akaike Information Criterion



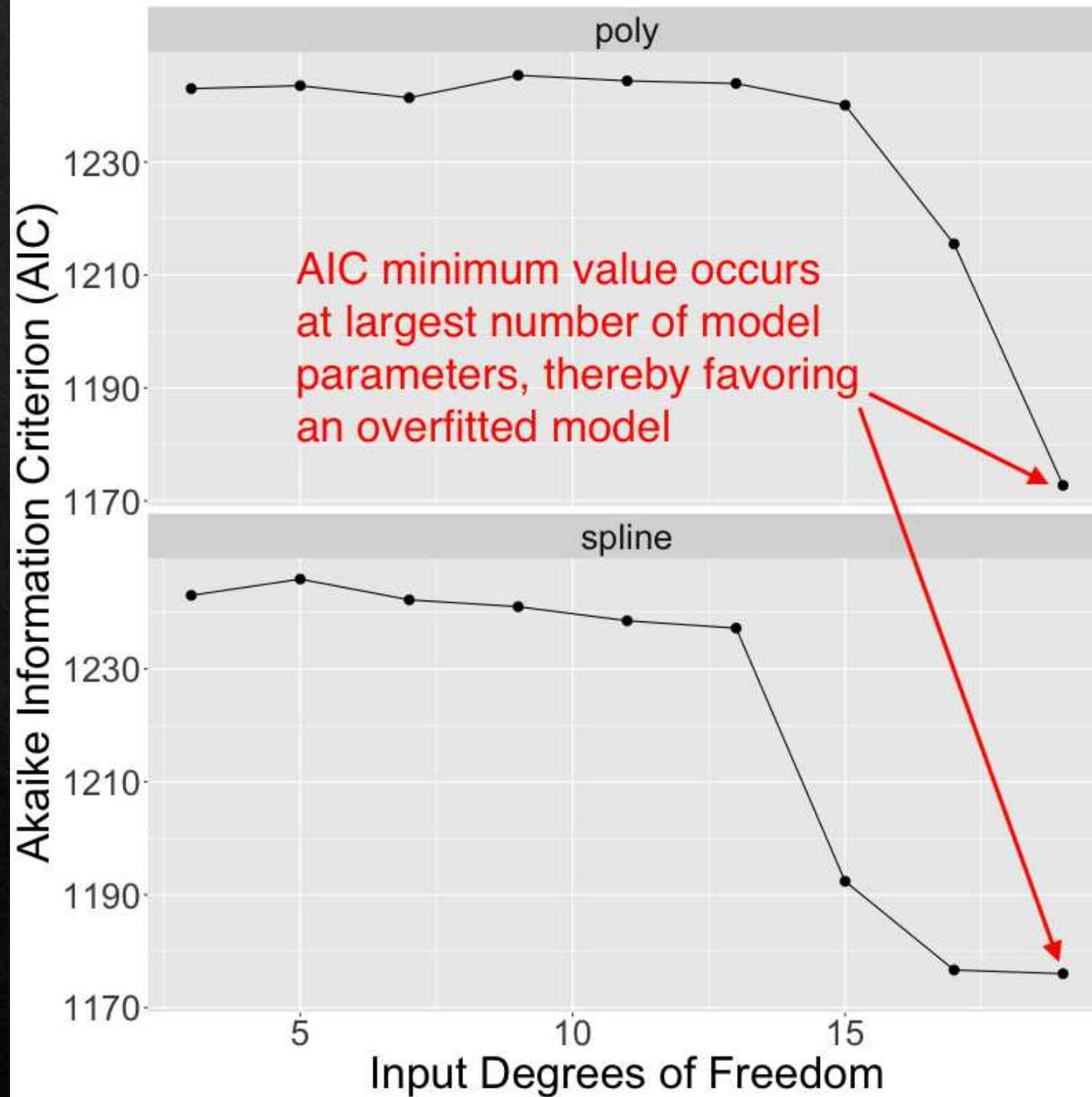
Find a balance between model accuracy and complexity

Akaike Information Criterion, AIC

Number of model
parameters

$$AIC = 2k - 2\ln(\mathcal{L})$$

$\mathcal{L} = \mathcal{L}(\hat{\theta})$ = maximum value of the
likelihood function of the model



Bayesian Information Criterion



A statistical
method



Helps identify the best
model among a set of
available options



Used in hypothesis
testing and model
selection

Bayesian Information Criterion

Bayesian Information Criterion (BIC) is a criterion for model selection among a finite set of models. It is based, in part, on the likelihood function, and it is closely related to Akaike information criterion (AIC).

When fitting models, it is possible to increase the likelihood by adding parameters, but doing so may result in overfitting. The BIC resolves this problem by introducing a penalty term for the number of parameters in the model. The penalty term is larger in BIC than in AIC.

Bayesian Approach and BIC

Like AIC used in when fitting by max log-likelihood

Bayesian Information Criterion (BIC):

$$BIC = -2\loglik + (\log N)d$$

Assuming Gaussian model: σ_ε^2 known,

$$-2 \cdot \loglik \approx \sum_i (y_i - \hat{f}(x_i))^2 / \sigma_\varepsilon^2 = N \cdot \overline{err} / \sigma_\varepsilon^2$$

$$\text{then } BIC = \frac{N}{\sigma_\varepsilon^2} [\overline{err} + (\log N) \cdot \frac{d}{N} \sigma_\varepsilon^2]$$

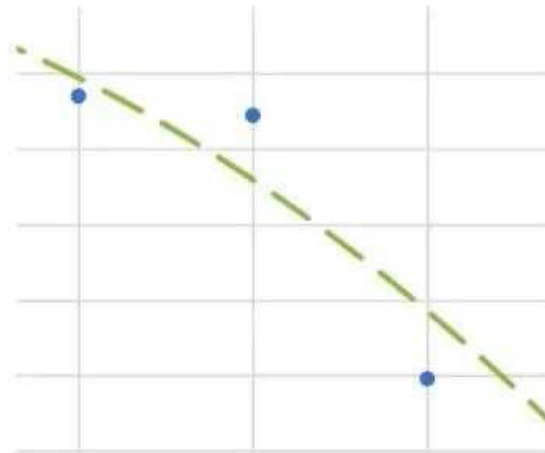
BIC proportional to AIC except for $\log(N)$ rather than factor of 2. For $N > e^2$ (approx 7.4), BIC penalizes complex models more heavily.

Adjusted R²

$$R^2 = 1 - \frac{SS_E}{SS_T}$$

$$R^2_{adj} = 1 - \frac{SS_E/(n-p)}{SS_T/(n-1)}$$

Preferred, penalizes for more terms



To calculate SS_E , we need the residuals

$$e_i = \hat{y}_i - y_i$$

$$SS_E = \sum_{i=1}^n e_i^2$$

Playlist

General Linear Models: Regression

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k + \epsilon$$

$$C_p = \frac{(n - (k + 1))s^2}{\hat{\sigma}_0^2} - n + 2(k + 1)$$

Mallow's C_p

Mallow's Cp

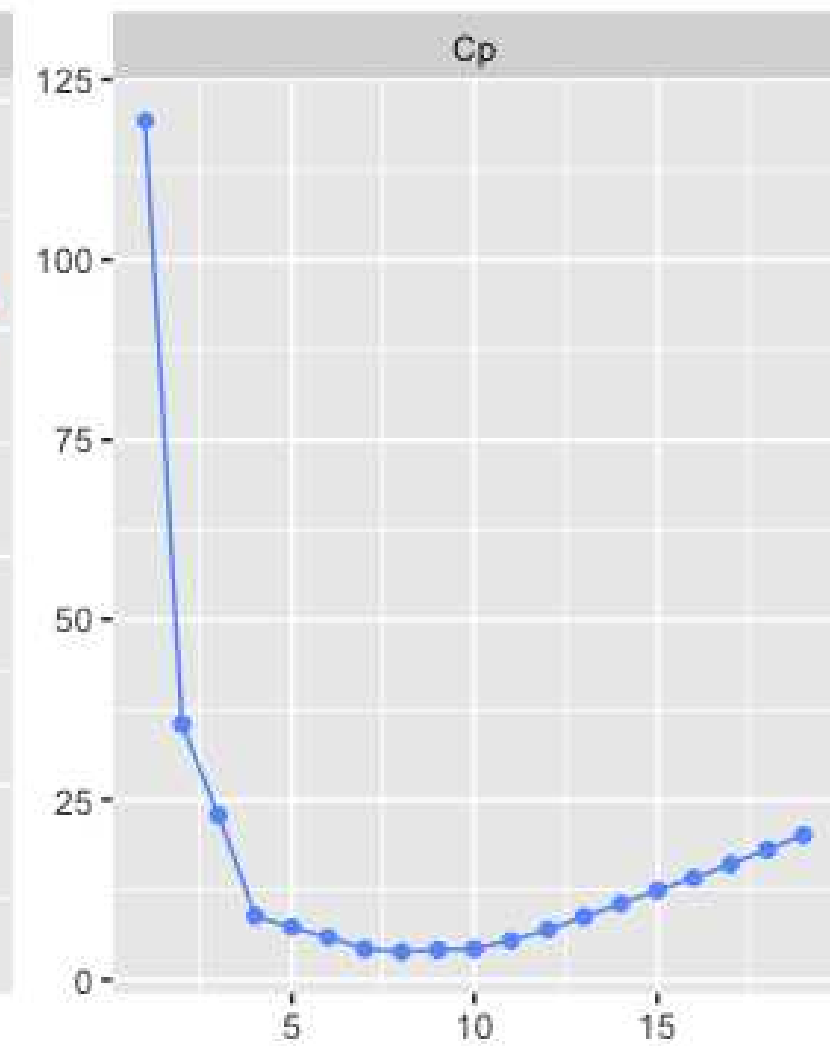
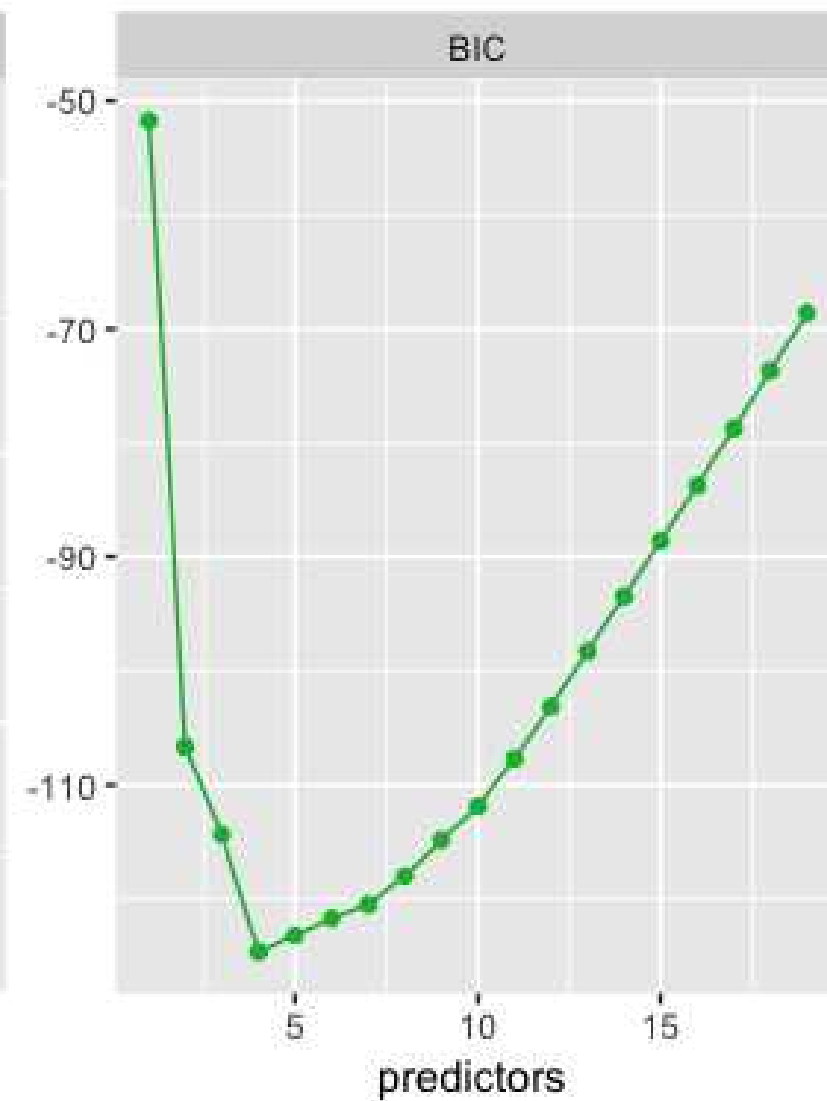
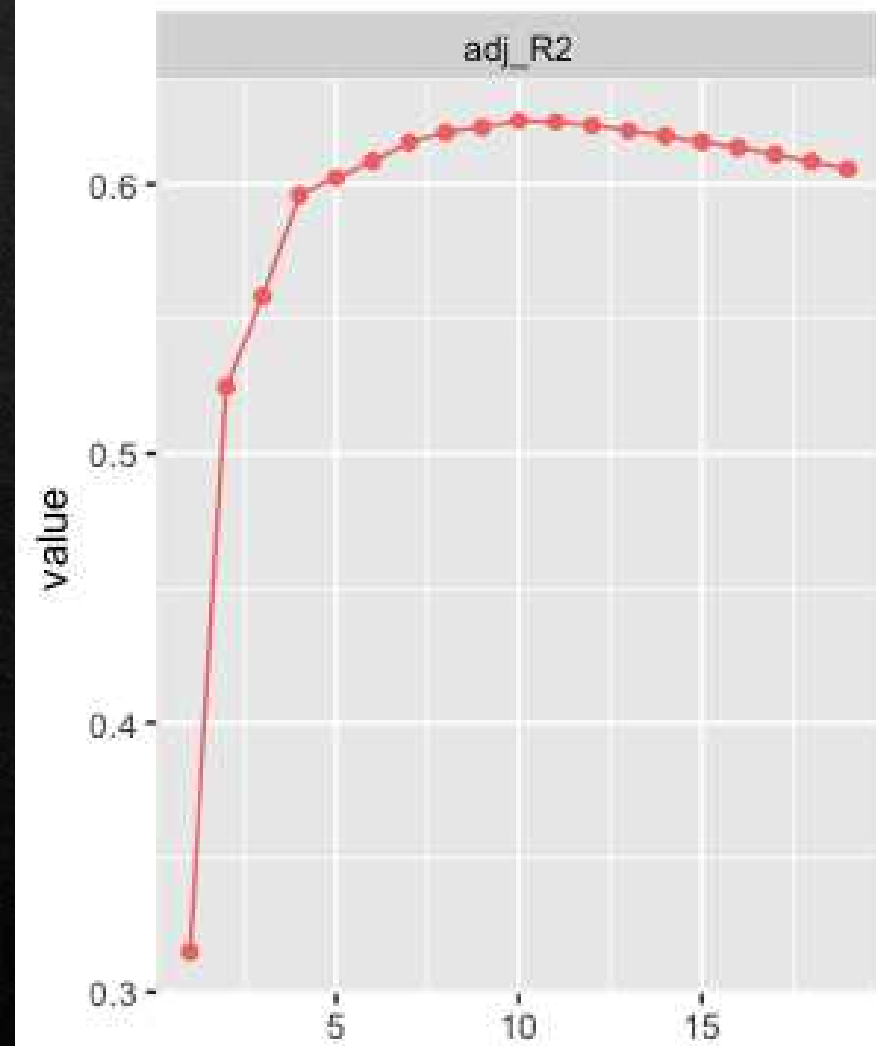
$$Cp = \frac{SSE_p}{MSE_F} - (N - 2p)$$

SSE_p : The sum of squared errors for the reduced or potential model.

MSE_f : The mean squared error for the full model.

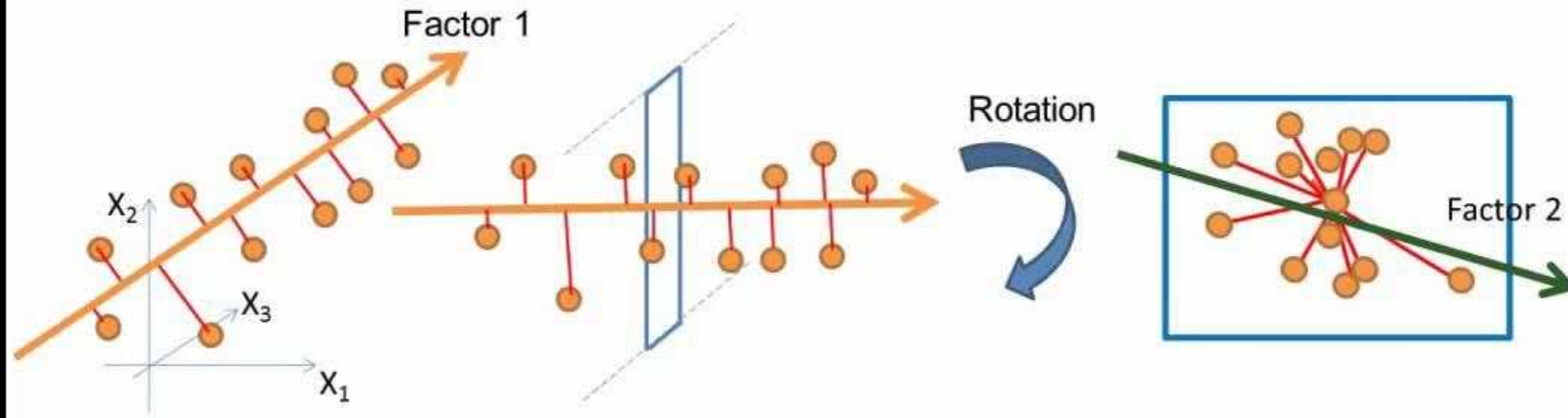
p : The count of predictors. The penalty term " $n-2p$ " imposes a cost on models that incorporate a higher number of predictors.

N : Number of observations.



What is Principal Component regression?

- Step 1 = Principal Component Analysis on the X table





Principal Component Regression

Principal Components Regression (PCR) is a regression analysis technique that is based on Principal Components Analysis (PCA).

Principal Components Regression is a technique for analyzing multiple regression data that suffer from multicollinearity.

When multicollinearity occurs, least squares estimates are unbiased, but their variances are large so they may be far from the true value. By adding a degree of bias to the regression estimates, principal components regression reduces the standard errors.



This can be avoided by using the selected Principal Components in place of the original variables since the Principal Components are uncorrelated.

The main purpose of Principal Component Regression is to estimate the values of a response variable at the basis of selected Principal Components of the explanatory variables.

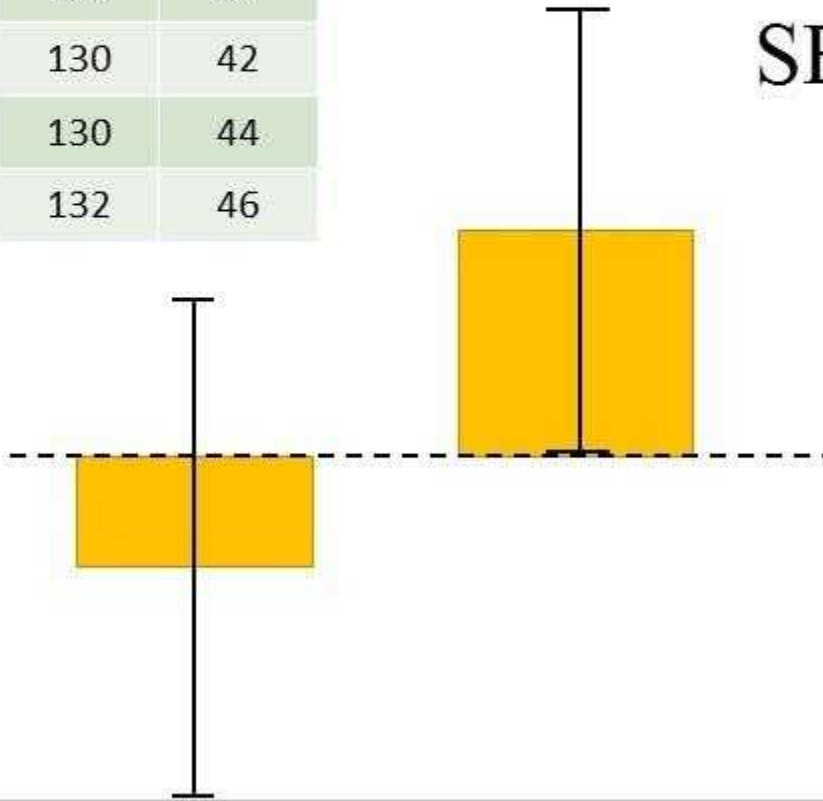
Secondly, the dimensionality of the regressors is reduced by taking only a subset of Principal Components for prediction.

Principal component regression

	SBP	Chol	Age
1	120	126	38
2	125	128	40
3	130	128	42
4	121	130	42
5	135	130	44
6	140	132	46

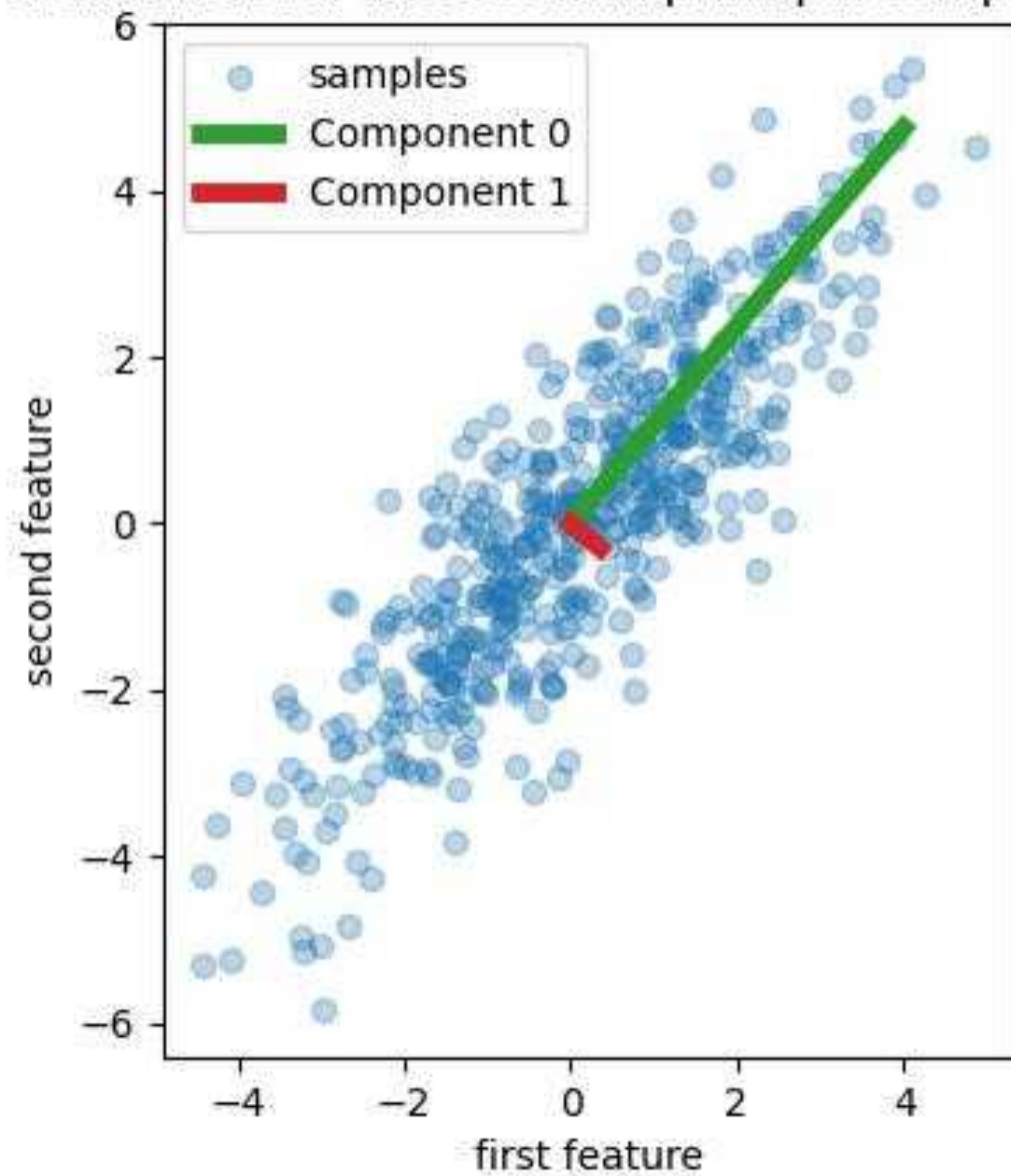
$$\text{SBP} = \text{intercept} + \underbrace{\text{Chol} + \text{Age}}$$

$$\text{SBP} = \text{intercept} + \boxed{\text{PC1}}$$



TileStats

2-dimensional dataset with principal components



Principal Component Regression

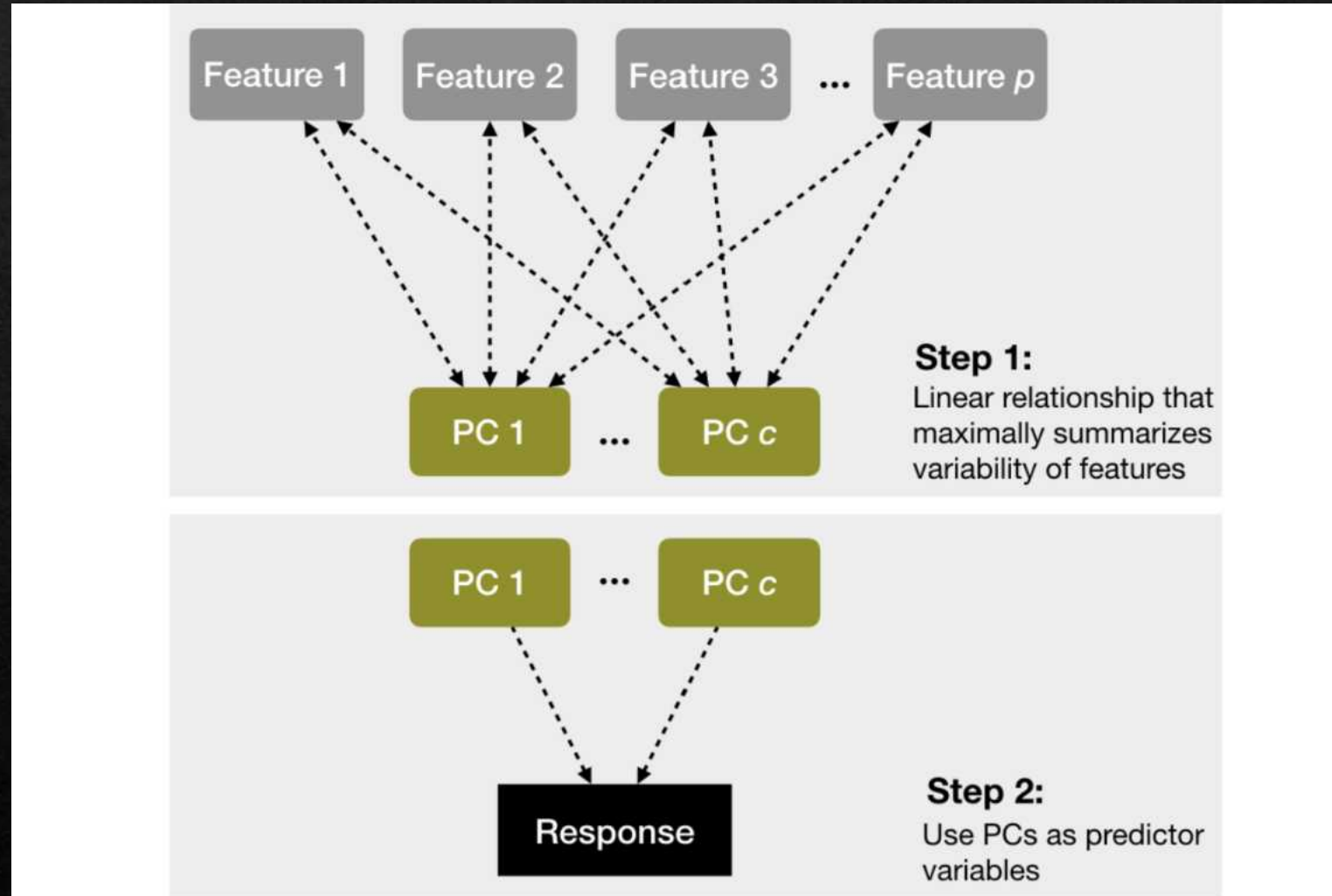
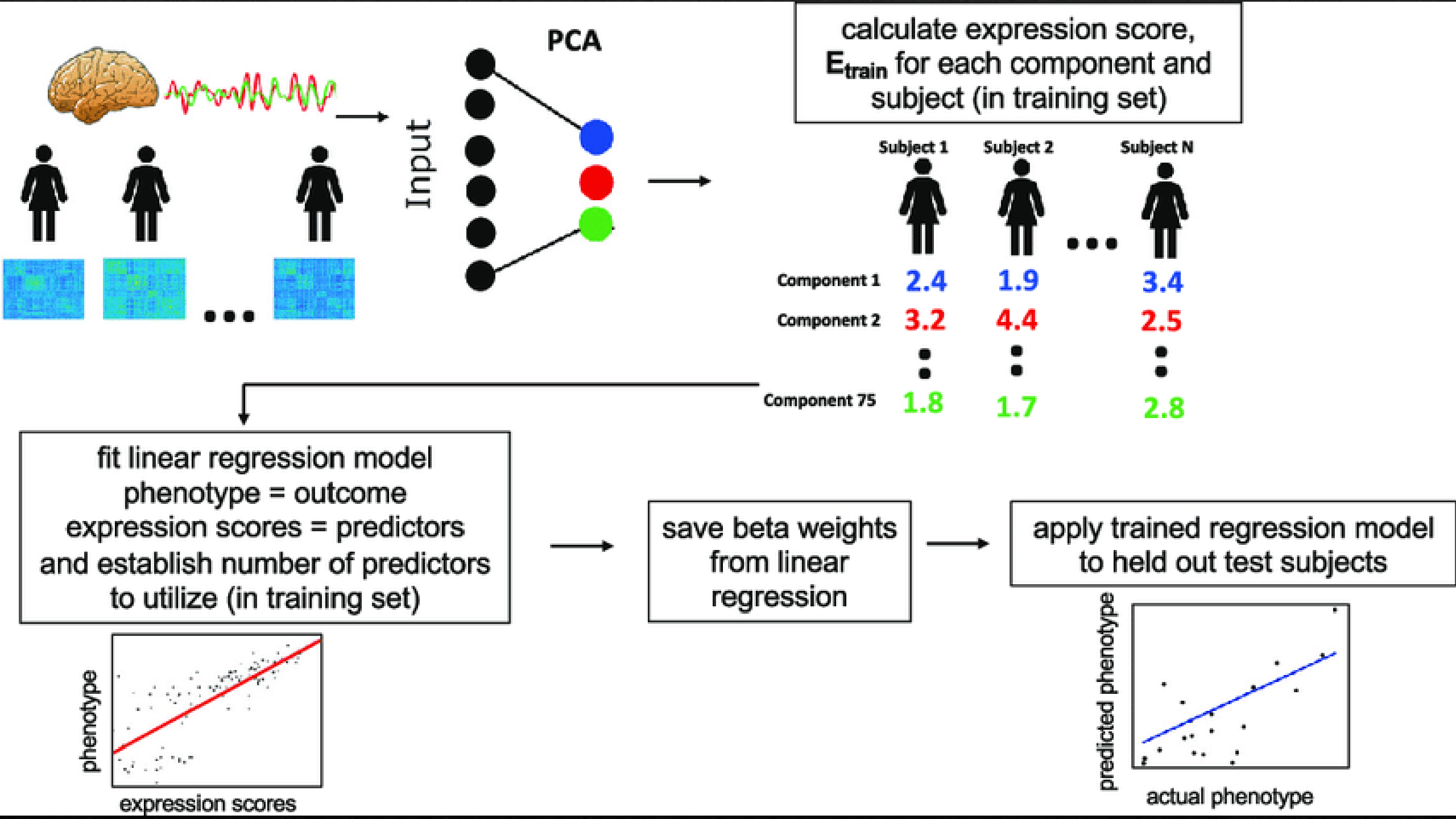
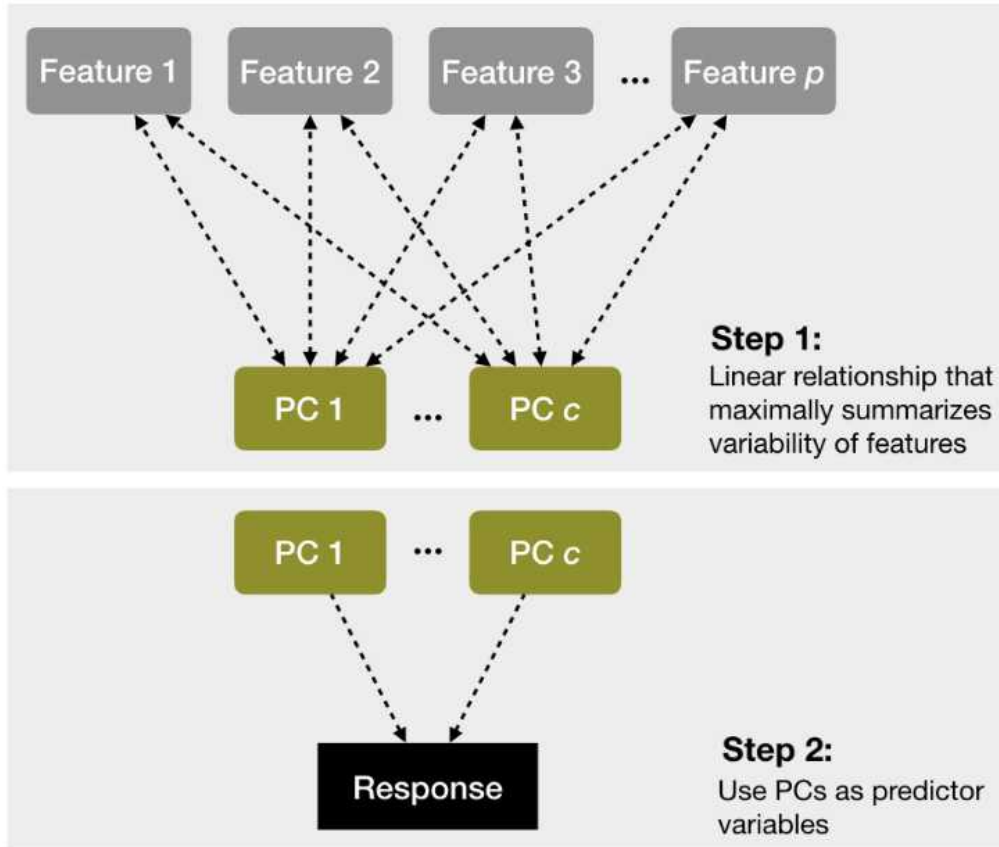


Figure 4.6: A depiction of the steps involved in performing principal component regression.

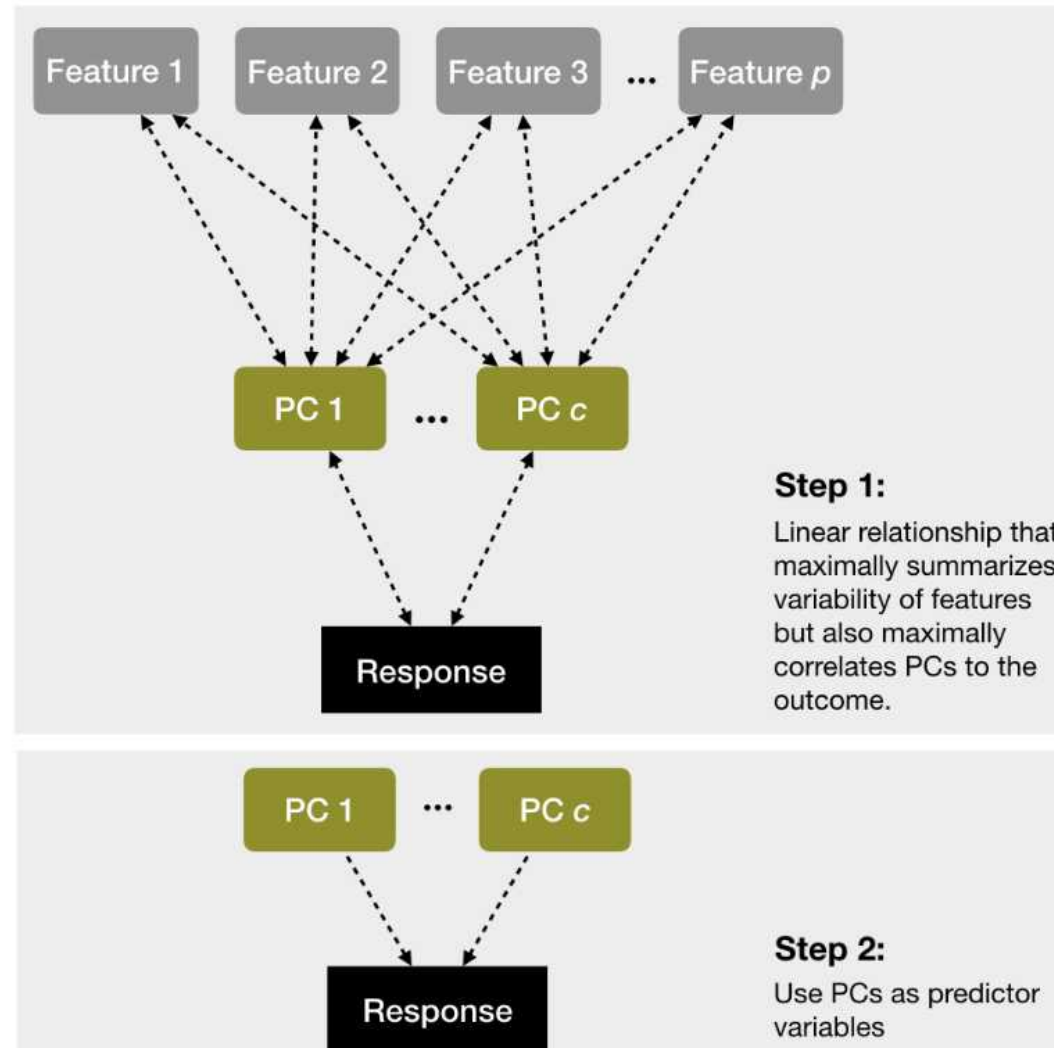


Partial Least Squares

(a) Principal Components Regression



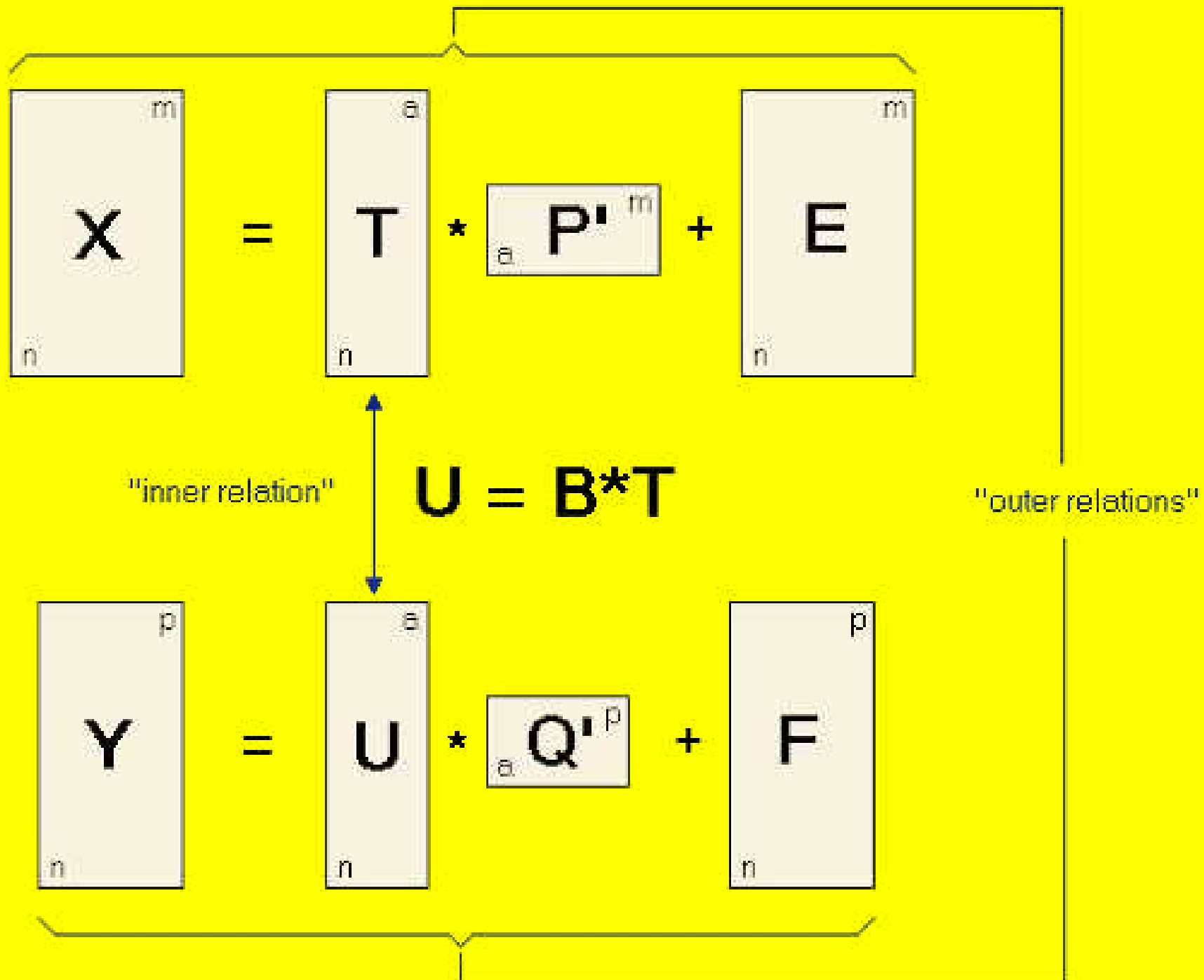
(b) Partial Least Squares Regression



Advantages of PLS

- PLS regression:
 - Deals with multicollinearity
 - Allows taking into account the data structure
 - Provides visual results that help the interpretation
 - Can model several response variables at the same time taking into account their structure



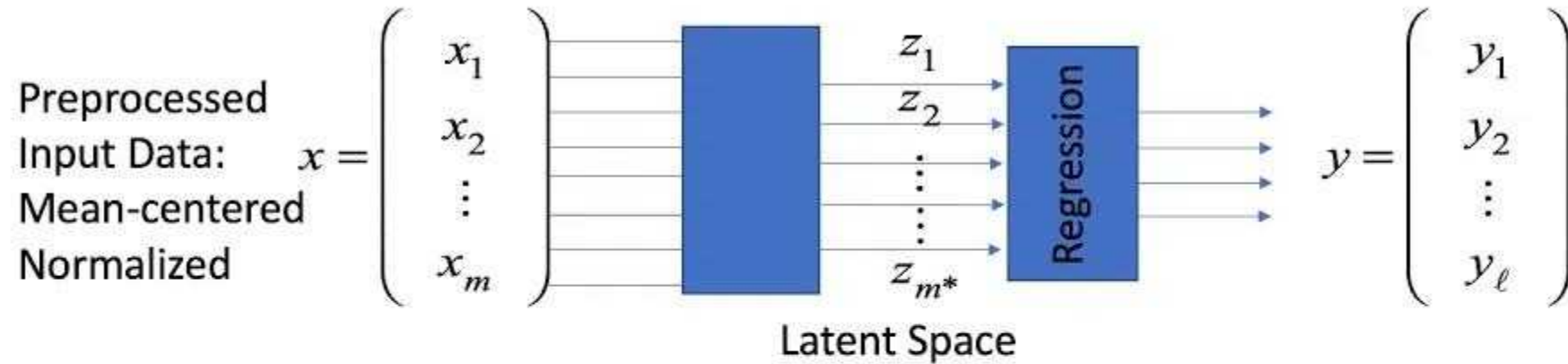


Partial Least Squares

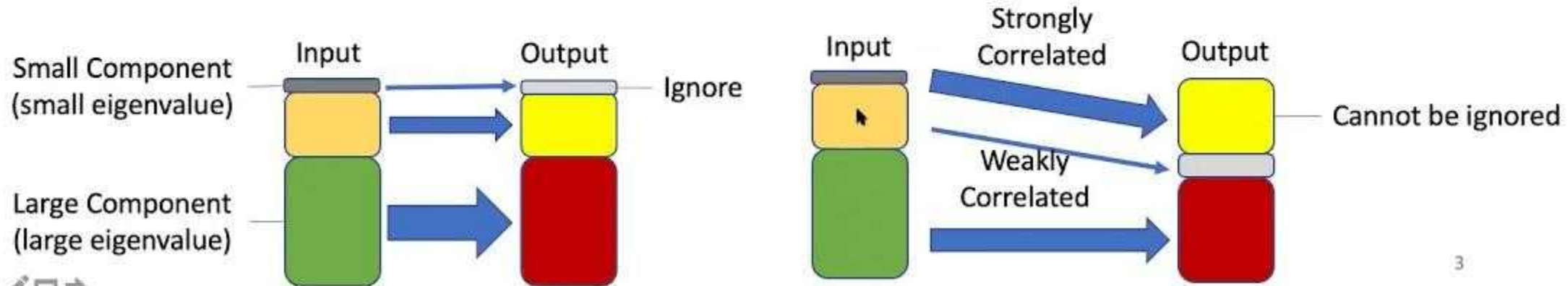
- Partial Least Squares is just like PC Regression except in how the component scores are computed
- PC regression = weights are calculated from the covariance matrix of the predictors
- PLS = weights reflect the covariance structure between predictors and response
 - While conceptually not too much of a stretch, it requires a more complicated iterative algorithm
 - Nipals and SIMPLS algorithms probably most common
- Like in regression, the goal is to maximize the correlation between the response(s) and component scores



Latent Modeling



- ❑ Principal Component Regression: Characterizes the input data space, reduces the input dimension based on Principal Component Analysis, and regresses on principal components.
- ❑ Caveat! Small principal components, which are ignored, may be highly correlated with outputs. Those components must not be neglected.

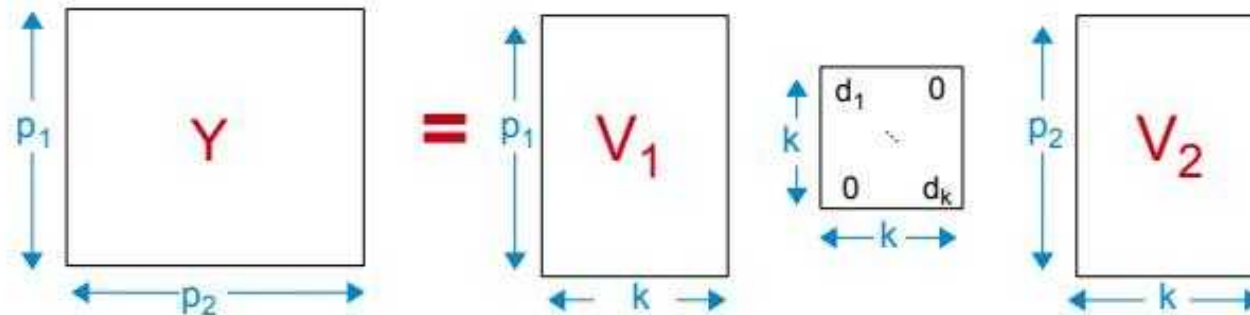


Singular Value Decomposition, SVD

- ◆ SVD is a matrix factorization technique that decomposes a matrix into three other matrices.
- ◆ SVD is a powerful mathematical tool in data science and machine learning.
- ◆ SVD is primarily used for dimensionality reduction, information extraction, and noise reduction.
- ◆ It has interesting algebraic properties and conveys important geometrical and theoretical insights about linear transformation.

Matrix Factorization - SVD

- ▶ **SVD:** Decompose $p_1 \times p_2$ matrix $Y := V_1 D V_2^T$
 - ▶ V_1 are k eigenvectors of YY^T
 - ▶ V_2 are k eigenvectors of $Y^T Y$
 - ▶ $D := \sqrt{\text{eig}(\text{diag}(YY^T))}$



Properties:

- ▶ Method from linear algebra for arbitrary Y
- ▶ Tool for descriptive and exploratory data analysis
- ▶ Decomposition optimizes the Frobenius norm with orthogonality constraints

Nonlinear Iterative Partial Least Squares

Fact 13. There is another iterative method for finding the principal components and scores of a matrix $\mathbf{X}$ called the Nonlinear Iterative Partial Least Squares (NIPALS) algorithm. It is built on the observation that a principal component $\mathbf{p}$ in $\mathbf{P}$ and a vector $\mathbf{t}$ in the scores $\mathbf{T}$ satisfy:

$$\mathbf{X}^T \mathbf{X} \mathbf{p} = \lambda_p \mathbf{p} \quad (8)$$

$$\mathbf{t} = \mathbf{X} \mathbf{p} \quad (9)$$

$$\mathbf{p} = \frac{1}{\lambda_p} \mathbf{X}^T \mathbf{t}, \quad (10)$$

where (10) is obtained by substituting (9) into (8). The algorithm works by initially setting $\mathbf{t}$ to an arbitrary $\mathbf{x}_i$ in $\mathbf{X}$ and then iteratively applying (10) and (9) until the $\mathbf{t}$ vector stops changing. This gives us the first principal component $\mathbf{p}$ and scores vector $\mathbf{t}$. To find the subsequent components/vectors, we set

$$\mathbf{X} := \mathbf{X} - \mathbf{t} \mathbf{p}^T$$

and then repeat the same steps. The NIPALS algorithm will turn out to be important in understanding the Partial Least Squares algorithm.

V. Esposito Vinzi
W.W. Chin
J. Henseler
H. Wang
(Editors)

Springer
Handbooks of
Computational
Statistics

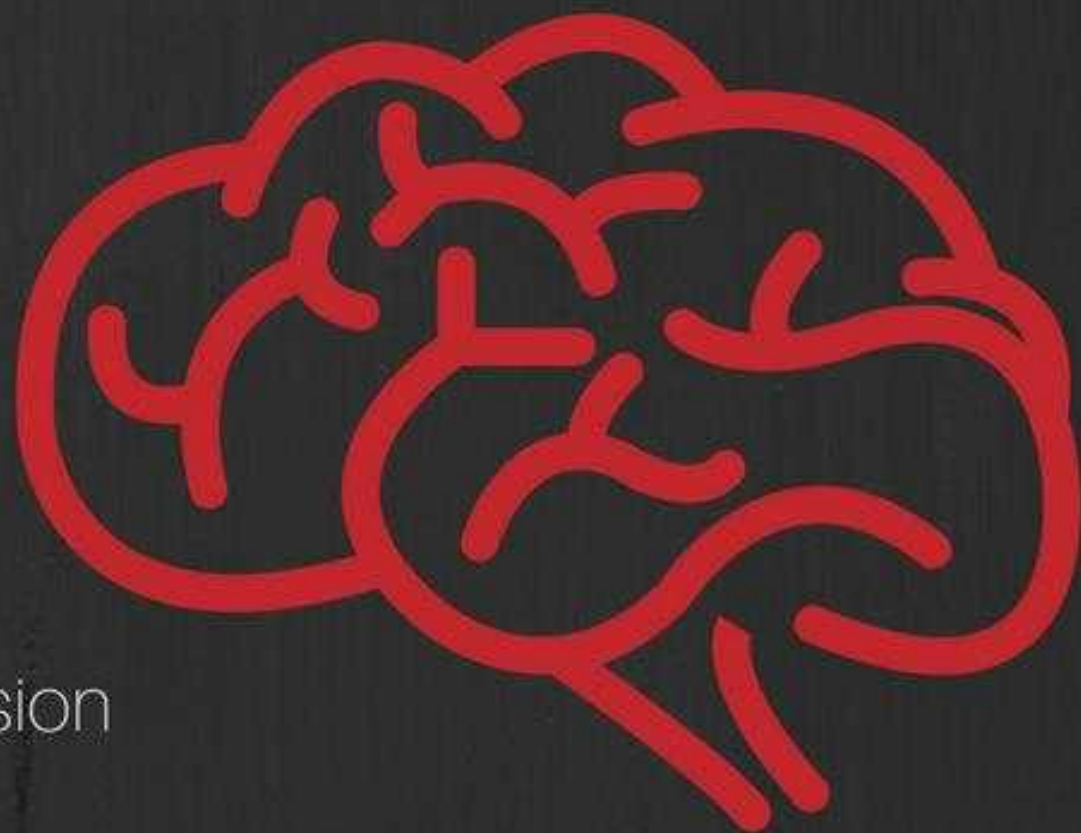
Handbook of Partial Least Squares

Concepts, Methods
and Applications

 Springer

Practical Machine Learning

Regularized linear regression



Ridge regression

Lasso

Tackling Multicollinearity Using

***Ridge
Regression***



***Lasso
Regression***



***Elastic-Net
Regression***

Regularized Regression

The objective function of a regularized regression model is similar to OLS, albeit with a penalty term P .

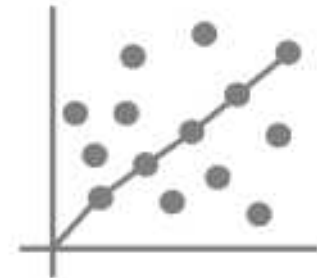
$$\text{minimize } (SSE + P) \quad (6.2)$$

There are three common penalty parameters we can implement:

1. Ridge;
2. Lasso (or LASSO);
3. Elastic net (or ENET), which is a combination of ridge and lasso.

What is Ridge regression?

A **specialized technique** that is used to **analyze multiple regression data which is multicollinear in nature**. Ridge regression is a fundamental regularization technique, but it is not used very widely



Ridge Regression

- *Ridge regression* penalize the size of the regression coefficients based on their l^2 norm:

$$\operatorname{argmin}_{\beta} \sum_i (y_i - \beta' x_i)^2 + \lambda \sum_{k=1}^K \beta_k^2$$

- The tuning parameter serves λ to control the relative impact of these two terms on the regression coefficient estimates.
 - Selecting a good value for λ is critical; cross-validation is used for this.

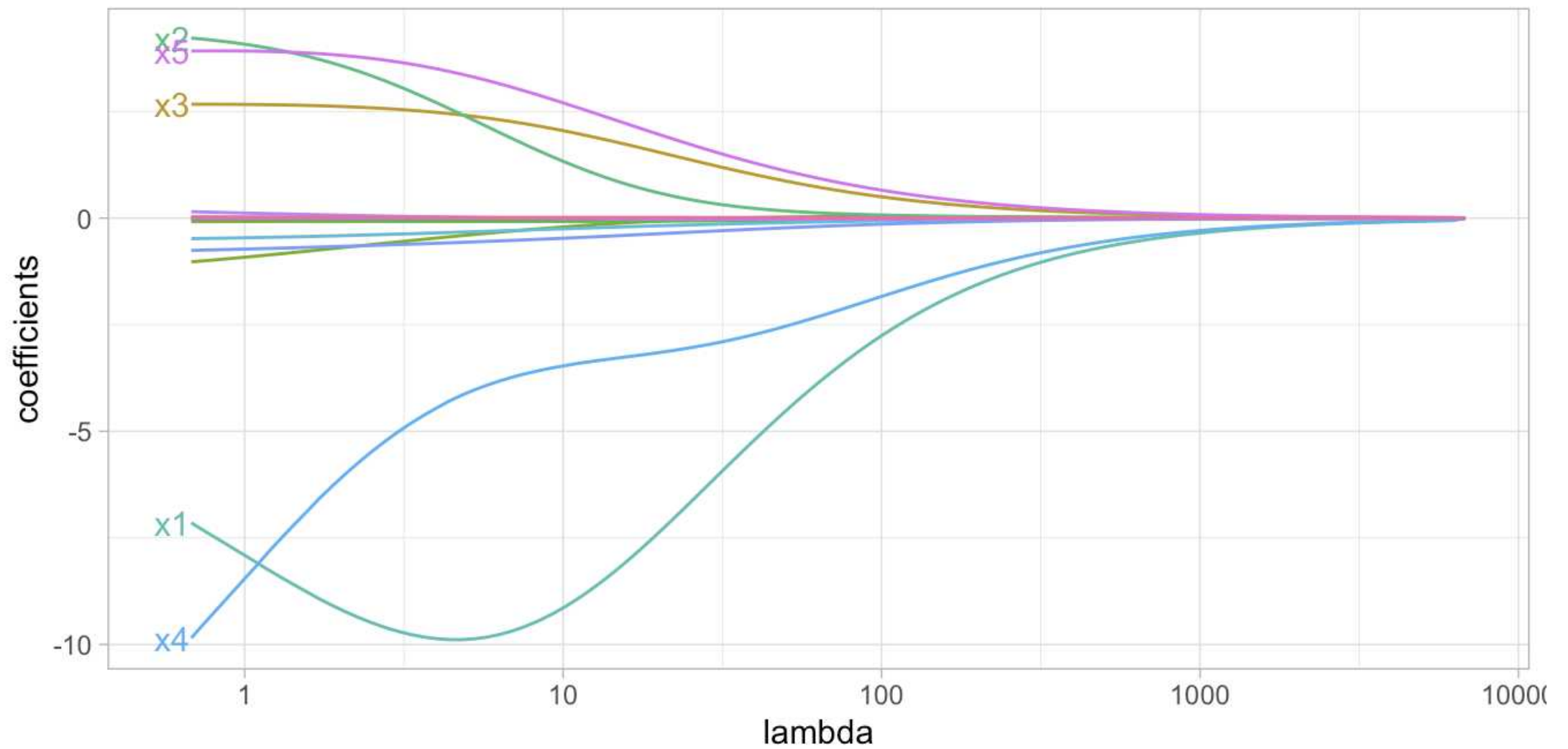
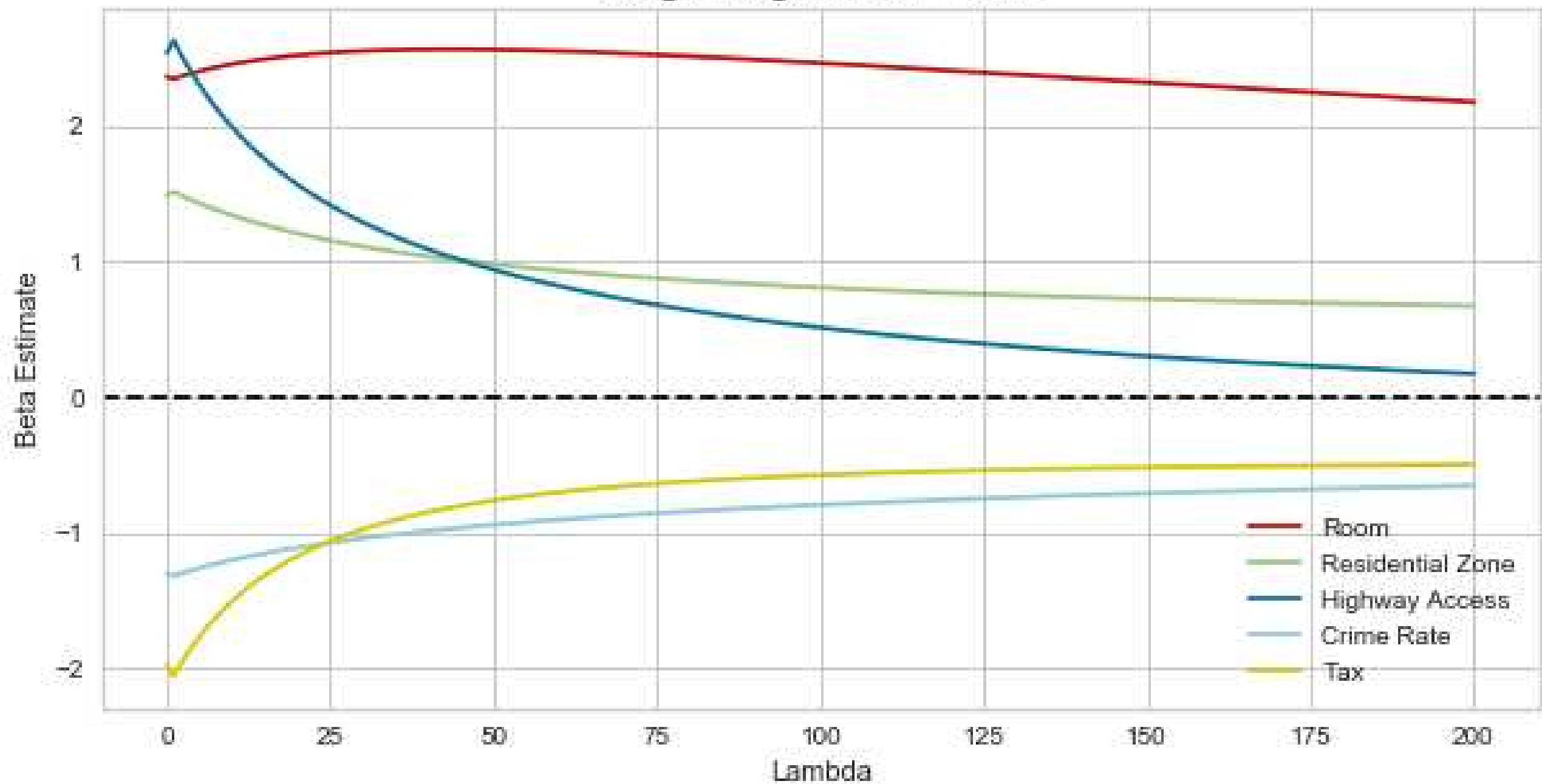


Figure 6.2: Ridge regression coefficients for 15 exemplar predictor variables as λ grows from $0 \rightarrow \infty$. As λ grows larger, our coefficient magnitudes are more constrained.

Ridge Regression Trace





Optimum Ridge Regression Parameter Using R-Squared of Prediction as a Criterion for Regression Analysis

Akbar Irandoukht^{*}

Catalysis Division, Research Institute of Petroleum Industry, Tehran, 1485733111, Iran

ARTICLE INFO

Article History

Received xx xxx xxxx
Accepted xx xxx xxxx

Keywords

Ridge parameter
PRESS
Maximization of R^2 -prediction
Model prediction power

ABSTRACT

The presence of the multicollinearity problem in the predictor data causes the variance of the ordinary linear regression coefficients to be increased so that the prediction power of the model not to be satisfied and sometimes unacceptable results be predicted. The ridge regression has been proposed as an efficient method to combat multicollinearity problem long ago. In application of ridge regression the researcher uses the ridge trace and selects a value of ridge parameter in such a manner that he thinks the regression coefficients have stabilized; this leads the ridge regression to be subjective technique. The purpose of this paper is the conversion of the ridge regression method from a qualitative method to a quantitative one meanwhile to present a method to find the optimum ridge regression parameter which maximizes the R -squared of prediction. We examined four well-known case studies on this regard. Significant improvements at all of the cases demonstrated the validity of our proposed method.

© 2021 The Authors. Published by Atlantis Press B.V.
This is an open access article distributed under the CC BY-NC 4.0 license (<http://creativecommons.org/licenses/by-nc/4.0/>).

1. INTRODUCTION

In many scientific and engineering disciplines, the researchers face to study the effect of some input or predictor variables on an output or response variable. After performing experiments, which are usually done either in classical one-factor-at-time method or using experimental design patterns, there is a need to relate the input variables to output variable via a characterizing mathematical relationship. Hence, if there is a theoretical relationship between Y and X , then the form of the relating function will be specified. If the theoretical functionality has not been developed, then the researcher may use empirical linear models using some parameter β . The values of these parameters can be estimated by using linear regression technique [7,18,30,36,39,41].

Sometimes the number of predictor variables is large or predictors are not really independent and there may be some correlations between input variables. In these cases, the relationship $X_i X_j = 0$ does not satisfy. The presence of correlation between predictor variables produces the multicollinearity problem. The effects of collinearity have been discussed by above mentioned researchers and also by Gunst [19], Jensen and Ramirez [29].

The significant number of methods has been proposed to overcome multicollinearity problem (e.g., see [3,14,15,17,42]).

One of the efficient methods to combat the multicollinearity problems is ridge regression developed by Hoerl [23], Hoerl and Kennard [24,25] for the first time. Many researchers used it in practice and found it useful on this regard for example see Marquardt [34], Hemmerle [20], Marquardt and Snee [35], Khuri and Myers [31], Dorugade [5], Fitrianto and Yik [13], Yahya and Olaifa [49].

Hoerl and Kennard [26] themselves published the first review paper on ridge regression and then other reviews appeared in the literature, among them, the works of Singh [44], Dube and Isha [9], Duzan and Shariff [10], Goktas and Sevinc [16] can be mentioned.

Some of the researchers tried to show the application and usefulness of ridge regression using simulation studies [6,11,27,33,43].

On the other hand some researchers criticized the ridge regression method from different points of view for example see Smith and Campbell [45], and also the comments of Thisted *et al.* were appeared in the same issue of *Journal of American Statistical Association* (1980); Draper and Smith [7], Singh [44], Chatterjee and Hadi [3].

One of the main weaknesses of the ridge regression is that the ridge regression parameter can be selected by examining the ridge trace (Marquardt and Snee [35] points out that for using ridge regression 25 values of k in the interval of $[0, 1]$ need to be examined).

^{*} Corresponding author. Email: irandoukht@ripi.ir



An Application of Proposed Ridge Regression Methods to Real Data Problem

N S M Shariff^{1*}, H M B Duzan²

^{1,2} Faculty of Science and Technology, Universiti Sains Islam Malaysia (USIM),

Bandar Baru Nilai 71800, Nilai, Negeri Sembilan, Malaysia

^{*}Corresponding author E-mail: nurulshima@usim.edu.my

Abstract

The Ordinary Least Squares (OLS) is a common method to investigate the linear relationship among variable of interest. The presence of multicollinearity will produce unreliable result in the parameter estimates if OLS is applied to estimate the model. Due to such reason, this study aims to use the proposed ridge estimator as linear combinations of the coefficient of least squares regression of explanatory variables to the real application. The numerical example of stock market price and macroeconomic variables in Malaysia is employed using both methods with the aim of investigating the relationship of the variables in the presence of multicollinearity in the data set. The variables on interest are Consumer Price Index (CPI), Gross Domestic Product (GDP), Base Lending Rate (BLR) and Money Supply (M1). The obtained findings show that the proposed procedure is able to estimate the model and produce reliable result by reducing the effect of multicollinearity in the data set.

Keywords: Macroeconomic Variables; Multicollinearity; OLS; Ridge Regression.

1. Introduction

Multicollinearity refers to high strength correlation among the explanatory variables in regression analysis. This situation commonly occurs due to large number of explanatory variables are incorporated in the analysis. The well-known method of estimation that is Ordinary Least Squares (OLS) is unreliable because the violation of the independency assumption of explanatory variables. The high dependency in explanatory variables will cause huge value in the standard errors of parameter estimates and thus, OLS is no longer appropriate to be employed in modeling the data.

Several methods of estimation have been suggested in the literatures to overcome the multicollinearity problem. The most popular method is ridge regression that has been introduced by [1, 2]. The ridge regression is proposed by introducing a positive value of k to the diagonal of the matrix $X^T X$ (where X is matrix of explanatory variables) with the aim of minimizing the biased estimates and mean squared error (MSE) of the model. The k is also known as ridge estimator and there are a variety of methods to estimate k (see [3 - 5]). Some of them make comparison with OLS and find ridge estimators outperform OLS and conclude that the generalized ridge regression is the best model among all methods. On the other hand, an alternative method of the existing ridge method is proposed in [6] where k is estimated by using coefficient of determination in the regression of an explanatory variable. This method has shown able to produce reliable results as in existing ridge methods in such away the dispersion of the standard error of the parameter estimates can be minimized. Due to such interest, the method will be applied to financial data. In the financial data, some explanatory variables maybe correlated due to high dependency among each other. This problem will have a vital influence on the results of statistical inference in the regression

analysis. In preliminary analysis, the descriptive analysis and the correlation measures are analyzed to describe the data and illustrate the presence of multicollinearity. The next step is the modeling procedure to obtain the results of the parameter estimates and the inference of the model. The remaining section of this paper is organized as follows: Section 2 presents the data and model used in this study while Section 3 discusses the results followed by the conclusion is written in Section 4.

2. Data and model

The data of macroeconomic variables represented by interest rate (base lending rate (BLR)), inflation (consumer price index (CPI)), gross domestic product (GDP) and monetary supply (M1) and stock price are on interest in this study. The macroeconomic variables refer to explanatory variables which can be indicators to Malaysian's economic that might affect the stock market movement. As such, the response (dependent variable) is stock price index (Kuala Lumpur Composite Index (KLCI)) with the length of the study is 15 years (2000-2015) and quarterly basis that yield 654 total number of observations.

2.1. Ridge regression model

Suppose that the data consists of n observations. Let Y_i be a scalar response and x_i be a vector of p explanatory variables. Then, a linear regression model is considered as follows

$$Y_i = x_i^T \beta + e_i \quad (1)$$

Lasso Regression



Lasso Regression for Regularization

$$L_{\text{lasso}}(\hat{\beta}) = \sum_{i=1}^n (y_i - x_i^s \hat{\beta})^2 + \lambda \sum_{i=1}^m |\hat{\beta}_i|$$

$$SSE_{L_1} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

Setting Parameter
(λ)

Punishment Value

LASSO

LASSO = least absolute shrinkage and selection operator

Find coefficients $\hat{\beta}$ that minimize the **penalized sum of squared residuals (RSS)**:

$$\text{minimize} \quad \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i)^2}_{\text{RSS}} + \underbrace{\lambda \sum_{j=1}^p |\hat{\beta}_j|}_{\text{penalty term}}$$

$p = \#$ coefficients for predictors
 λ is a tuning parameter ≥ 0

Sum of coefficients' absolute values:

Lots of predictors with tiny effects $\Rightarrow$ worry about overfitting

"A little x_1 , a bit of x_2 , some of x_3 can help the slightest bit more with RSS!"

$\approx$

"Fleck in the right ear, orange tint of nose, discoloration of the 4th stripe help explain just a few more pixels of this cat!"



Source: [Wikipedia](#)

$\uparrow$ when a predictor has a large effect

$\uparrow$ when lots of predictors have tiny effects

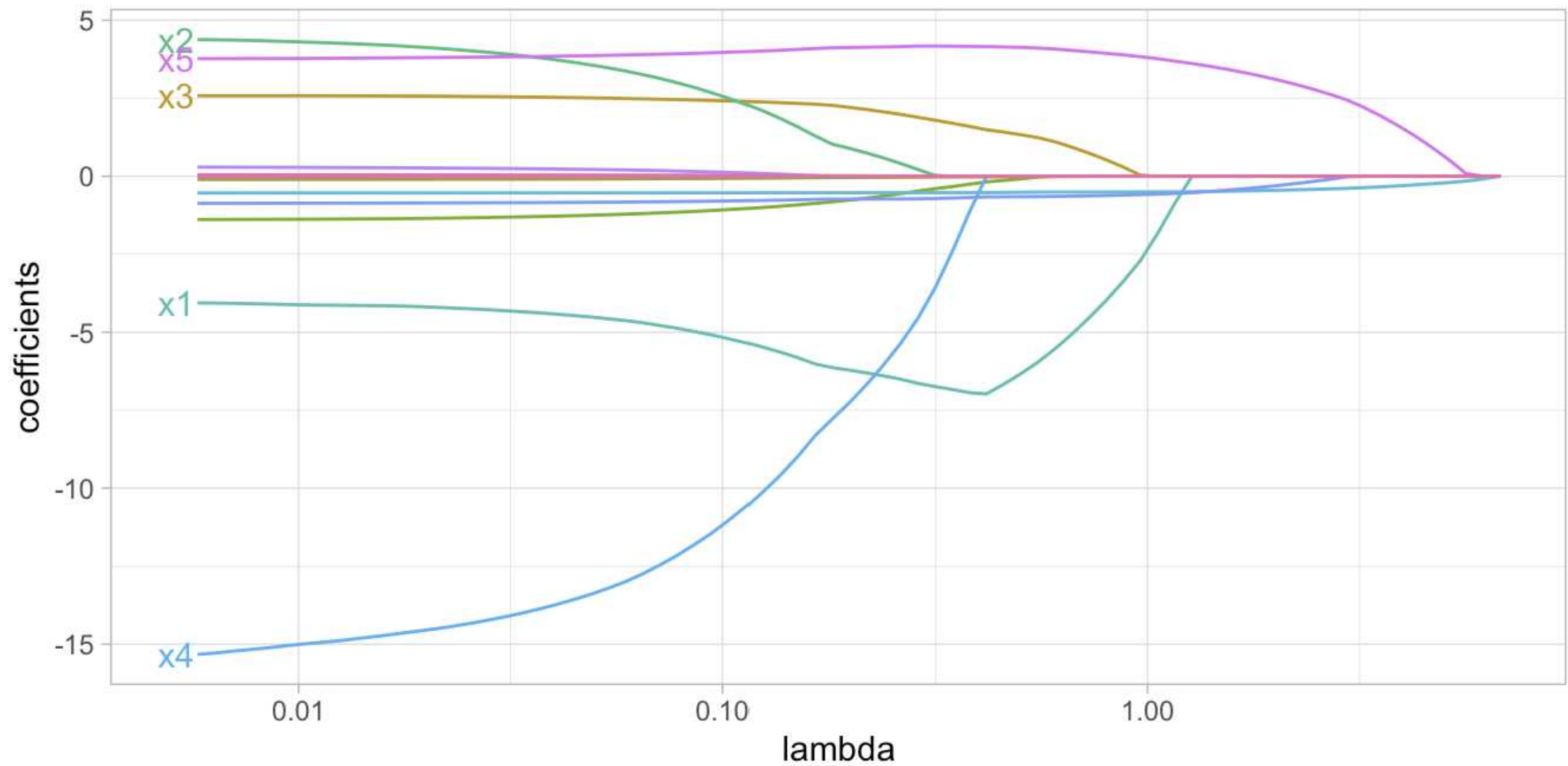


Figure 6.3: Lasso regression coefficients as λ grows from $0 \rightarrow \infty$.

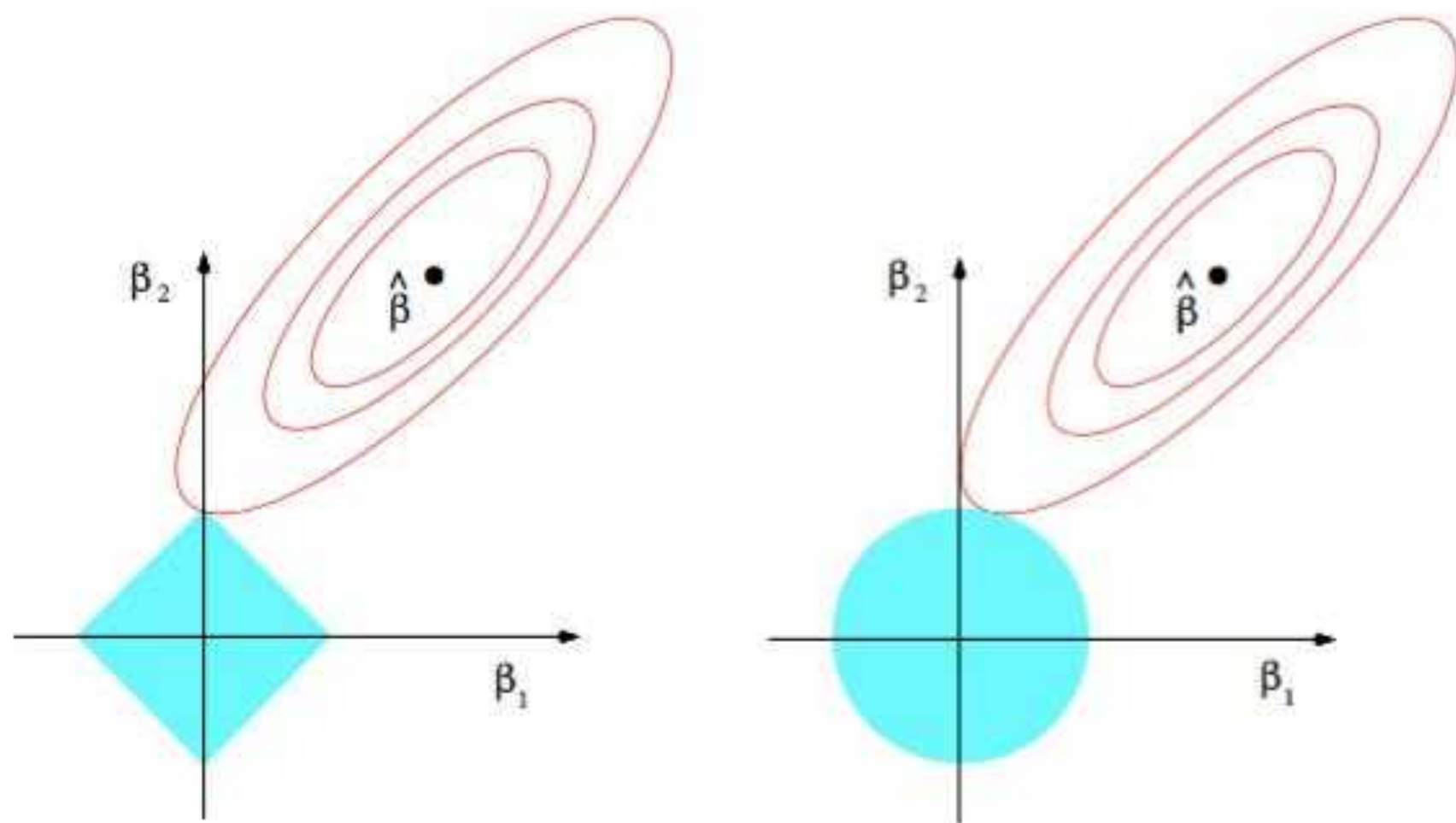


FIGURE 3.11. Estimation picture for the lasso (left) and ridge regression (right). Shown are contours of the error and constraint functions. The solid blue areas are the constraint regions $|\beta_1| + |\beta_2| \leq t$ and $\beta_1^2 + \beta_2^2 \leq t^2$, respectively, while the red ellipses are the contours of the least squares error function.

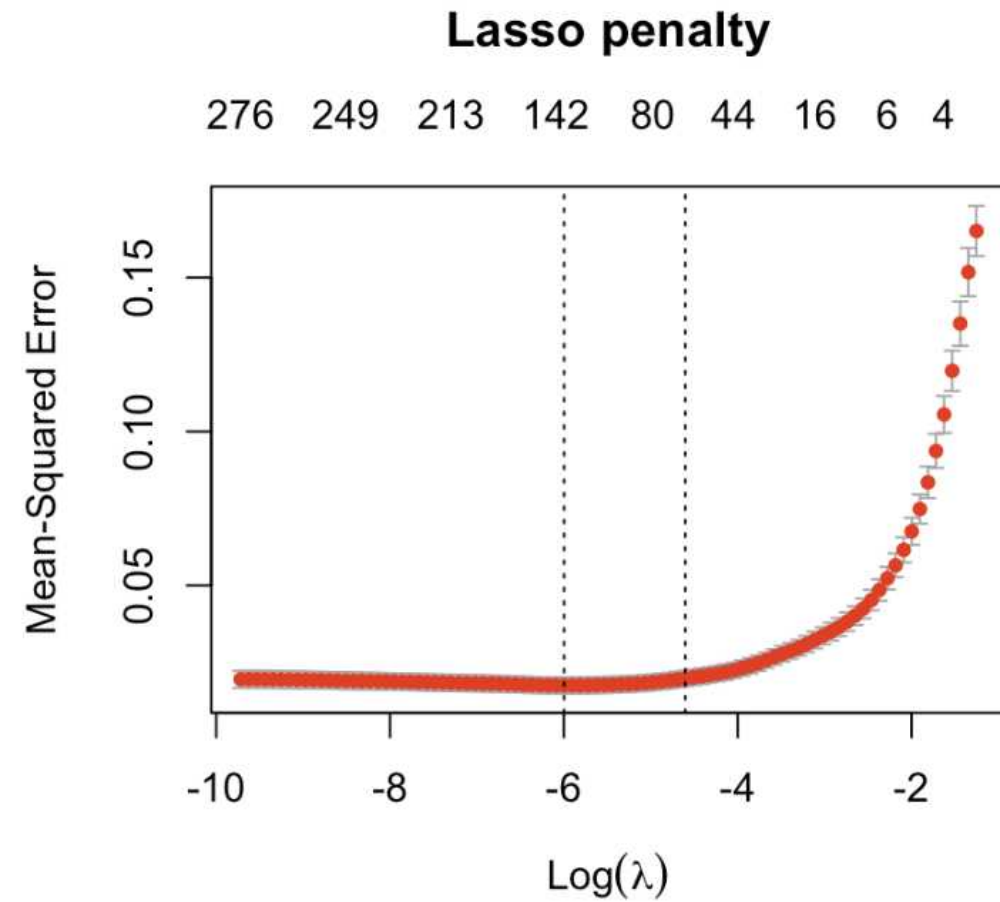
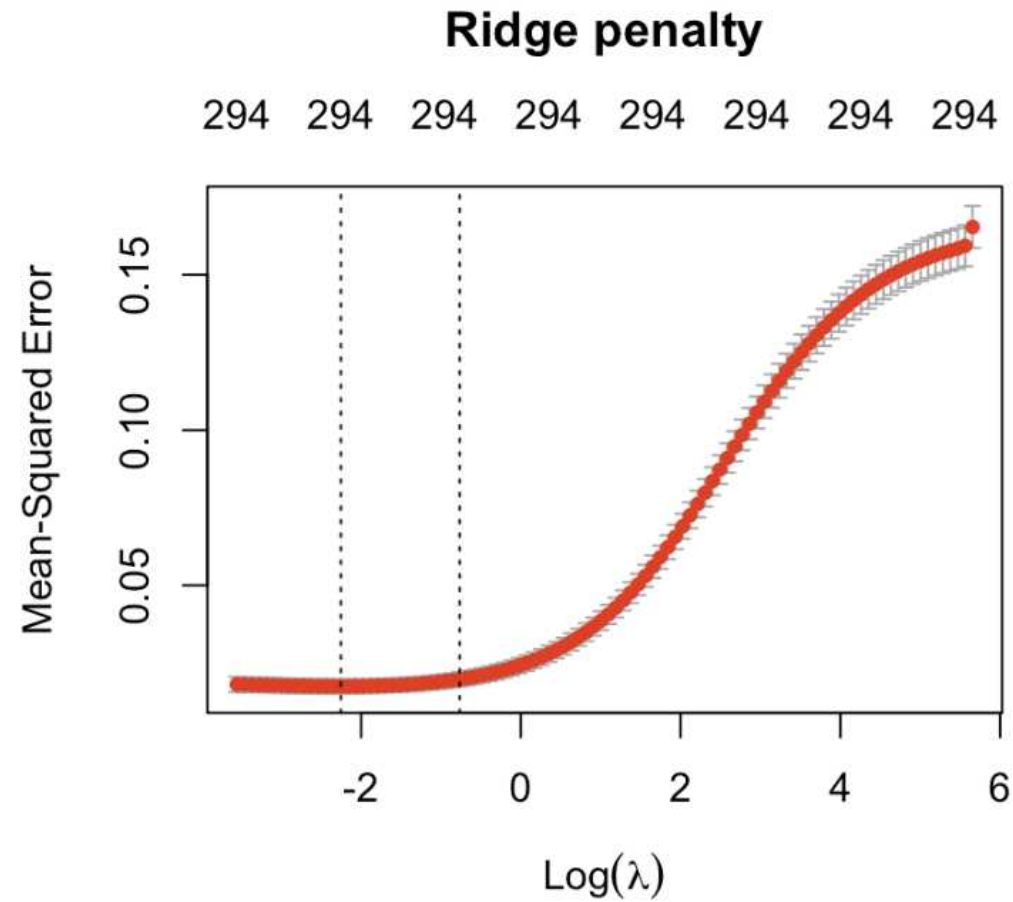


Figure 6.6: 10-fold CV MSE for a ridge and lasso model. First dotted vertical line in each plot represents the λ with the smallest MSE and the second represents the λ with an MSE within one standard error of the minimum MSE.

Lasso Regression Vs Ridge Regression

Lasso Regression uses L1 regularization (absolute value of coefficients).	Ridge Regression uses L2 regularization (square of coefficients).
Lasso Regression can force them to be exactly zero.	Ridge Regression shrinks coefficients of less significant features towards zero.
Lasso Regression performs both regularization and feature selection, making it more suitable for high-dimensional datasets.	Ridge Regression does not perform feature selection and can only shrink the coefficient values. This makes it more suitable for datasets with highly correlated predictors since it avoids including all of them in the model.
Lasso Regression may be more effective in situations where only a subset of features contribute significantly to the output	Ridge Regression generally works better in scenarios where there are fewer significant features.
Lasso Regression can lead to a sparse model, which means it can create a model with fewer features.	Ridge Regression does not produce sparse models.



Modelling House Price Using Ridge Regression and Lasso Regression

Seng Jia Xin¹, Kamil Khalid^{2*}

^{1,2}Department of Mathematics and Statistics, Faculty of Applied Sciences and Technology, Universiti Tun Hussein Onn Malaysia, Pagoh Educational Hub, 84600 Pagoh, Muar, Johor, Malaysia

*Corresponding author E-mail: kamil@uthm.edu.my

Abstract

House price prediction is important for the government, finance company, real estate sector and also the house owner. The data of the house price at Ames, Iowa in United State which from the year 2006 to 2010 is used for multivariate analysis. However, multicollinearity is commonly occurred in the multivariate analysis and gives a serious effect to the model. Therefore, in this study investigates the performance of the Ridge regression model and Lasso regression model as both regressions can deal with multicollinearity. Ridge regression model and Lasso regression model are constructed and compared. The root mean square error (RMSE) and adjusted R-squared are used to evaluate the performance of the models. This comparative study found that the Lasso regression model is performing better compared to the Ridge regression model. Based on this analysis, the selected variables includes the aspect of house size, age of house, condition of house and also the location of the house.

Keywords: Adjusted R-squared; Lasso Regression; Ridge Regression; Root mean square error (RMSE)

1. Introduction

In the year of 1990s, the house price in United State has increased sharply and from the year onward the price has been increasing for seven percent at the nation level annually [1]. The bubble of house price has brought up a negative growth of the economy in United State since the people do not afford to buy a house. Therefore, it is significant to study the growth of the house price as the pattern and the fluctuation of the data of house price which will provide the important information to the people who related [2]. The house price often can be given a great impact by the factors. It is a multidimensional study in factors that contribute a great impact on the house pricing [3]. Since there are numerous factors affect the prices of the house, the relationship between those factors is complex and a mathematical model can be seen to summarize the relationship of the influential factors and the house price. With the house price model, it will contribute important information to the real estate market which will indirectly influence the economic growth of the country [4]. Multicollinearity is commonly occur in the high dimensional data where it will give a serious problem whenever analysis using multiple regression [5, 6, 7]. Therefore, Ridge regression and Lasso regression can be used to cope the multicollinearity problem [8, 9]. In this study, Ridge regression model and Lasso regression model are constructed to predict the house price in the United State. This will provide the important information to government, financial firm such as bank, or even a homeowner. Besides that, the constructed house price model can improve the growth of the real estate market [10]. There are quite considerable studies were carried out to study the application of statistics [11, 12, 13].

2. Material and Method

Two methods are used to construct the house price model which are Ridge regression and Lasso regression. Before constructing the model, data pre-processing is necessary to be carried out. Data pre-processing is used to clean up the outliers and handle the missing value in the data. The outliers will be removed from the observations and the missing value for the numeric variables will be replaced by zero while for the missing value of categorical variables will be replaced by none. Then the data has been split into two part which train data set for constructing the models and test data set is used for the model validation. The performance of the Ridge regression and Lasso regression are evaluated based on the root mean square error and adjusted R-squared.

2.1. Ridge Regression

Ridge regression is modifying the least squares method which to allow to have biased estimators of the regression coefficients in the regression model. Ridge regression put a particular form of constraints on parameters. β^{ridge} which to be used in minimizing the penalized sum of squares in the equation 3.1 [14].

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^k x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^k \beta_j^2 \quad (3.1)$$

where y_i is the value of the response variable in the i^{th} trial, β_0 is the intercept coefficient, β_j is the regression coefficient for $j = 1, \dots, k$, x_{ij} is the j^{th} component of x_i which x_i is a known con-

RESEARCH



Prediction of compressive strength of GGBFS and Flyash-based geopolymer composite by linear regression, lasso regression, and ridge regression

Ujjwal Sharma¹ · Nakul Gupta¹ · Manvendra Verma¹

Received: 8 May 2023 / Accepted: 16 May 2023 / Published online: 29 May 2023
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2023

Abstract

This study aims to propose an efficient technique for improving the compressive strength of geopolymer concrete using Flyash and GGBS. Various regression models, including linear regression, lasso regression, and ridge regression, were compared to predict the compressive strength. The research holds significance in understanding the strengths and limitations of these regression techniques for real-world data scenarios, aiding data scientists and promoting advancements in machine learning. The outcomes can have practical implications in various domains, enhancing knowledge and comprehension of regression techniques and their applications. The dataset consisted of 147 samples with 11 independent variables and 1 dependent variable, and statistical measurements, such as MSE, MAE, and RMSE, were used to evaluate the accuracy of the models. These findings underscore the importance of selecting the appropriate regression model tailored to the specific dataset to ensure accurate predictions and reliable insights. Based on these findings, it can be concluded that the Linear Regression model is the most suitable for predicting compressive strength based on the given dataset. However, further research and analysis may be required to explore additional factors that could improve the accuracy of the models and enhance the understanding of the relationship between the independent variables and compressive strength.

Keywords Geopolymer concrete · Mechanical properties · Machine learning · GGBFS · Regression analysis

Introduction

In essence, concrete is composed of a blend of aggregates, paste, and, on occasion, an additional substance. Once the cement and water in the paste react chemically, the resulting hardened paste binds the aggregate together, creating a firm substance (Kosmatka, Kerkhoff and Panarese, 2002; Thiagarajan, R and Doddurani, 2013). Similarly, the amalgamation of Flyash and GGBS, consisting of several mineral additives, can act as a substitute for cement as a binder in the formation of geopolymer concrete composite (Verma &

Dev, 2021, 2022a, 2022b, 2022c; Verma, Dev, et al., 2022). Geopolymer is produced through the polymerization process that occurs when source materials rich in Si (Silicon) and (Al) Aluminum are combined with an alkaline liquid or by-products like fly ash and rice husk ash. This chemical reaction leads to the formation of a strong binding material, which replaces traditional cement in construction (Provis & Van Deventer, 2014; Provis et al., 2009).

GGBFS, a residual material from the blast furnaces used in iron production, is commonly referred to as Ground Granulated Blast-Furnace Slag. The primary constituents of GGBFS are silicate and aluminosilicate derived from molten calcium, which necessitates its intermittent elimination from the blast furnace. The chemical structure of GGBFS, akin to fly ash, is liable to differ based on the type of raw materials utilized in iron production, while its physical properties are contingent on the cooling process employed to solidify the molten material (Praveen Kumar Gupta, Rakesh Kumar, Y. K. Gupta, 2017). The molten material can be cooled via water quenching or pelletization, producing glassy granulated particles of varying sizes. GGBFS is ground to < 45 µm

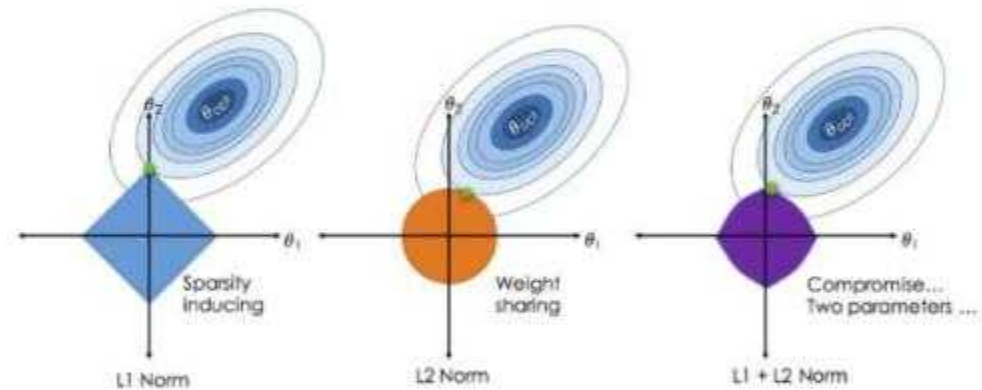
✉ Manvendra Verma
Manvendra.verma@glu.ac.in
Ujjwal Sharma
Ujjwal1205@gmail.com
Nakul Gupta
Nakul.gupta@glu.ac.in

¹ Department of Civil Engineering, GLA University, Uttar Pradesh, Mathura, India



What is Elastic net regularization?

Elastic Net Regression



Elastic Net

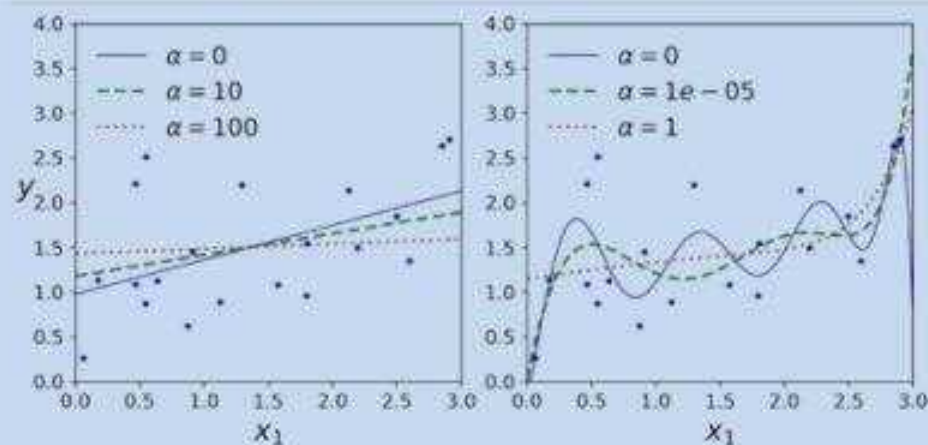
A generalization of the ridge and lasso penalties, called the *elastic net* (Zou and Hastie [2005](#)), combines the two penalties:

$$\text{minimize} \left(SSE + \lambda_1 \sum_{j=1}^p \beta_j^2 + \lambda_2 \sum_{j=1}^p |\beta_j| \right) \quad (6.5)$$

Elastic Net Regularization

Elastic Net combines both L1 (Lasso) and L2 (Ridge) regularization techniques to overcome the limitations of each technique.

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k$$



$$\text{Elastic Net : Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{i=1}^k \left(r |\theta_i| + \frac{1-r}{2} \theta_i^2 \right)$$

Note that bias term θ_0 is not regularized and in 2nd term the sum starts at $i=1$, not 0

The regularization term is a simple mix of both Ridge and Lasso's regularization terms, and you can control the mix ratio r . When $r = 0$, Elastic Net is equivalent to Ridge Regression, and when $r = 1$, it is equivalent to Lasso Regression.

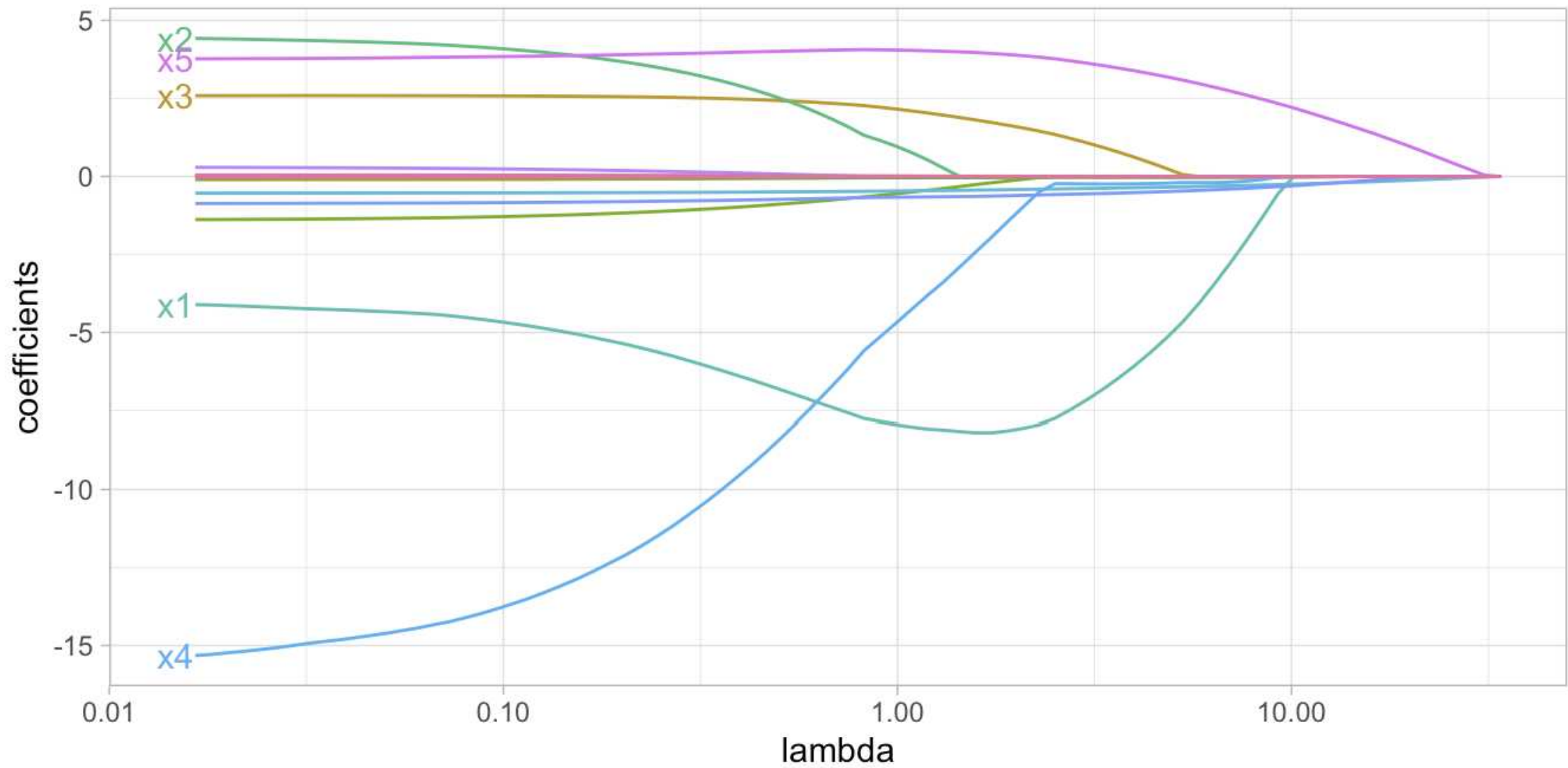


Figure 6.4: Elastic net coefficients as λ grows from $0 \rightarrow \infty$.

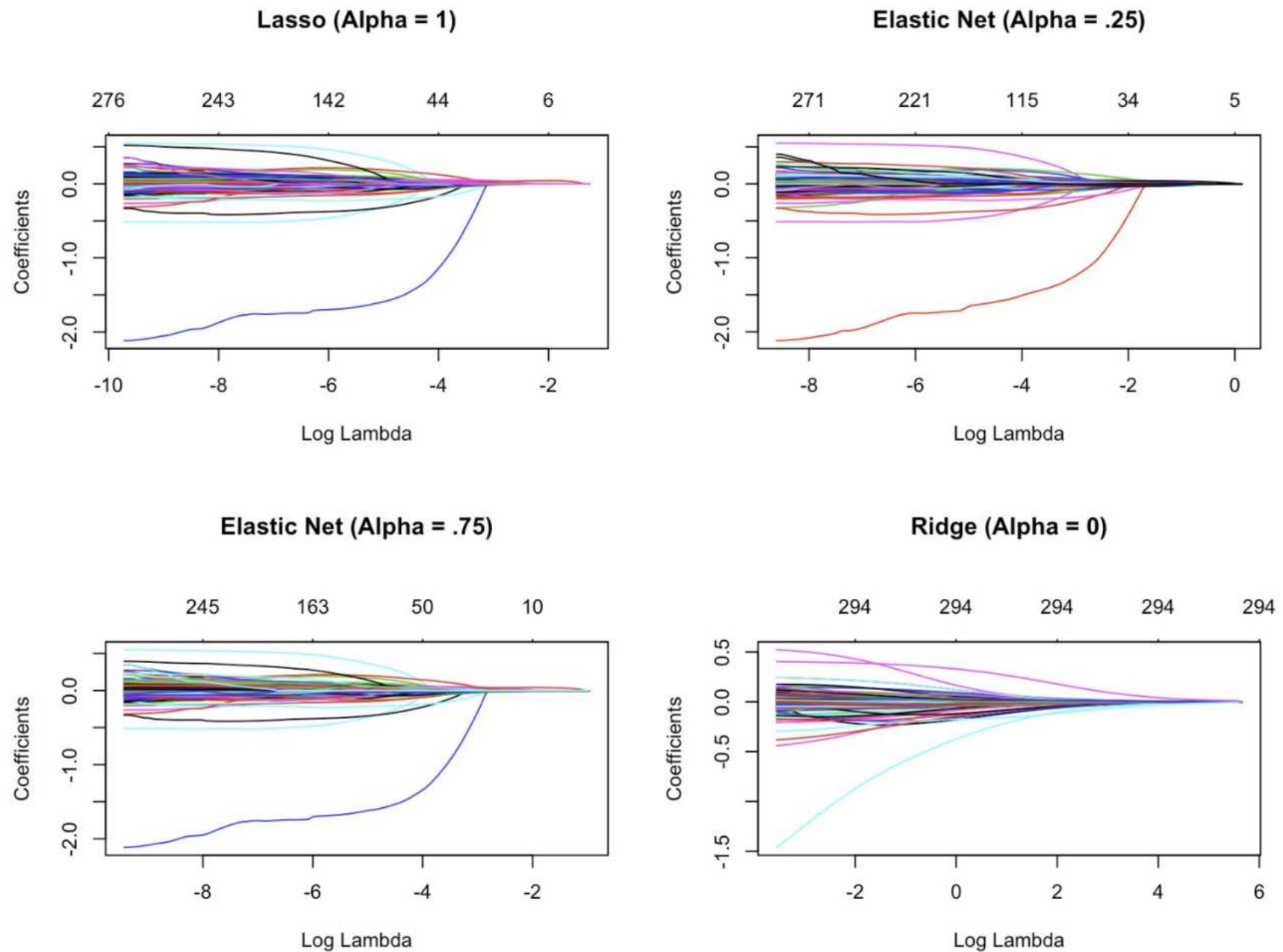


Figure 6.8: Coefficients for various penalty parameters.

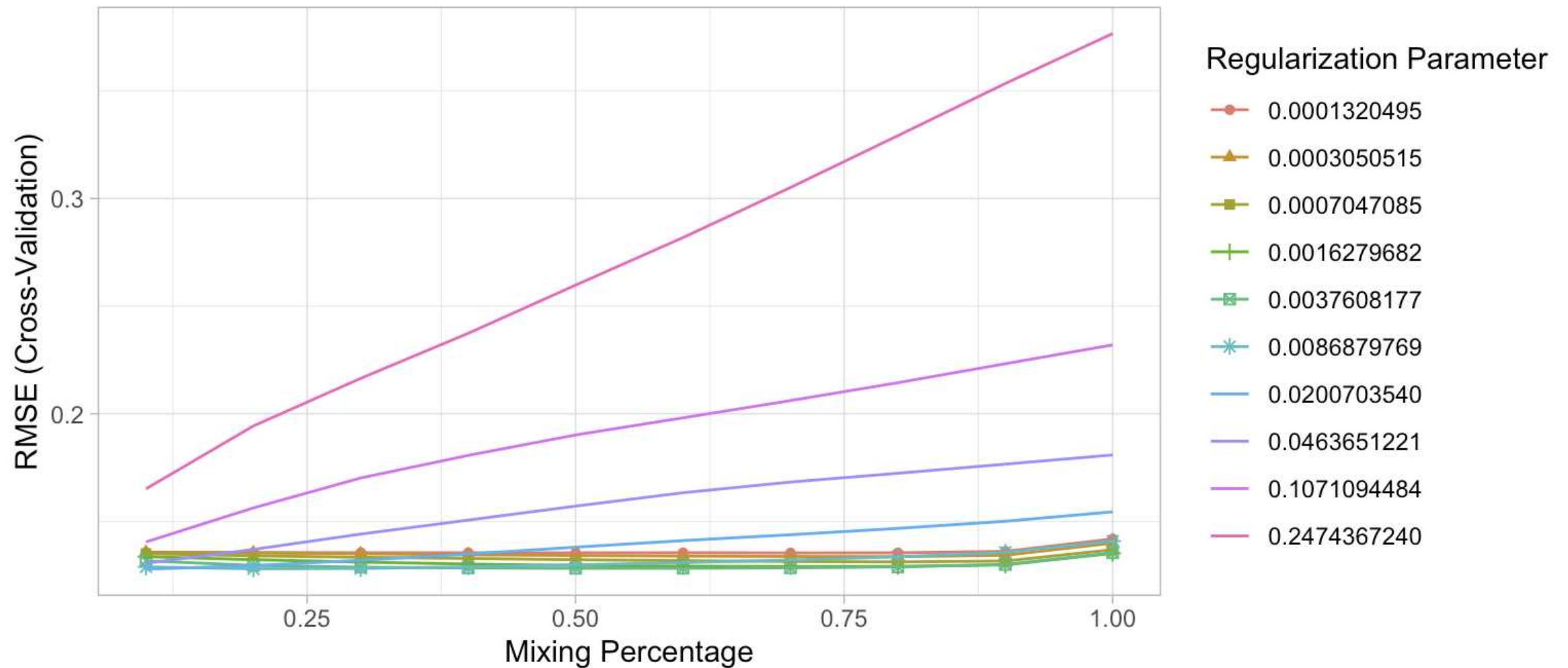
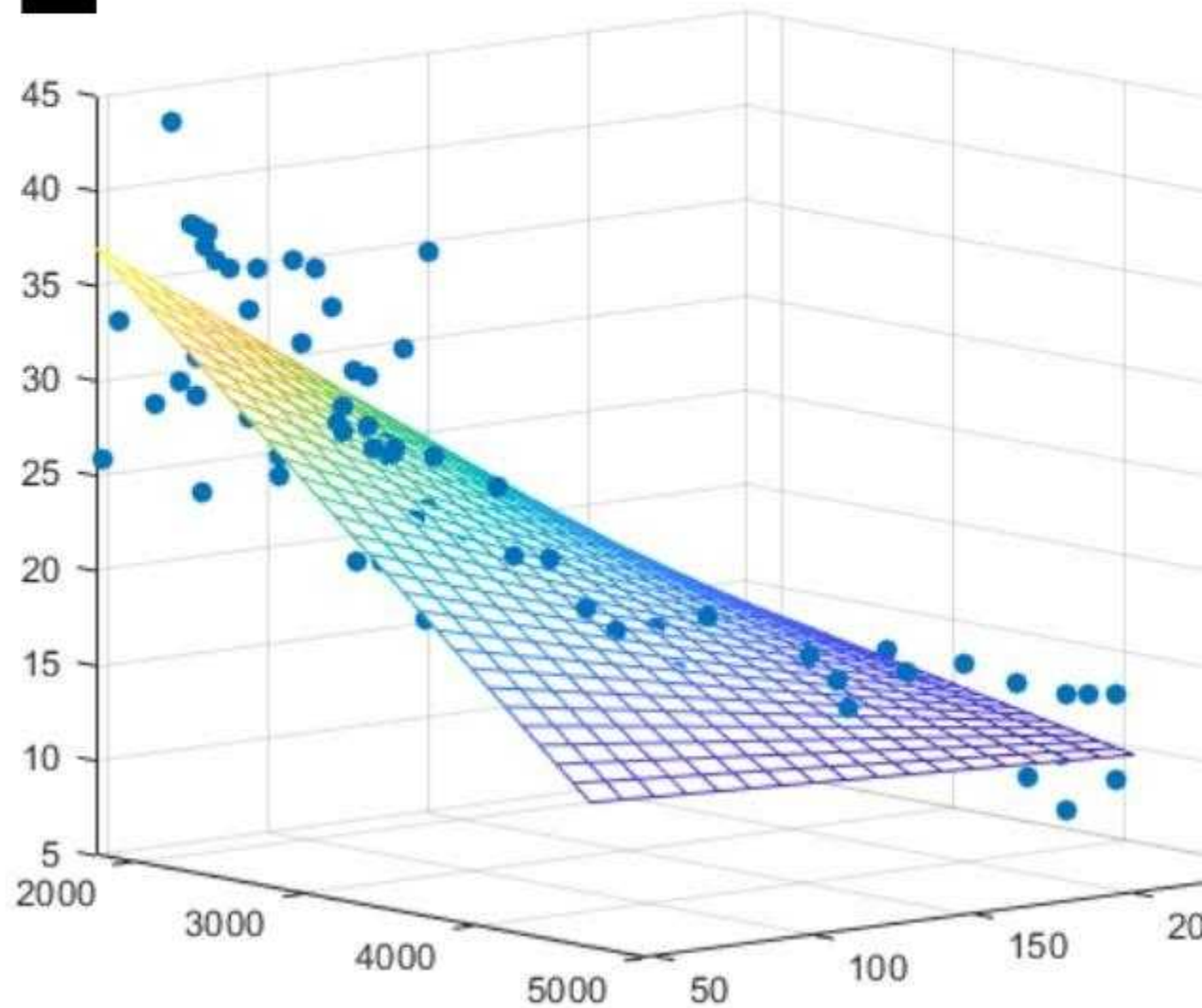


Figure 6.9: The 10-fold cross validation RMSE across 10 alpha values (x-axis) and 10 lambda values (line color).

MULTIVARIATE ADAPTIVE REGRESSION SPLINE



Multivariate Adaptative Regression Spline, MARS

- ◆ MARS is a nonparametric regression technique that can be seen as an extension of linear models.
- ◆ It automatically models nonlinearities and interactions between variables.
- ◆ https://en.wikipedia.org/wiki/Multivariate_adaptive_regression_spline

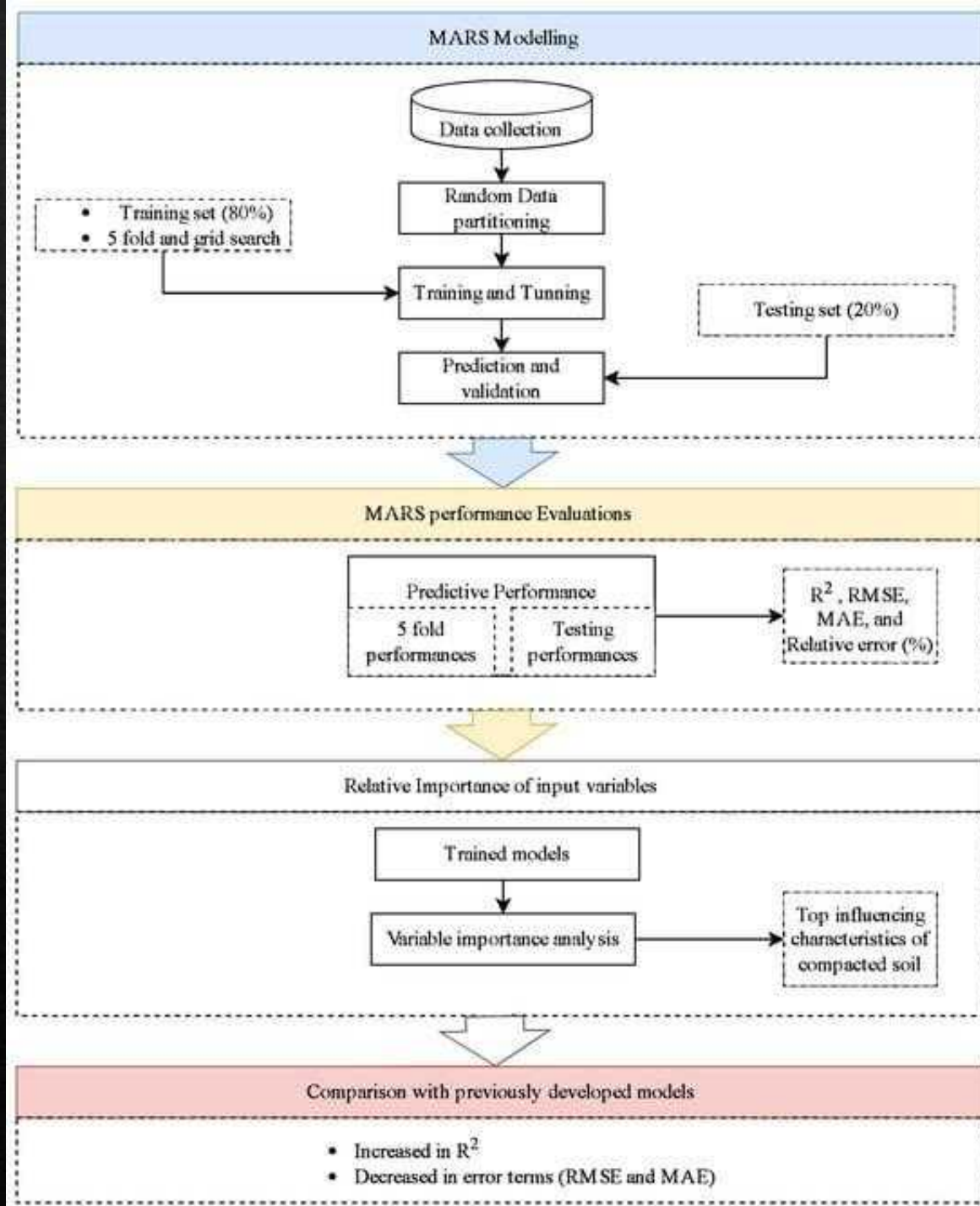
MARS: Multivariate Adaptive Regression Splines - model selection

MARS functions typically overfit the data and so a backward deletion procedure is applied

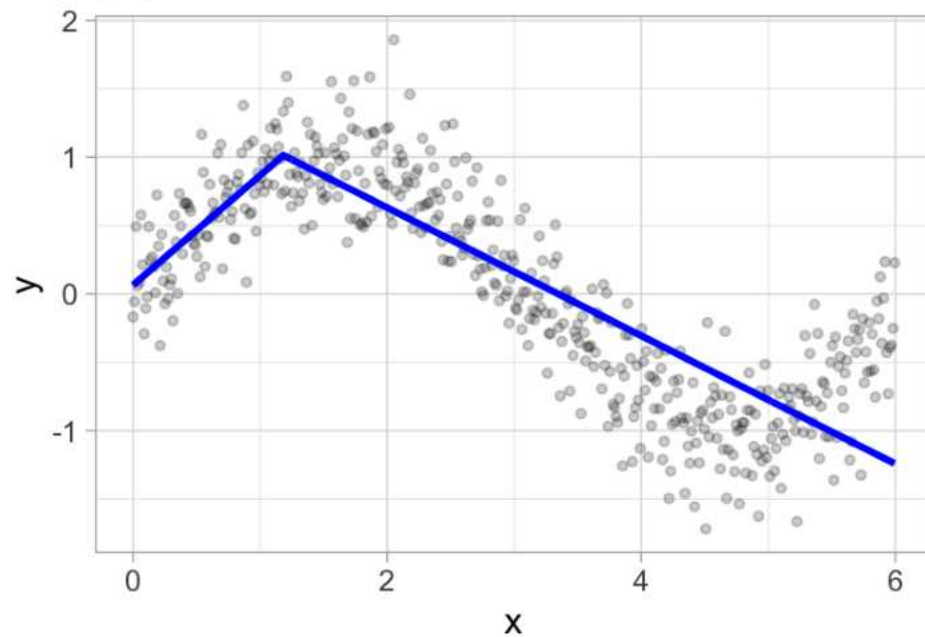
The size of the model is determined by Generalized Cross Validation

An upper limit can be set on the order of interaction

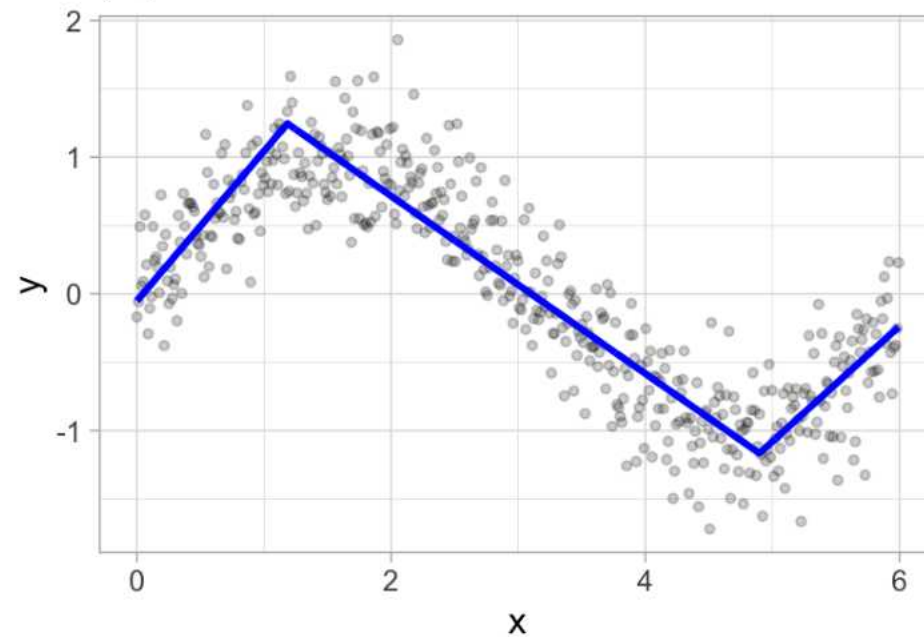




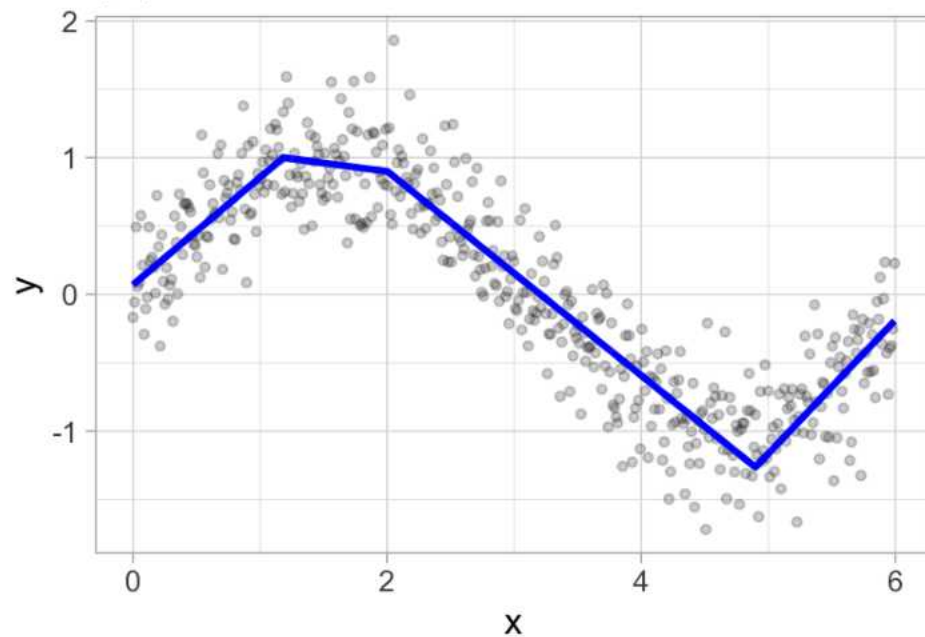
(A) One knot



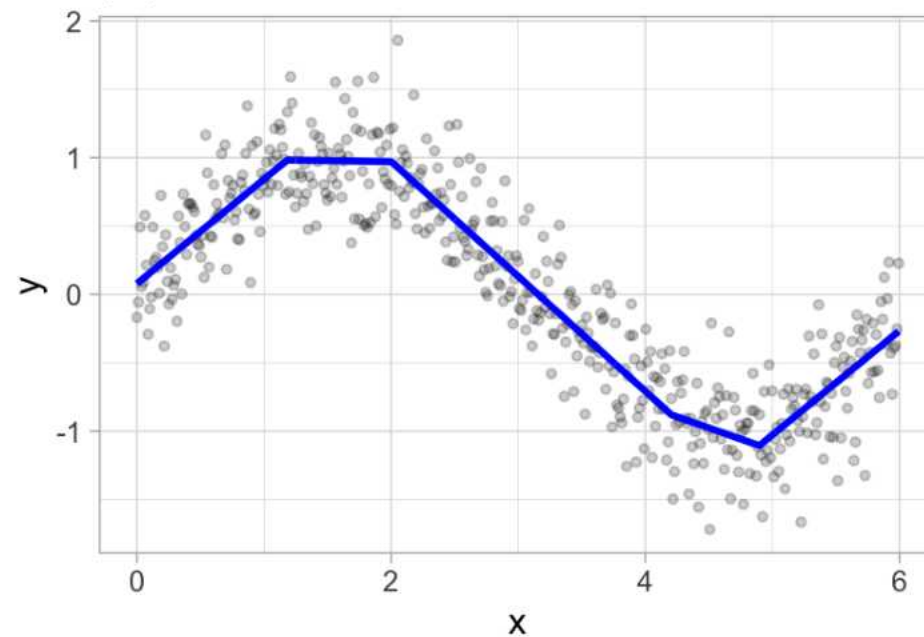
(B) Two knots



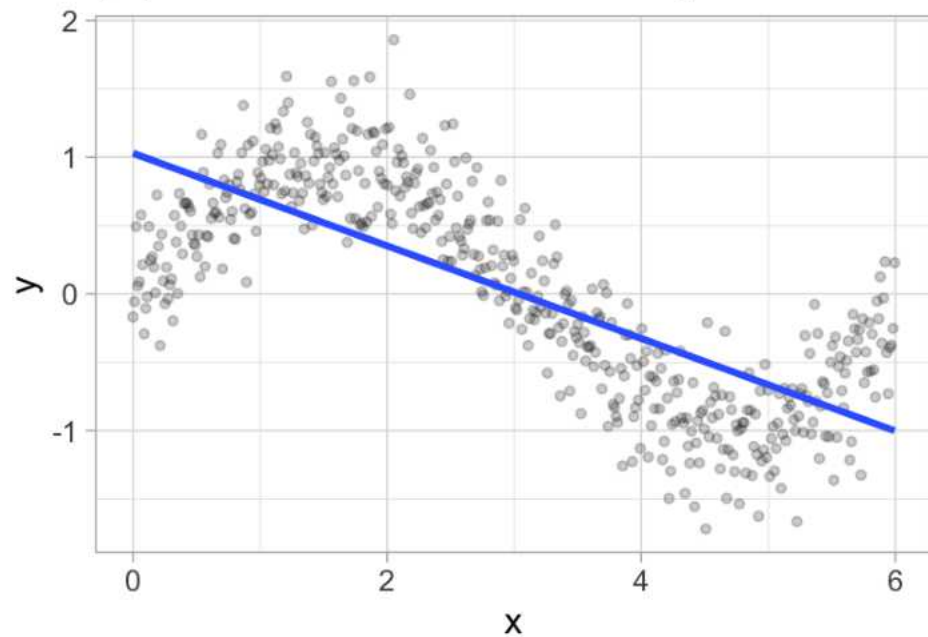
(C) Three knots



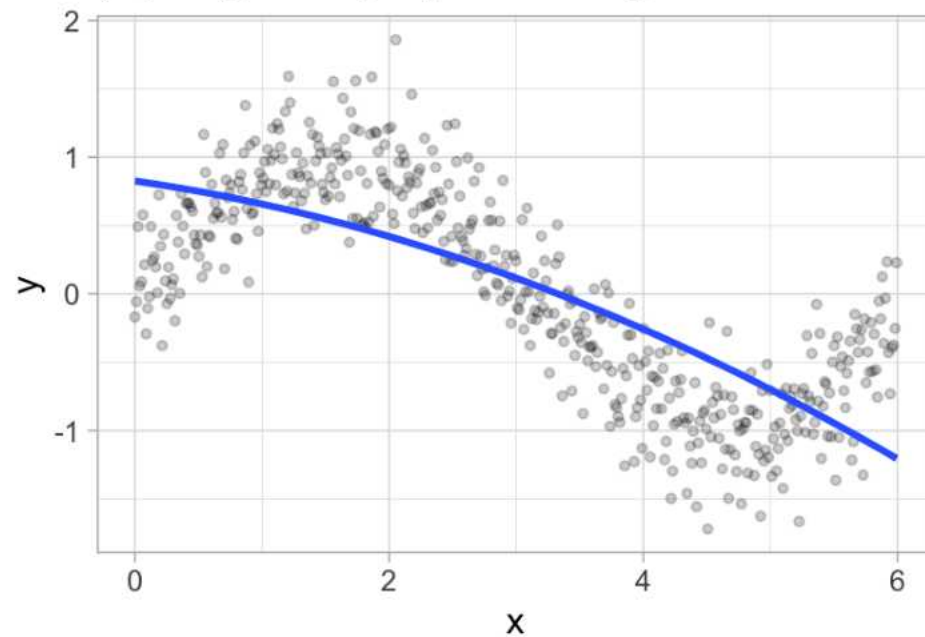
(D) Four knots



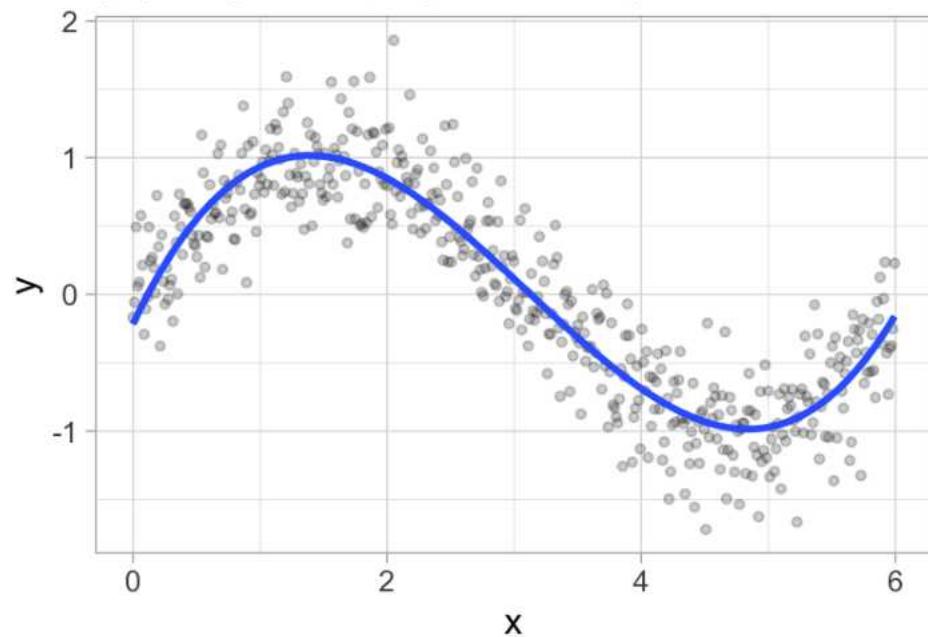
(A) Assumed linear relationship



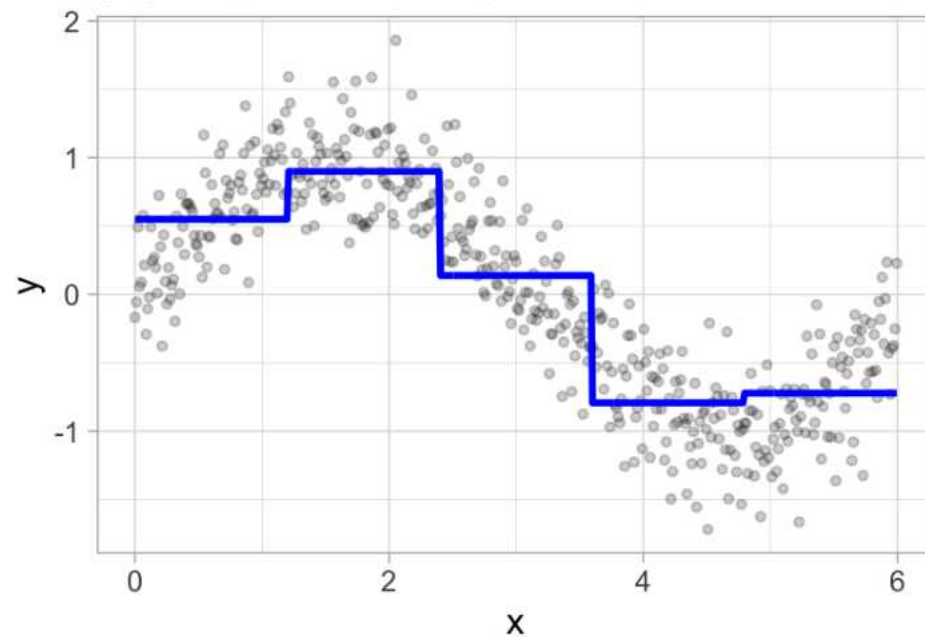
(B) Degree-2 polynomial regression

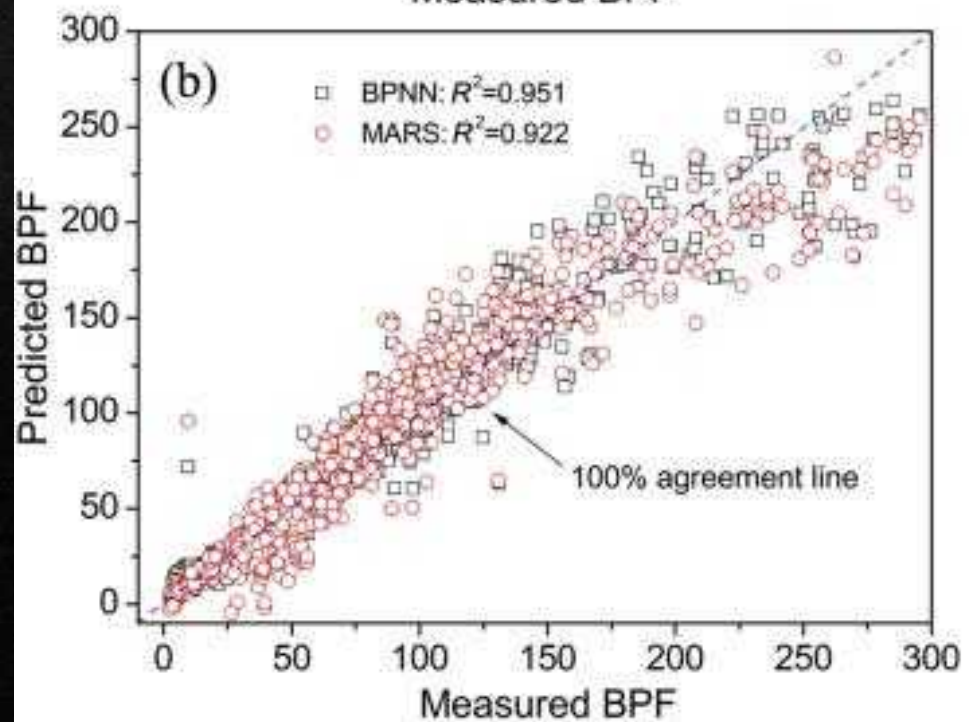
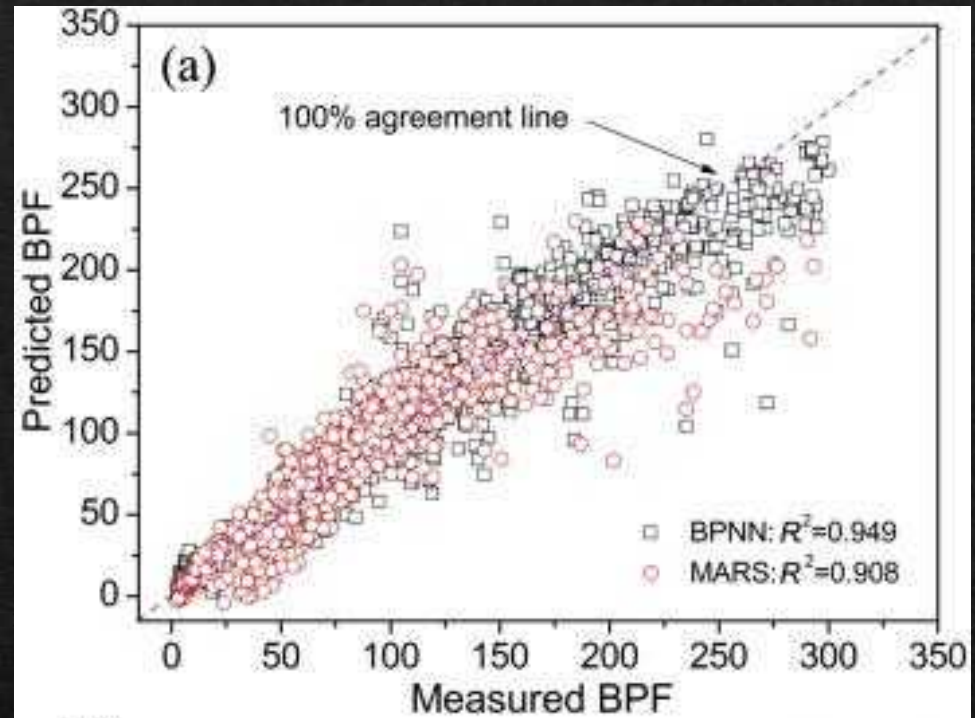


(C) Degree-3 polynomial regression

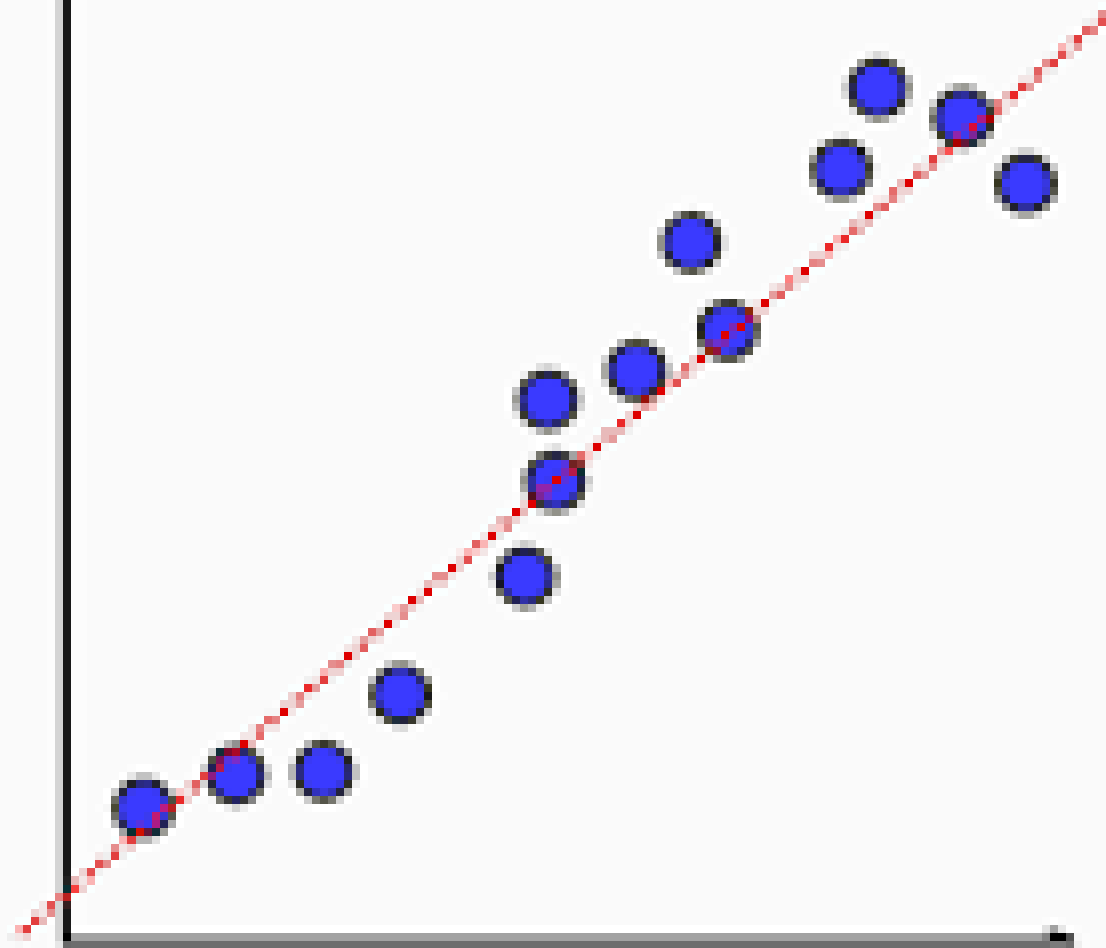


(D) Step function regression

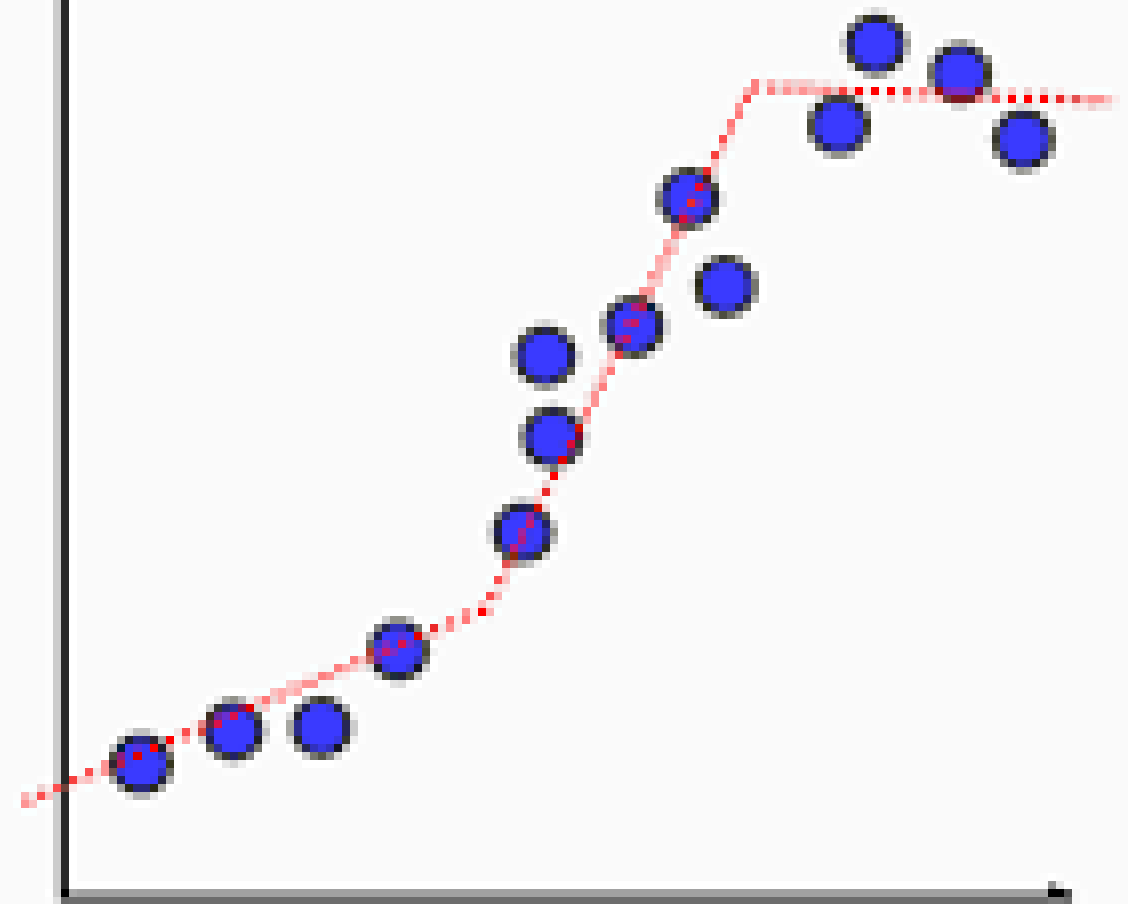


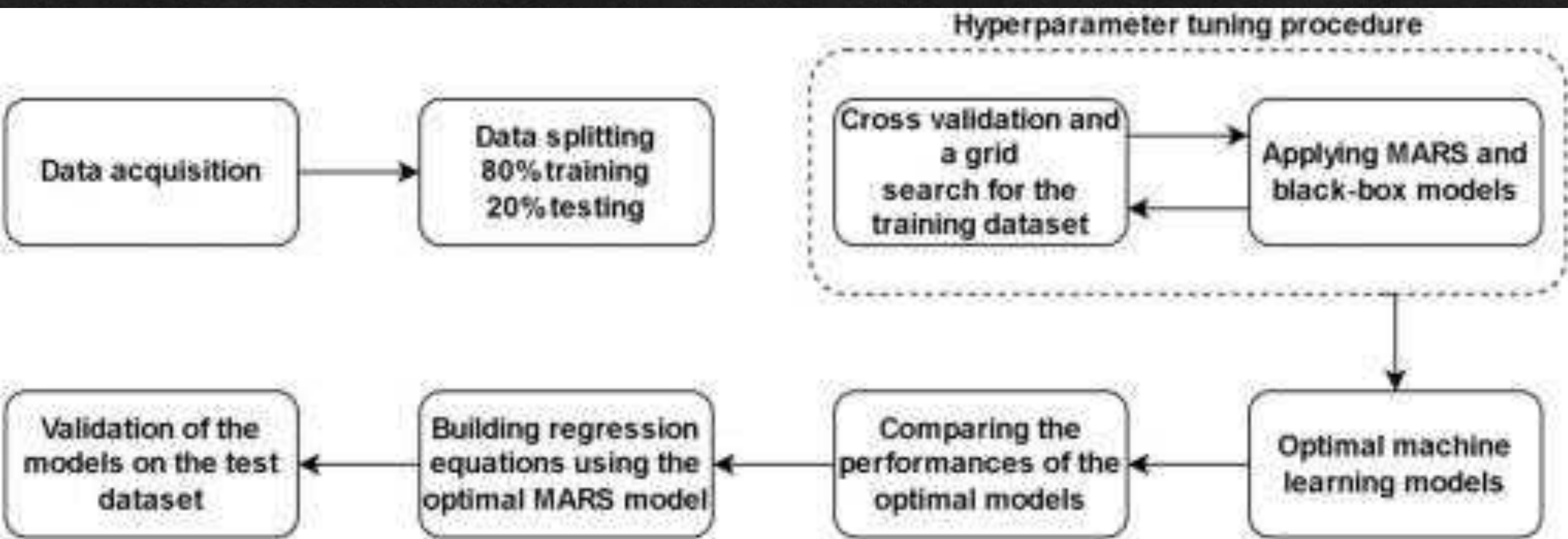


Ordinary linear regression



MARS





Multivariate Adaptive Regression Splines as a Tool to Improve the Accuracy of Parts Produced by FSPIF

Johan Verbert^a, Amar Kumar Behera^b, Bert Lauwers^c, Joost R. Duflou^d

Katholieke Universiteit Leuven, Department of Mechanical Engineering, Celestijnenlaan 300B, B-3001 Leuven, Belgium

^ajohan.verbert@gmail.com, ^bamarkumar.behera@cib.kuleuven.be,

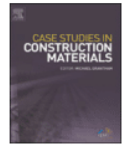
^cbert.lauwers@mech.kuleuven.be, ^djoost.duflou@mech.kuleuven.be

Keywords: incremental forming, features, regression splines, accuracy, SPIF

Abstract. Feature Assisted Single Point Incremental Forming (FSPIF) is a technique to increase the accuracy of the SPIF process. FSPIF generates an optimized toolpath based on the features detected in the workpiece geometry and using knowledge of the behavior of these features during incremental forming. Using this optimized toolpath, parts can be formed with higher accuracy. The prediction of the dimensional deviations occurring in different features during forming as a function of their type (e.g. planar, ruled, freeform or ribs) and various process parameters, such as sheet thickness, wall angle, tool diameter, rolling direction, etc., is an important step in the FSPIF method. Due to the great number of parameters and combinations that are possible, a mathematical tool should be used in order to automate the prediction process. One such tool is MARS or Multivariate Adaptive Regression Splines, a fast, non-parametric multivariate regression technique with automatic variable selection, which generates continuous surfaces as a response function. In this paper, the authors describe and validate the use of MARS as a tool to predict deviations in uncompensated tests by training the MARS model using only a limited number of experiments. Using this validated model, compensation strategies are developed and implemented, which have shown significant improvements in accuracy in new test cases.

Introduction

Feature Assisted Single Point Incremental Forming (FSPIF) is an optimized toolpath generation method for incremental forming of sheet metal parts. This method uses feature detection to split a workpiece into a configuration of planes, edges, freeform and ruled features [1]. For each of these features an optimized toolpath is determined depending on the knowledge of the behavior of each individual feature [1] and its interaction with neighbouring features [2]. Using this method, a single pass toolpath is generated which leads to more accurate parts compared to a standard contouring toolpath. However, in order to generate the compensated single pass toolpath, it is important to predict the deviations that occur in an uncompensated forming test. These deviations, in turn, depend on the type of feature, and various process parameters such as: sheet thickness, wall angle, tool diameter, rolling direction, etc. In order to automate the prediction of these deviations, a mathematical tool is needed, which takes into account all potentially relevant process parameters and combinations thereof. One possible tool to do this is finite element analysis [3]. However, the finite element simulations require long computation times for convergence. Depending on the size and geometry of the part, and the choice of elements, typical computation times can range from 1 day to 2 weeks [4]. An alternative to FEM is the use of MARS or Multivariate Adaptive Regression Splines. MARS is a regression tool that enables automatic selection of variables and generation of continuous surfaces as a response function. MARS is especially suited for flexible regression modeling of high dimensional data. The model takes the form of an expansion in product spline basis functions, where the number of basis functions as well as the parameters associated with each one (product degree and knot locations) are automatically determined by the data [5]. Using the



Short communication

Application of Multivariate Adaptive Regression Splines (MARS) approach in prediction of compressive strength of eco-friendly concrete

Ali H. Naser^a, Ali H. Badr^a, Sadiq N. Henedy^b, Krzysztof Adam Ostrowski^{c,*}, Hamza Imran^{d,*}

^a Department of Reconstruction & Projects, University of Thi-Qar, Nasiriya City, Thi-Qar governorate 64001, Iraq

^b Department of Civil Engineering, Mazaya University College, Nasiriya City, Thi-Qar governorate 64001, Iraq

^c Faculty of Civil Engineering, Cracow University of Technology, 24 Warszawska Str., 31-155 Cracow, Poland

^d Department of Construction and Project, Al-karkh University of Science, 10081 Baghdad, Iraq

ARTICLE INFO

Keywords:

Machine learning
Multivariate Adaptive Regression Splines (MARS)
Compressive strength of concrete
Ground granulated blast-furnace slag
Recycled concrete aggregate
Eco-friendly concrete

ABSTRACT

Concrete is the most often used material in the building sector. The use of ground-granulated blast-furnace slag (GGBFS) and recycled concrete aggregates (RCA) in concrete reduces its negative environmental impact. However, increasing the number of ingredients in concrete complicates the prediction of its compressive strength. This research aims to develop a machine learning model for predicting the compressive strength of eco-friendly concrete using Multivariate Adaptive Regression Splines (MARS). MARS is a well-known technique for creating predictive modeling equations using experimental data. A data set of 161 concrete specimens was obtained to train and evaluate the machine learning algorithms for this work. The findings of the MARS prediction algorithm were compared to those of the Support Vector Machine and the Random Forest regression algorithms (black-box models) using the training set. A hyperparameter tuning procedure was applied to find the optimal MARS, RF, and SVM model parameters. The results of 5-fold cross-validation indicate that the MARS optimal model has the highest coefficient of determination (R^2), with a value equal to 0.889, and the lowest Root Mean Squared Error (RMSE) measure, with a value equal to 4.110 MPa. For our case, the performance metrics measures proved the superiority of the MARS model when compared with RF and SVM. The final MARS optimal prediction equations were built using a full training set and were then evaluated using unseen data (testing set). Our model has the potential to help civil engineers in designing durable infrastructures.

1. Introduction

Portland cement is the most widely used building material and the most commonly used industrial commodity. However, the massive environmental impact of its exploitation and the increased regional scarcity of natural aggregates (NA) are international issues connected with the manufacturing of Portland cement concrete. In 2015, the global consumption of NAs was predicted to be 48.3 billion tonnes, with a 5-year growth rate of more than 5 % [1]. Furthermore, the growing volume of construction and demolition waste

* Corresponding authors.

E-mail addresses: krzysztof.ostrowski.1@pk.edu.pl (K.A. Ostrowski), hamza.ali1990@kus.edu.iq (H. Imran).

<https://doi.org/10.1016/j.cscm.2022.e01262>

Received 17 March 2022; Received in revised form 11 June 2022; Accepted 21 June 2022

Available online 23 June 2022

2214-5095/© 2022 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).